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**Internet Technology, Web Design and Web Technology: Internet Technology and Protocol, Internet Connectivity, Internet Network, Services on Internet, Electronic Mail, Current Trends on Internet, Web Publishing and Browsing, HTML Programming Basics, Interactivity Tools Internet. Security Management Concepts, Information Privacy and Copyright Issues, Web Technology: protocols, development strategies, applications, Web project and team. Web Page Designing, Scripting, Server Site Programming.**

## Chapter - 1

**Internet Technology, Web Design and Web Technology: Internet Technology and Protocol, Internet Connectivity**

# What is the internet

**Internet** is a global system of interconnected computer networks that use the standard Internet protocol (often called TCP/IP) to serve billions of users worldwide. It is a network of networks that consists of millions of private, public, academic, business, and government networks, of local to global scope, that are linked by networking technologies.

# INTERNET

Internet is a worldwide network of networks that uses the standard Internet protocol suite (TCP/IP) to link several billion devices worldwide.

## Hardware and software Requirements

To connect to the Internet we need the following four things:

1. A computer
2. A modem and telephone line (if you are using dial up access) A data line of some sort (if you are not using dial up access)
3. An Internet browser (software) and software to connect you to the ISP
4. An account with an Internet Service Provider (ISP)

# Applications of Internet:

- Sending and receiving email
- Searching and browsing information archives
- Copying files between computers
- Conducting financial transactions
- Navigating (in your car, smart scooter, smart bike, or other)
- Playing interactive games.
- Video and music streaming
- Chat or voice communication (direct messaging, video conferencing) etc.

# History of the internet

- The **history of the Internet** begins with the development of electronic computers in the 1950s.
- The US Department of Defense awarded contracts as early as the 1960s, including for the development of the ARPANET project.
- The first message was sent over the ARPANET in 1969 from computer science laboratory at University of California, Los Angeles to the second network at Stanford Research Institute (SRI).

# Who Owns the Internet?

There are many organizations, corporations, governments, schools, private citizens and service providers that all own pieces of the infrastructure, but there is no one body that owns it all. There are, however, organizations that oversee and standardize what happens on the Internet and assign IP addresses and domain names, such as the National Science Foundation, the Internet Engineering Task Force, ICANN , InterNIC and the Internet Architecture Board.

# Intranet

An intranet is a computer network for sharing information, collaboration tools, operational systems, and other computing services within an organization, usually to the exclusion of access by outsiders.

## Uses of the intranet:

- Streamlining everyday activities by making repeated tasks more feasible.
- Centralizing and managing important information and company data in a single database.
- Making collaboration easier since information can be shared across the entire network.
- Providing personalized content to employees based on their role within the company.
- Improving internal communication by making employee directories, company news and organization charts readily available.
- Providing fast and easy access to information about company policies, benefits and updates.

## Extranet:

The extranet is a private network that uses the internet that allows people outside a business partners, vendors or authorized customers to access business information.





| Parameter      | Internet                                                                                                                                                                 | Intranet                                                           |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Usage          | Public                                                                                                                                                                   | Private                                                            |
| User Types     | Any user having dial up of Internet access line.                                                                                                                         | Organization employees and Internal company departments            |
| Usage          | Access all kind of information                                                                                                                                           | Internal employee communication , telephone directories etc.       |
| Security       | Low security. Configured under 0 security level in firewall                                                                                                              | High security. Configured under 100 security level in firewall     |
| Regulated by   | Internet Architecture Board (IAB): Oversees the technical and engineering development of the IETF and IRTF. Internet Corporation for Assigned Names and Numbers (ICANN). | It is regulated by an organization.                                |
| Coverage       | Wide Area                                                                                                                                                                | Within an organization                                             |
| Access         | Large number of users                                                                                                                                                    | Limited number of users                                            |
| System failure | Unpredictable                                                                                                                                                            | System availability is high since system is monitored by authority |

## INTERNET SERVICE PROVIDER

An Internet service provider (ISP) is an organization that provides a myriad of services for accessing, using, or participating in the Internet.

### Factors to consider when choosing ISP

- **Bandwidth** : Data transferring speed provided by ISP Company.
- **Availability** : Availability of Network & performance to its users.
- **COST** : Refers pricing of the connection as well as services
- **Network security** : It is an important issue related to the network over the Internet. Everyone has its own private information being safe on their servers & nodes.
- **Customer Services** : Better Customer service is highly required an ISP.
- **Location & they need for speed** : It is an also a important factor, when we looking an Internet provider is the location where we live or work. A better location refers a good level of customers support.

### Example

IP address 213 . 23 . 47 . 37

Mask 255 . 255 . 255 . 240

Subnetwork address 213 . 23 . 47 . 32



# TCP/IP Protocol Suite

TCP/IP Suite consists of Four layer

## 1. Network Interface layer:

- It include the function of physical layer and data link layer
- It is the bottom layer of TCP/IP Model lies below the Internet Layer.
- Function of this layer is to connect the Host To Network & inform the upper layers so that they could start sending the data packets.
- This layer varies from network to network.
- Host To Network Layer protocols
  - SLIP(Serial Line IP)
  - PPP( Point To Point Protocol)

TCP/IP

Application

Transport

Internet

Network Interface

## 2. Internet Layer :

- The internet layer is exactly same to the network layer of OSI model
- IP is the primary protocol operating at this layer and it provides data encapsulation routing, addressing and fragmentation services to the protocols at the transport layer above it.
- The five network layer protocols are as follows:-
- The five network layer protocols are as follows:-
  - ARP
  - RARP
  - IP
  - ICMP
  - IGMP
- IP is an important protocol in this layer.

## **ARP ( Address resolution protocol )**

- Every machine on internet has one address, these address cannot actually be used for sending packets data because data link layer does not understand the internet address.
- ARP Provides an essential services when TCP/IP is running in LAN.

## **RARP ( Reverse Address Resolution Protocol )**

- The Reverse Address Resolution Protocol is used to obtain the IP address of a Host based on Its physical address. That is , it preform a job that is exactly opposite to that of ARP.
- RARP works in a very similar way to ARP. but in the exactly opposite direction , as shown in the side figure.

## **Internet Protocol ( IP )**

- Internet Protocol is very important protocol present in this network layer.
- IP is the protocol responsible for carrying data, generated by nearly all the other TCP/IP protocol, from the source system to its destination.

- **IP Features :** -

- **Unreliable :** IP is unreliable , it means that it does not provide a guarantee that a datagram send from a source computer definitely will arrive at the destination.
- **Connectionless :** IP services is similar to the postal service.
- It is possible that the order in which the message are sent and the order in which they are received is different.

### 3. Transport Layer

- The transport layer runs on the top of internet layer and is mainly concerned with transport of packets from the source to the destination.
- The main function of transport layer is to deliver packets between the end points.
- In TCP/IP, the transport layer include to protocols : TCP and UDP.

# Comparison between TCP And UDP

| Protocol                  | UDP                                                   | TCP                                                           |
|---------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| Data Transfer             | Data is sent in discrete packages by the application. | Data is sent by the application with no particular structure. |
| Transmission speed        | Very High                                             | High but not as high as UDP                                   |
| Protocol connection setup | Connectionless                                        | Connection oriented                                           |
| Used                      | UDP is useful when speed of delivery is critical      | TCP is useful for transmission of data without error.         |



## 4. Application Layer

- The TCP/IP Model Does Not have Session or presentation layer on the top of the transport layer.
- It is just has the application layer. It contains all higher level protocols.
- Higher Level Protocols Used in application layer are as Follows:
- **TELNET:** - the virtual terminal protocol allows a user on one machine to log onto a distant machine and work there.
- **FTP:** - File Transfer protocol provides a way to move data efficiently from one machine to another.
- **SMTP:** - Simple mail transfer protocol developed for email transfer.
- **DNS:** -Domain Name System Protocol is used for mapping the host names onto their network address.
- **HTTP:** -hyper Text transfer Protocol is used for fetching pages on the world wide web (WWW) and many others.

# Internet Connections

There are many ways a personal electronic device can connect to the internet. They all use different hardware and each has a range of connection speeds:

1. Dial-Up (Analog 56K).
2. DSL
3. Cable
4. Wireless
5. Satellite
6. Cellular

# 1. Dial-Up

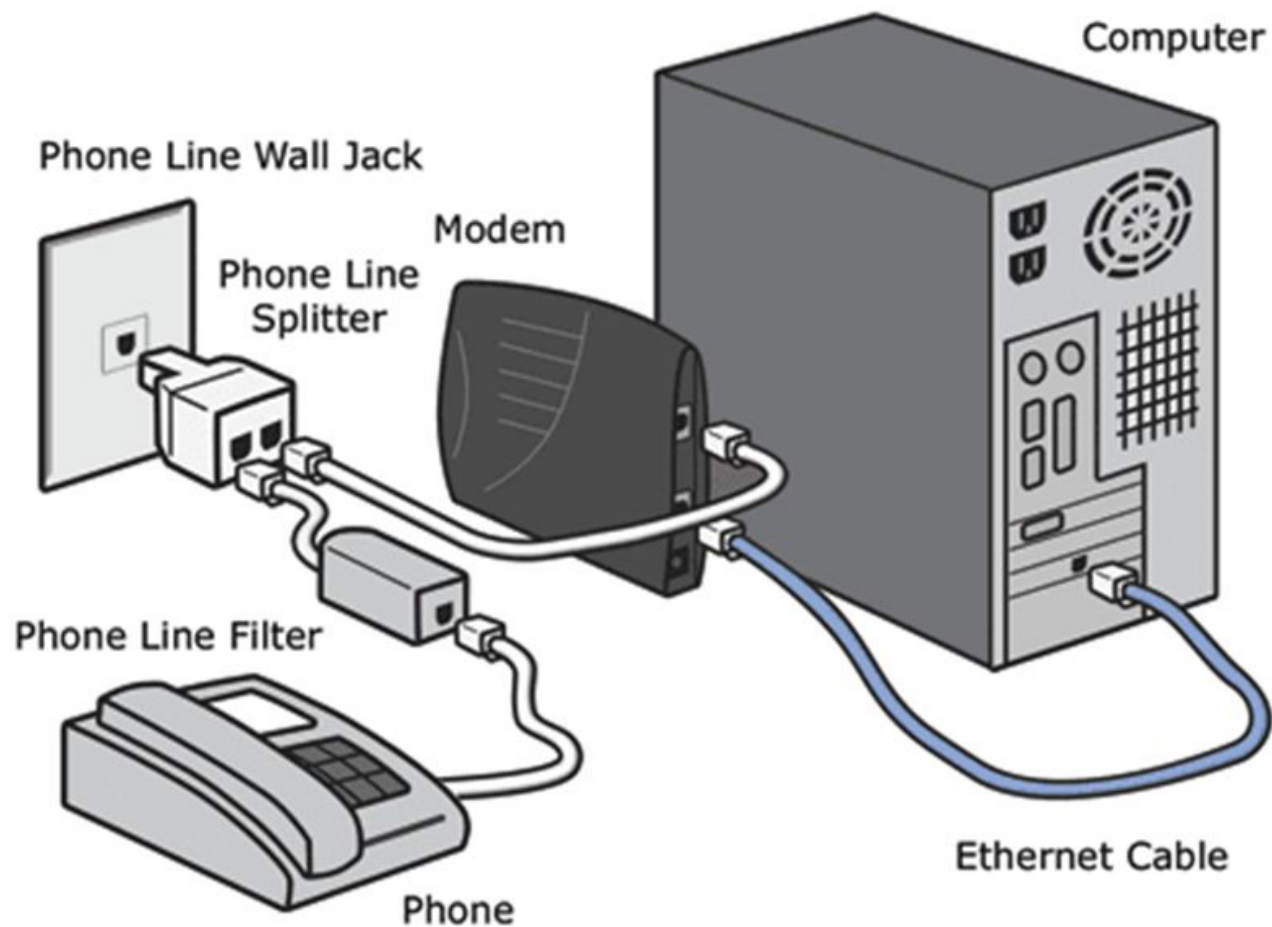
- Dial-up access is cheap but slow.
- A modem connects to the Internet after the computer dials a phone number.
- This analog signal is converted to digital via the modem and sent over a land-line serviced by a public telephone network.
- Telephone lines are variable in quality and the connection can be poor at times.
- The lines regularly experience interference and this affects the speed.
- Since a computer or other device shares the same line as the telephone, they can't be active at the same time.
- Speed: 28K to 56K

# Dial-Up Access



## 2. DSL

- DSL stands for Digital Subscriber Line.
- It is an internet connection that is always “on”.
- Uses 2 lines so your phone is not tied up when your computer is connected.
- There is also no need to dial a phone number to connect.
- DSL uses a router to transport data.
- Speed: 128K to 8 Mbps



GATE

### 3. Cable

- Cable provides an internet connection through a cable modem and operates over cable TV lines.
- There are different speeds depending on if you are uploading data transmissions or downloading.
- The coax cable provides a much greater bandwidth over dial-up or DSL telephone lines.
- Speed: 512K to 20 Mbp.

Inside Your Home



Cable Inside Your Home



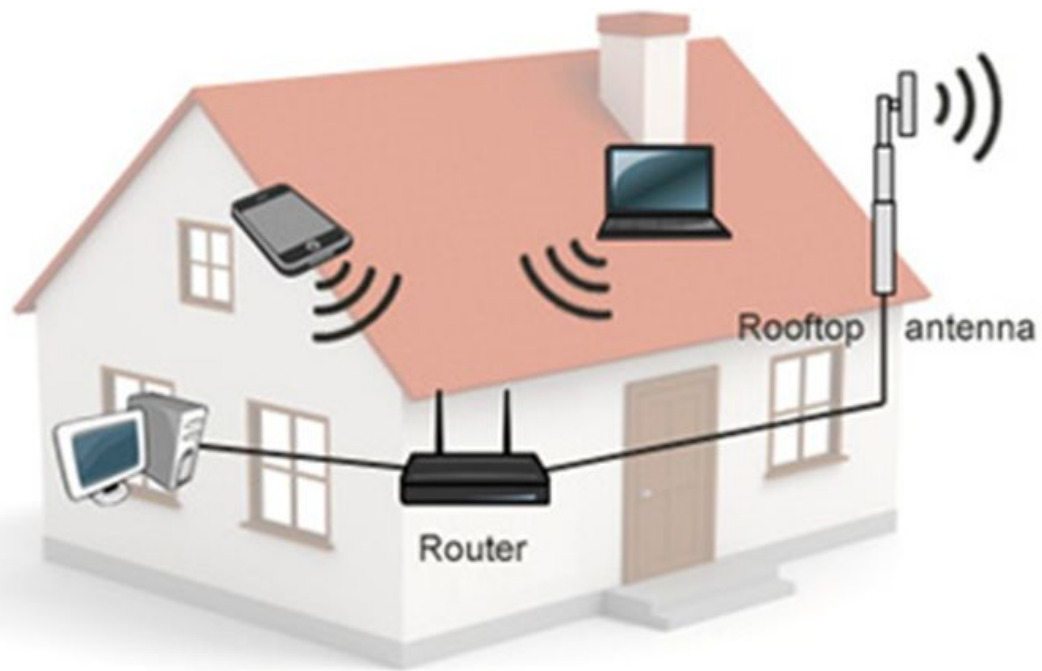
To/From  
Cable  
Company





## 4. Wireless

- Wireless, or Wi-Fi, does not use telephone lines or cables to connect to the internet.
- It uses radio frequency.
- Wireless is also an always on connection and it can be accessed from just about anywhere.
- Speed: 5 Mbps to 20 Mbps.



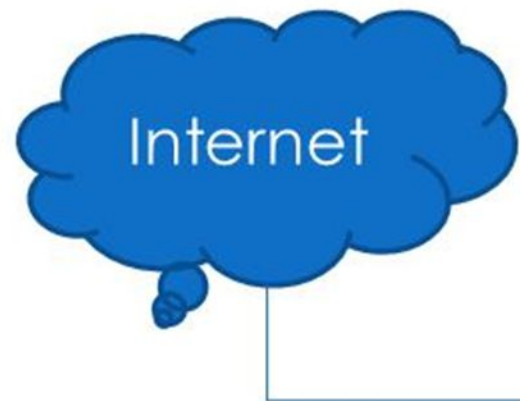
## 5. Satellite

- Satellite accesses the internet via a satellite in Earth's orbit.
- The enormous distance that a signal travels from earth to satellite and back again, provides a delayed connection compared to cable and DSL.
- Speed: 12K to 2.0 Mbps



## 6. Cellular

Cellular technology provides wireless Internet access through cell phones. The speeds vary depending on the provider, but the most common are 3G and 4G speeds. A 3G is a term that describes a 3<sup>rd</sup> generation cellular network obtaining mobile speeds of around 2.0 Mbps. 4G is the fourth generation of cellular wireless standards. The goal of 4G is to achieve peak mobile speeds of 100 Mbps but the reality is about 21 Mbps currently.



# MODEM

Modem, ( “modulator/demodulator”) is electronic device that convert digital data signals into modulated analog signals suitable for transmission over analog telecommunications circuits.

Working principle of MODEM.

## Modulator:

- This unit is used to convert the digital data from computer into analog data This process is called modulation
- This is done by adding a carrier signal to the digital signal.

## Demodulator:

- This unit is used to convert the analog data from telephone system into digital data
- This is done by eliminating the carrier signal from analog signal

# Types of modems

There are two types of modem

**Internal modem:** Internal Modem is modem that plugged directly into the CPU. Physically internal modem in the form of a card that is plugged into one of the expansion slots on the mainboard, usually on the ISA or PCI slot.

**External modem:** External Modem is modem that installed outside of the CPU. External modem connected to the CPU via the COM port or USB. This type of modem typically uses separate voltage source in the form of an adapter.



## Features of Modems:

- **Speed:** The speed at which the modem can send data in bps (bits per second). Typically modem speeds are: 300, 600, 1200, 2400, 4800, 9600, 14.4K, 19.2K, 28.8K bps
- **Auto Dial /Redial:** Smart Modems can dial the phone number and & auto redial if a busy signal is received.
- **Auto Answer:** Most modems can automatically answer the phone when an incoming call comes in. They have Ring Detect capability.
- **Self-Testing:** New modems have self-testing features. They can test the digital connection to the terminal /computer and the analog connection to a remote modem. They can also check the modem's internal electronics.
- **Voice over Data:** Voice over Data modems allow a voice conversation to take place while data is being transmitted. This requires both the source and destination modems to have this feature.
- **Synchronous or Asynchronous Transmission:** Newer modems allow a choice of synchronous or asynchronous transmission of data. Normally, modem transmission is asynchronous. We send individual characters with just start and stop bits. Synchronous transmission or packet transmission is used in specific applications.

## Functions of Modems:

- **Data Compression:** Data compression is the ability of the modem to take data in from the computer, reduce it in volume, and then send it out via the modem.
- **Error correction:** Error correction standards provide a way of correcting errors that result from outside interference, such as noise on the phone line. Error correction ensures that data coming out of the receiving modem is exactly the same as data going into the sending modem.
- **Flow Control:** Individual modems send information at different speeds. It's necessary for faster modems to slow down so that slower modems can catch up, otherwise the slower modem will receive more data than it can process.

**Q.** Which device is required to connect a computer to the Internet using a telephone line?

- A.** Modem
- B.** Router
- C.** Hub
- D.** Bridge

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- D.** Bridge

**Answer:** (A) Modem

**Explanation:** A modem (modulator-demodulator) converts digital data to analog for transmission over telephone lines.

**Which of the following Internet connection types uses optical fiber to deliver high-speed Internet?**

- (a) DSL
- (b) Cable modem
- (c) Fiber-to-the-Home (FTTH)
- (d) Dial-up

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- (d) Dial-up

**Answer:** (c) Fiber-to-the-Home (FTTH)

**Explanation:** FTTH provides broadband Internet using fiber optic cables, offering higher bandwidth and speed compared to DSL or cable connections.

# Disadvantages of the Internet

Although the Internet is one of the greatest creations, it also has many disadvantages. Because the internet is easily accessible to anyone, it can be a dangerous place. You have to know who you're dealing with or what you're getting into. Predators, cyber criminals, bullies, and corrupt businesses will try to take advantage of the unwary visitor.

## More Disadvantages of the Internet

- Bullying, trolls, stalkers, and crime
- Exploitation and pornography and violent images
- Addiction, time waster, and causes distractions
- Never being able to disconnect
- Identity theft, hacking, viruses, and cheating
- Spam and advertising
- Depression, loneliness, and social isolation
- Health issues and obesity
- Buying things not needed

# The Internet Protocols

The Internet protocols are required to transfer data over the networks and/or the Internet in an efficient manner. When various computers are connected through a computer network, it becomes necessary to use a protocol to efficiently use network bandwidth and avoid collisions.

A network protocol defines a language that contains rules and conventions for reliable communication between different devices over the network. For example, it includes rules that specify how to package data into messages, how to acknowledge a message and how to compress data.

A number of the Internet protocols are used nowadays. The most commonly used protocols are as follows

- Transmission Control Protocol/Internet Protocol (TCP/IP)
- HyperText Transfer Protocol (HTTP)
- File Transfer Protocol (FTP)
- Telnet

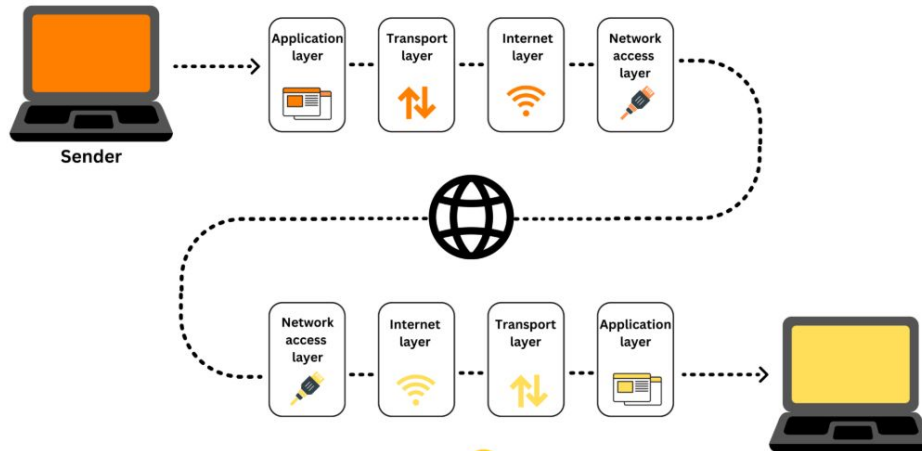


# Transmission Control Protocol/Internet Protocol or TCP/IP

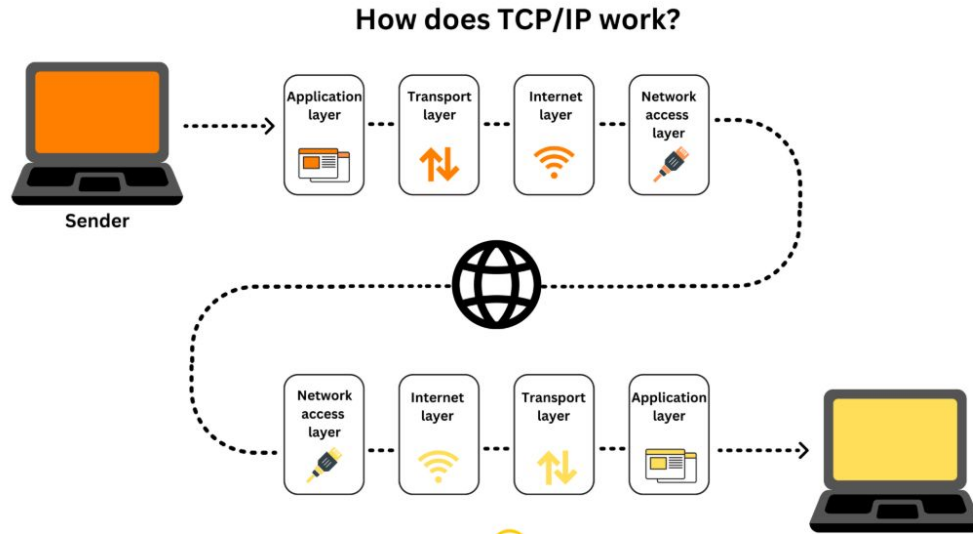
It is a protocol suite used to transfer data over the Internet. Two main protocols in this protocol suite are as follows:

**TCP:** It forms the higher layer of TCP/IP and divides a message or a file into smaller packets which are transmitted over the Internet. A TCP layer on the other side receives these packets and reassembles the data packets into the original message.

How does TCP/IP work?

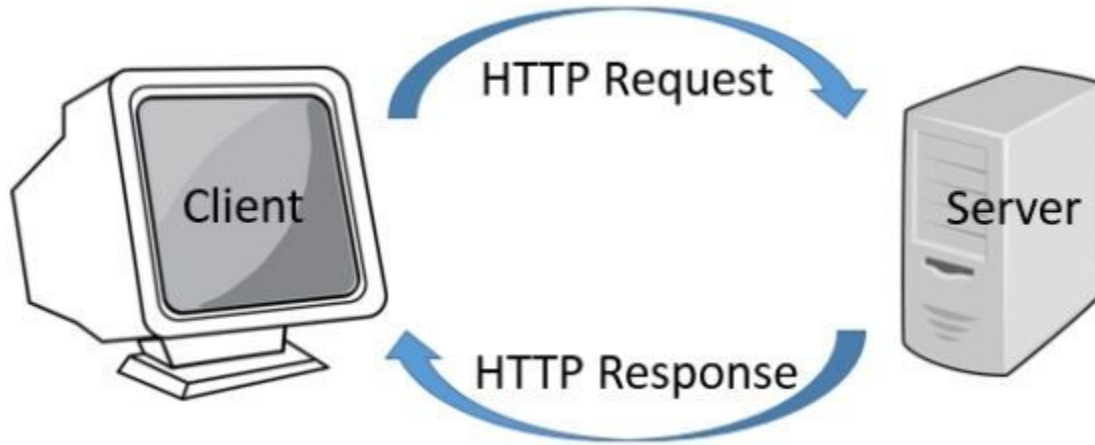


**IP:** It is the lower layer whose function is to handle the address part of each packet to enable it to be delivered to the right destination. Usually, this address is checked by each gateway computer on the network so that it is identified where to forward the message. This implies that all the packets of a message are delivered to the destination regardless of the route used for delivering the packets.



## HyperText Transfer Protocol or HTTP

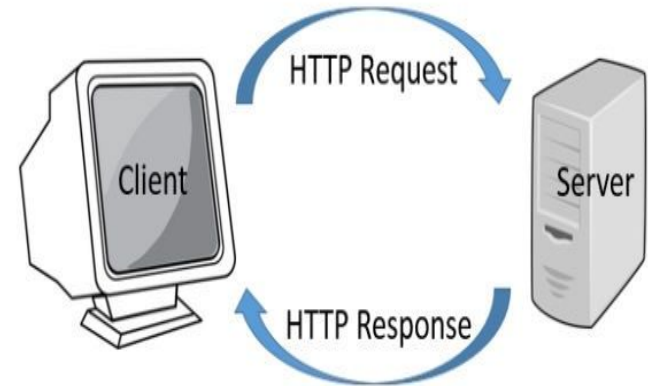
HTTP is a protocol for transferring files (text, image, sound, video and other multimedia files) using the Internet. It is an application protocol that runs on top of the TCP/IP protocol suite which is the foundation protocol of the Internet. It defines how messages are formatted and transmitted and what actions Web servers and Web browsers should take in response to the commands issued. HTTP is based on a client-server architecture where your Web browser acts as a HTTP client making requests to the Web server machines.



## HyperText Transfer Protocol or HTTP

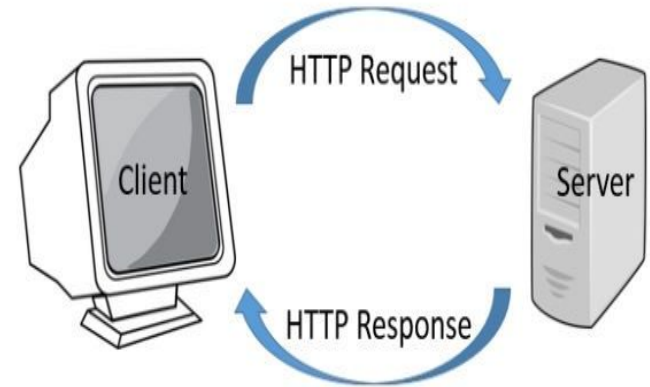
HTTP is based on request-response activity. A client, running an application called a browser, establishes a connection with a server and sends a request to the server in the form of a request method. The server responds with a status line, including the message's protocol version and a success or error code, followed by a message containing server information, entity information and possible body content. An HTTP transaction is divided into the following four steps:

1. The browser opens a connection.
2. The browser sends a request to the server.
3. The server sends a response to the browser.
4. The connection is closed



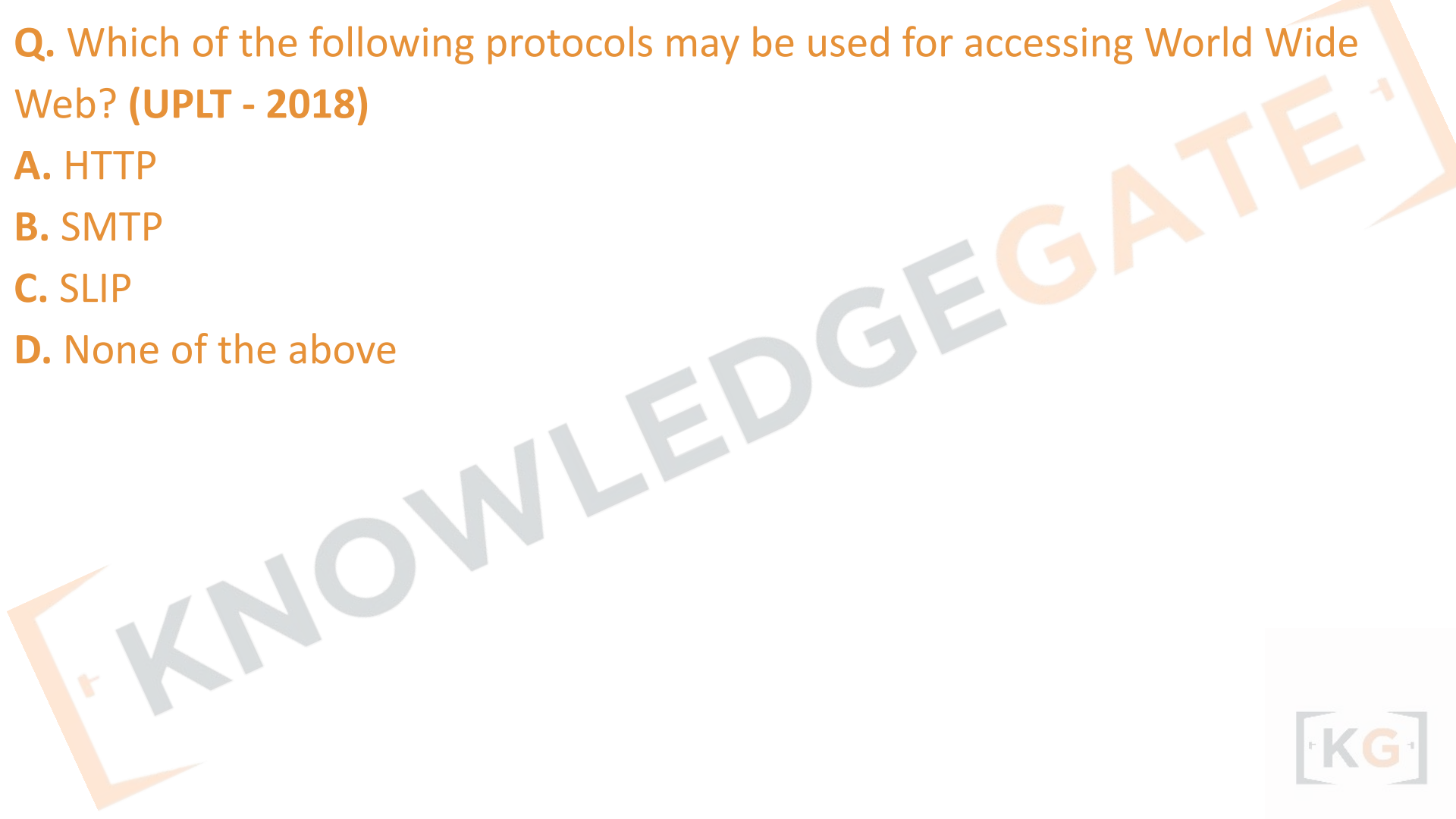
## HyperText Transfer Protocol or HTTP

HTTP is a stateless protocol because it does not keep track of the connections. To load a page including two graphics, for example, a graphic-enabled browser will open three TCP connections: one for the page and two for the graphics. Most browsers, however, are able to handle several of these connections simultaneously.



**Q. Which of the following protocols may be used for accessing World Wide Web? (UPLT - 2018)**

- A. HTTP**
- B. SMTP**
- C. SLIP**
- D. None of the above**



**Q. Which of the following protocols may be used for accessing World Wide Web? (UPLT - 2018)**

- A. HTTP**
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- C. SLIP**
- D. None of the above**

**Ans: A**

## Basic Features

There are three basic features that make HTTP a simple but powerful protocol:

**HTTP is connectionless:** The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client waits for the response. The server processes the request and sends a response back after which client disconnect the connection. So client and server knows about each other during current request and response only. Further requests are made on new connection like client and server are new to each other.

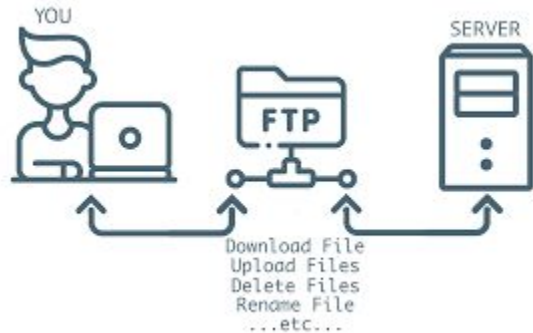
**HTTP is media independent:** It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.

**HTTP is stateless:** As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.



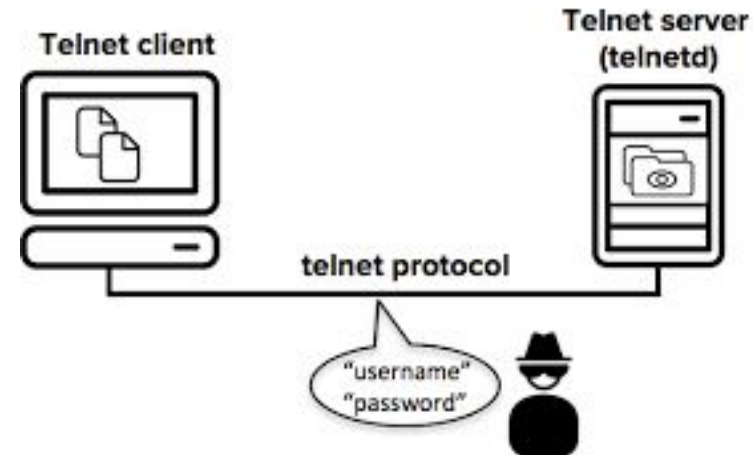
# File Transfer Protocol or FTP

FTP is an application protocol used for exchanging files between computers over the Internet. It is the simplest and most commonly used protocol for downloading/ uploading a file from/to a server. For example, downloading a document or an article from a Web site. Like other protocols, FTP also uses TCP/IP protocol suite for data transfer. FTP also works on a client-server architecture where an FTP client program is used to make a request to an FTP server (files can be stored on computers referred to as FTP servers). Basic FTP support is usually provided as a part of the TCP/IP suite of programs.



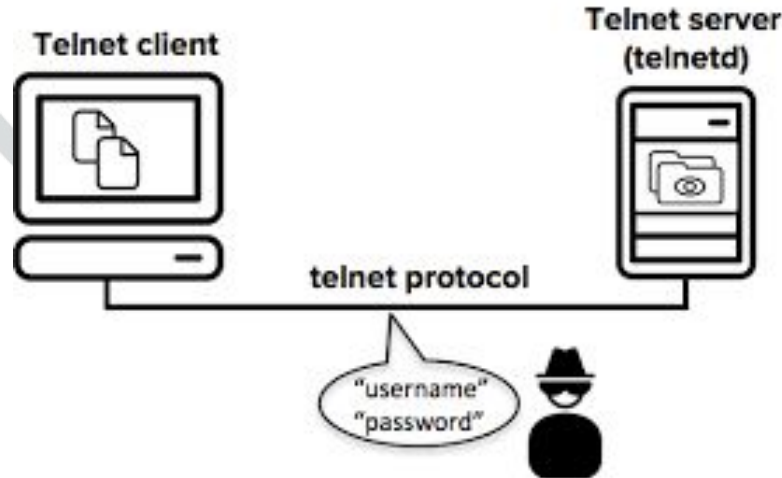
## Telnet

An IP address is a unique number associated with each computer making it uniquely identifiable amongst all the computers connected to the Internet. Telnet is a protocol that allows you to access a remote computer provided you have been given the permission to do so. It is typically referred to as 'remote login'. Telnet is based on a different concept from HTTP and FTP. HTTP and FTP allows you to request specific files from remote computers only while with Telnet you log in as a regular user on a remote machine with whatever privileges you may have been granted on that computer to access specific application and data.



# Telnet

A request for a connection to a remote host which may be a computer lying physically in a neighbouring room or miles away results in an invitation to log in with a user id and a password. If the request is accepted, the user can enter commands through the Telnet program and these would be executed as if they were being entered directly from the host machine. Once connected, the user's computer emulates the remote computer. Telnet is typically used by program developers or anyone who needs to access data and/or applications located at a particular host computer.



## The Internet Addressing

Since the Internet consists of a large number of computers connected with each other, it requires a proper addressing system to uniquely identify each computer in the network. Each computer connected to the Internet is associated with a unique number and/or a name called computer address. Before you wish to access any Web page on a computer, you would require the computer address.



## The Internet Addressing

**IP Address:** An IP address is a unique number associated with each computer making it uniquely identifiable amongst all the computers connected to the Internet. This is a 32-bit number and is divided into four octets, such as 00001010 00000000 00000000 00000110. For human readability, it is represented in a decimal notation, separating each octet with a period. The above number would therefore be represented as: 10.0.0.6. Each octet can range from 0 – 255, thus all IP addresses lie between 0.0.0.0 to 255.255.255.255 resulting in total 4294, 967,296 possible IP addresses. It may be worth noting that no two machines (or hosts) can have the same IP address.



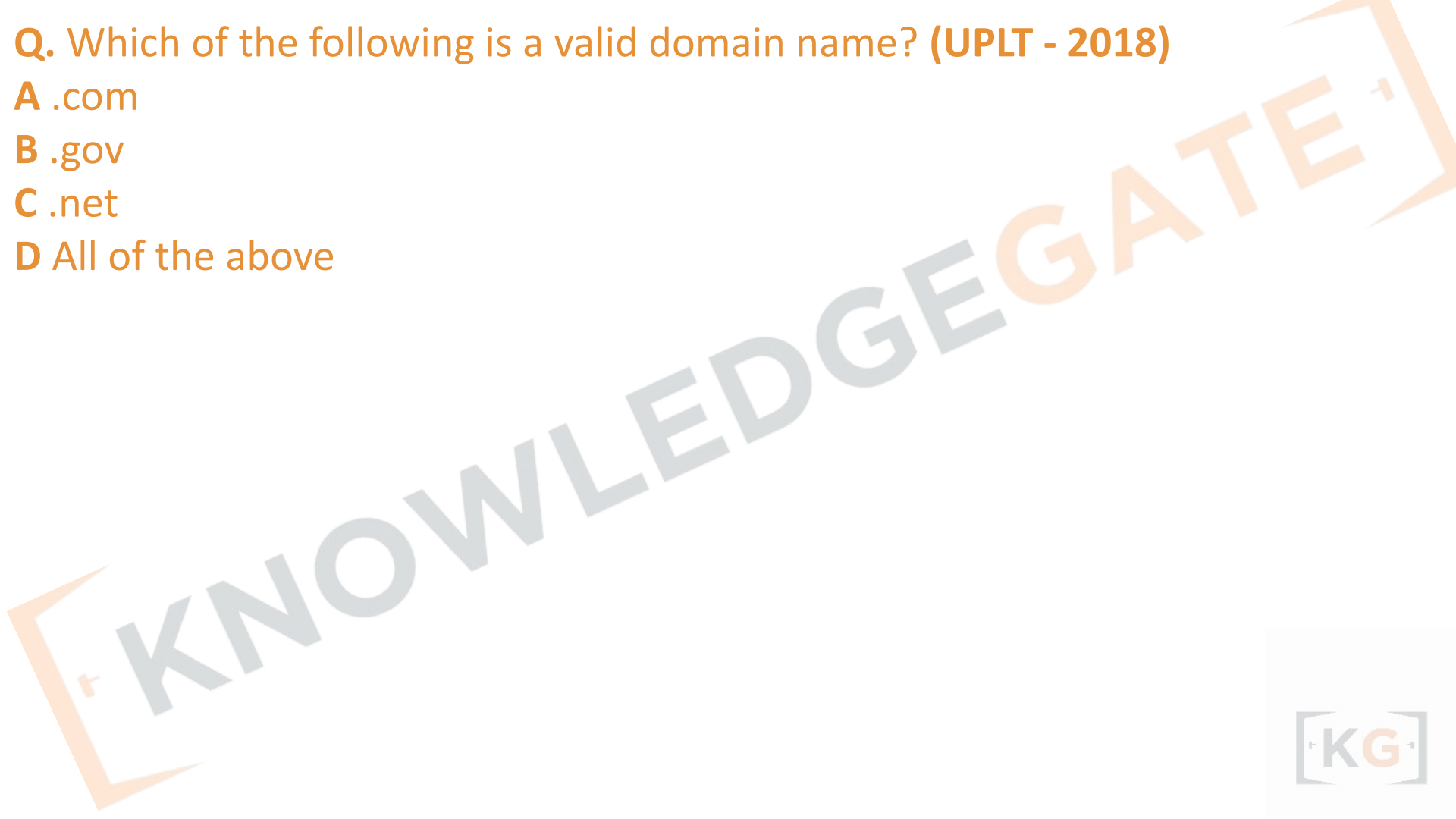
## Domain Naming System or DNS:

In a network, computers and devices can be grouped as a unit with common rules and procedures. Such a group is called a domain. Each domain name therefore corresponds to a group of IP addresses. Some examples of domain names include Yahoo.com, Microsoft.com and abcuniv.edu. The last portion of the domain name provides information on the kind of organization to which the address belongs. Commonly used abbreviations in domain names are summarized in Table .

| <i>Abbreviation</i> | <i>What it Represents</i> |
|---------------------|---------------------------|
| com                 | Commercial organization   |
| org                 | Non-profit organization   |
| edu                 | Educational institution   |
| net                 | Networking organizations  |
| gov                 | Government agencies       |

**Q. Which of the following is a valid domain name? (UPLT - 2018)**

- A .com**
- B .gov**
- C .net**
- D All of the above**



**Q. Which of the following is a valid domain name? (UPLT - 2018)**

- A .com**
- B .gov**
- C .net**
- D .All of the above**

Answer: D. All of the above.

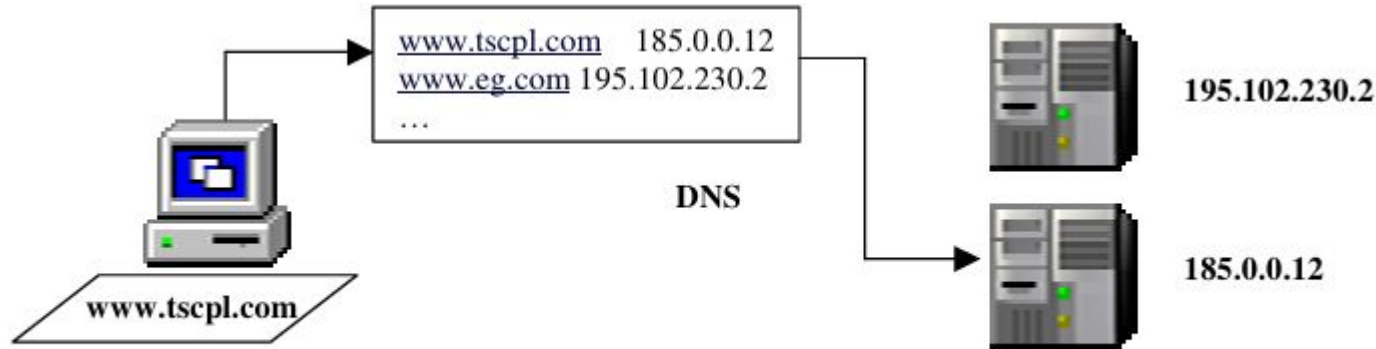
Explanation:

- **.com:** This is the most common top-level domain, widely used for commercial websites.
- **.gov:** This domain is reserved for government websites.
- **.net:** Originally intended for network-related organizations, but now used for various purposes.



## Domain Naming System or DNS:

Abbreviations used in Domain Names Abbreviation com What it Represents org Commercial organization Non-profit organization edu net Educational institution Networking organizations gov Government agencies 8 Self-Instructional Material DNS is an Internet service that translates domain names to or from IP addresses, which is the actual basis of addressing on the Internet. A DNS is typically a database containing information about domain names and their corresponding IP addresses (



*Fig. 1.2 DNS*

# Uniform Resource Locator (URL)

The URL is a standard for specifying any kind of information on the [Internet](#). The protocol is the client or server program which is used to retrieve the document or file. The protocol can be [FTP](#) and [HTTP](#).



# Uniform Resource Locator (URL)

The Host is the name of computer on which the information is located.

The URL can optionally contain the port number and it is separated from the host name by a colon.

Path is the pathname of the file where the file is stored.

<http://www.wlv.ac.uk:80/index.html>

Protocol

Host

Port number

Name of web page

# Uniform Resource Locator (URL)

Sometimes the URL is also referred to as 'Port', that is, it is a port number by which TCP connection is possible to the remote host machine. The default port for protocol is used if port is not specified. For instance, Port 80 is known as default port for HTTP. The two ports, Port 20 and Port 21 are used by ftp. The alternative port can be used in the following way: `http://aaa.bbb.edu:80/flower`.



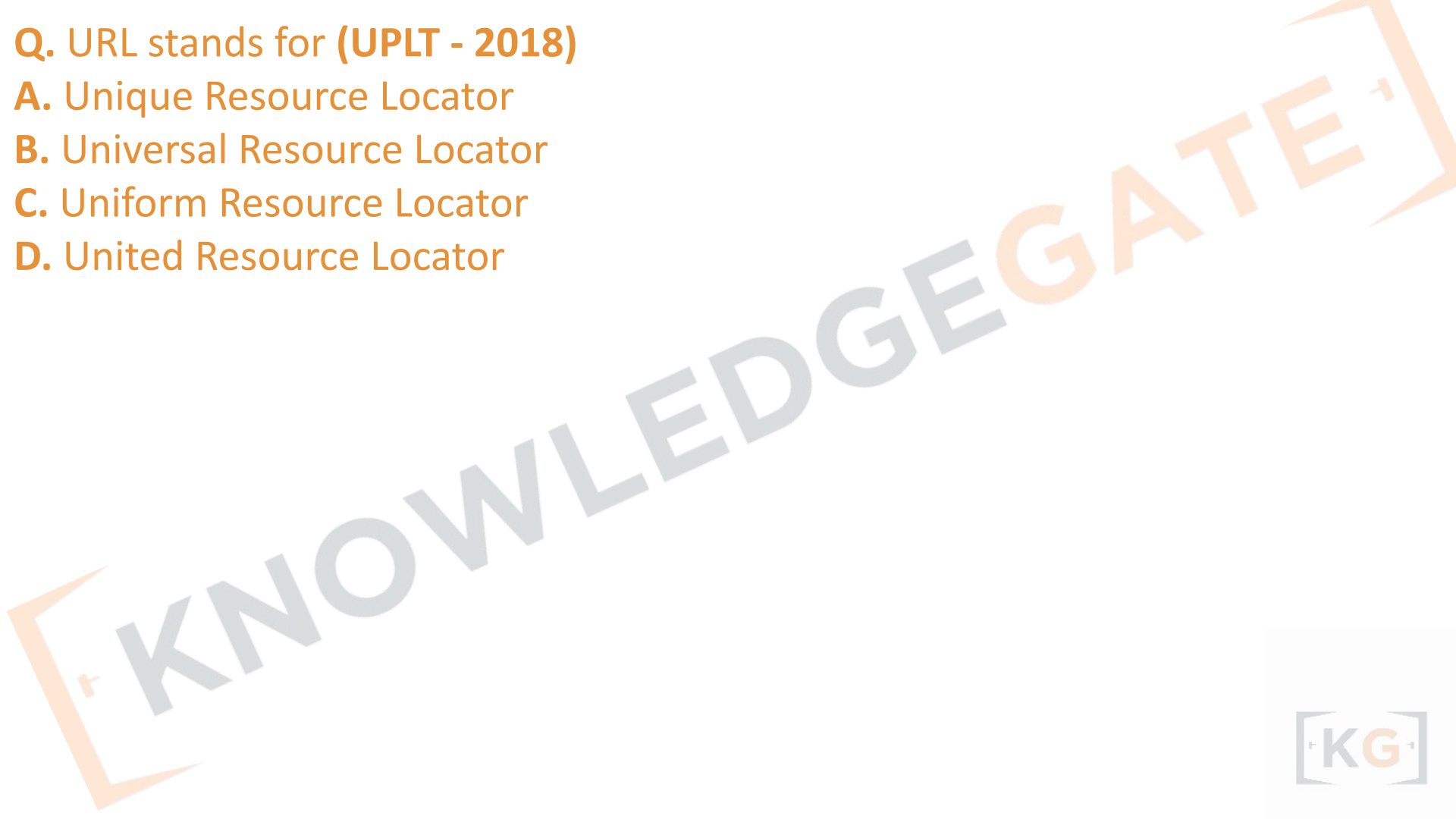
The diagram shows the URL `http://www.wlv.ac.uk:80/index.html` with four arrows pointing to its components: 'Protocol' points to 'http://', 'Host' points to 'www.wlv.ac.uk', 'Port number' points to ':80', and 'Name of web page' points to '/index.html'.

`http://www.wlv.ac.uk:80/index.html`

Protocol      Host      Port number      Name of web page

**Q. URL stands for (UPLT - 2018)**

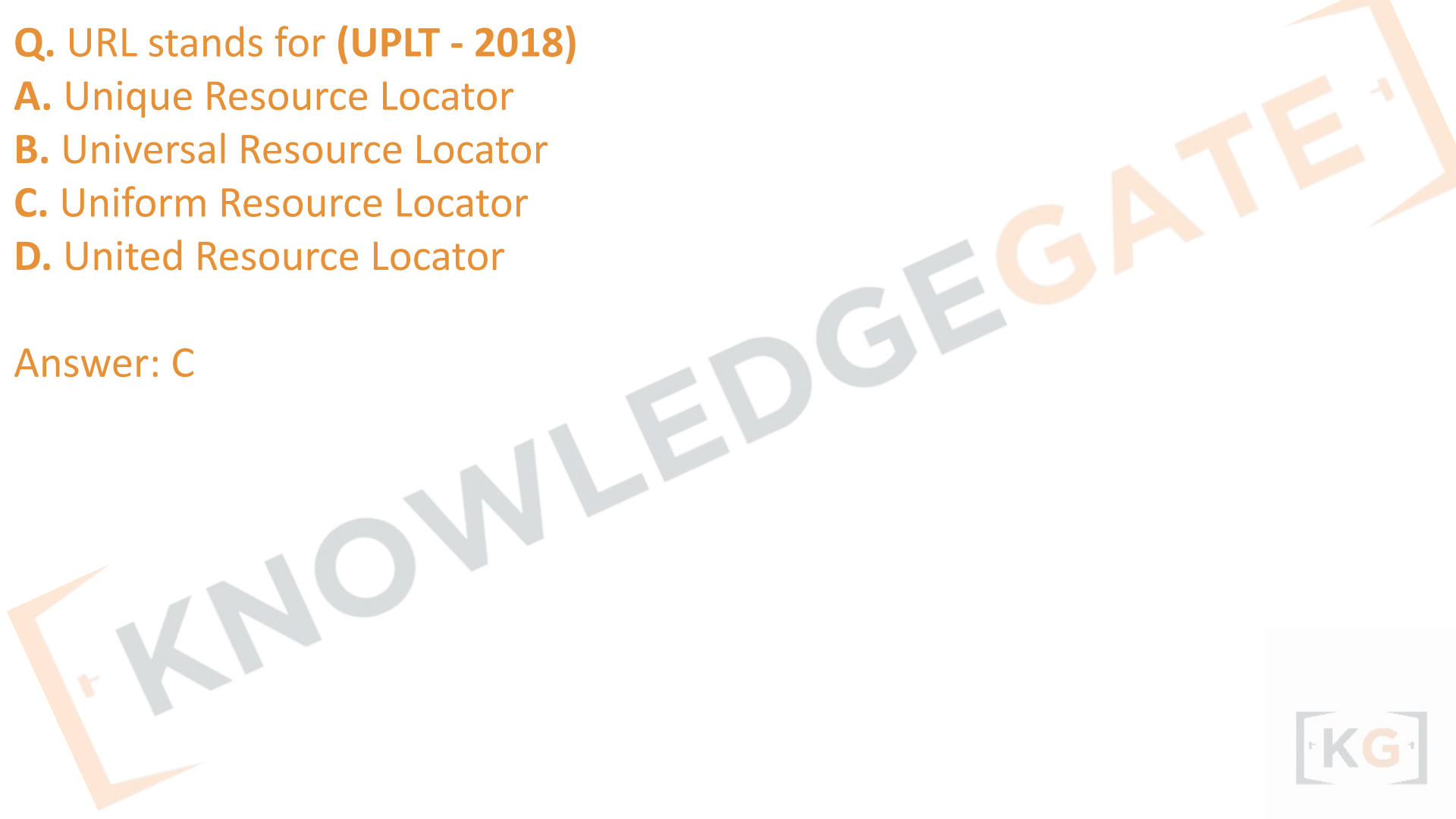
- A. Unique Resource Locator**
- B. Universal Resource Locator**
- C. Uniform Resource Locator**
- D. United Resource Locator**



**Q. URL stands for (UPLT - 2018)**

- A. Unique Resource Locator**
- B. Universal Resource Locator**
- C. Uniform Resource Locator**
- D. United Resource Locator**

**Answer: C**



**Q.**The Internet Protocol (IP) operates at which layer of the TCP/IP model?

- A.** Application Layer
- B.** Transport Layer
- C.** Internet Layer
- D.** Network Access Layer

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- D.** Network Access Layer

**Answer:** C) Internet Layer

**Explanation:**

IP (Internet Protocol) is responsible for logical addressing and routing of packets between networks. It functions at the **Internet Layer** of the TCP/IP model, which corresponds to the **Network Layer** in the OSI model.



**Q.** Which of the following protocols is connection-oriented and ensures reliable data transmission?

- A.** UDP
- B.** IP
- C.** TCP
- D.** ICMP

**Q.** Which of the following protocols is connection-oriented and ensures reliable data transmission?

- A.** UDP
- B.** IP
- C.** TCP
- D.** ICMP

**Answer:** C) TCP

**Explanation:**

**Transmission Control Protocol (TCP)** establishes a connection between sender and receiver, provides error detection, retransmission, and ordered delivery, making it a reliable protocol. UDP, on the other hand, is connectionless and unreliable.

**Q.** Which protocol is used to convert a domain name like `www.example.com` into an IP address?

- A.** ARP
- B.** DHCP
- C.** DNS
- D.** ICMP

**Q.** Which protocol is used to convert a domain name like www.example.com into an IP address?

- A.** ARP
- B.** DHCP
- C.** DNS
- D.** ICMP

**Answer:** C) DNS

**Explanation:**

**Domain Name System (DNS)** translates human-readable domain names into machine-readable IP addresses. It operates at the **Application Layer** of the TCP/IP model and typically uses port 53.



Q. Web Services are a mean of **(Punjab State Civil Supplies Corp. Ltd. 13.11.2011)**

- A. Improving security through encryption technology
- B. Distributing program functions through remote procedure calls
- C. Developing customer loyalty through improved technology
- D. Employing Java through the Web

Ans. (b)

**Q.**The values GET, POST are specified in----of HTTP message **APPSC Lect. 2017**  
**(Degree College) (CS)**

- A.** Request line
- B.** Header line
- C.** Status line
- D.** Entity body

Ans. (a)

Q.Which of the following protocol(s) is/are used by HTTP? (**MPPSC State Forest Service Examination 2014**)

- A. FTP
- B. SMTP
- C. POP
- D. All options are correct

Ans. (D)

**Q.** In HTTP, for sending data to server which of the following command(s) is used (**RPSC VPITI 2014 (IT)**)

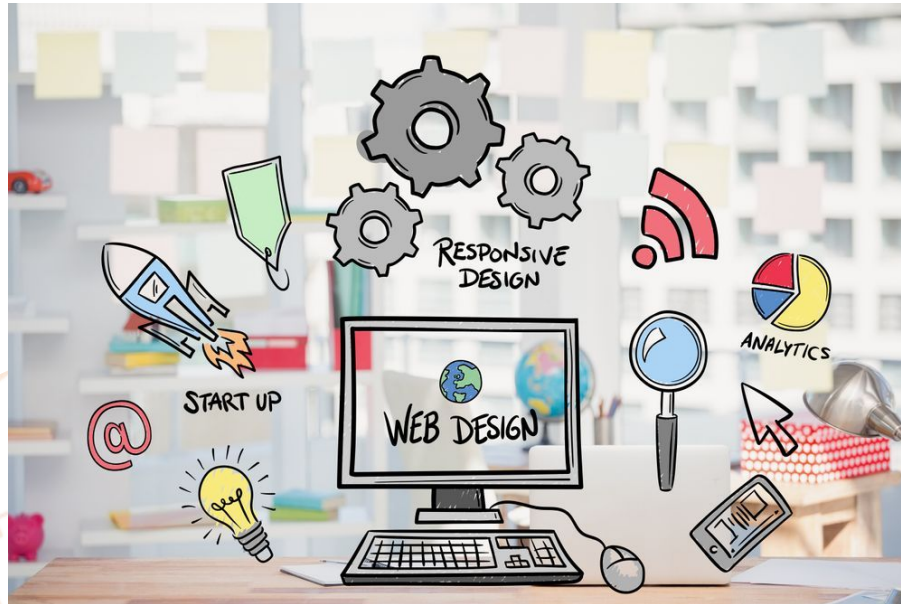
- A.** Only GET
- B.** Only POST
- C.** Either of GET or POST
- D.** Neither GET nor POST

**Ans.** (B)



## Web Design

Web design is the field of Designing Website interfaces. It deals with the looks part of the website rather than the coding part. Designing how the interface will look is called User Interface(UI) Design and Designing the flow of the user on the website, how the user will navigate through the website, etc are a part of User Experience(UX) Design.



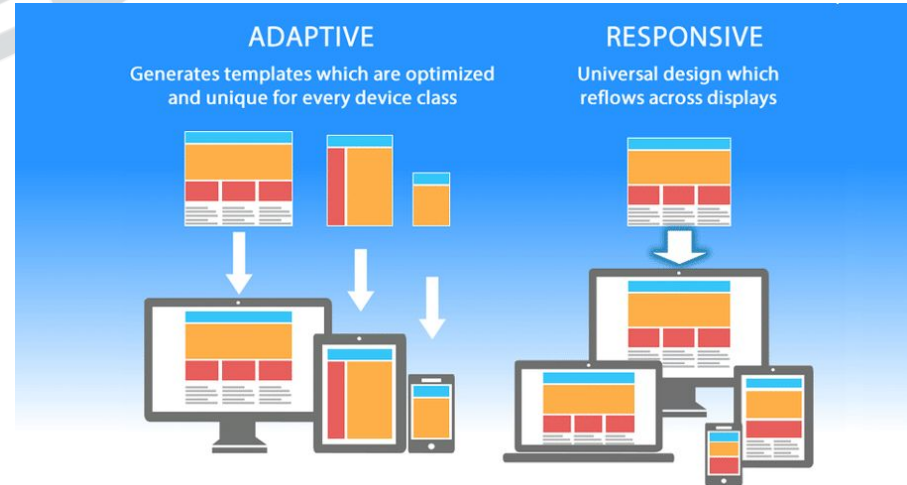
## Responsive Web Design

Responsive Web Design means designing a website that is compatible with every screen type. Responsive websites resize themselves depending on the screen size of the device. It automatically adjusts to fit the user's screen whether it's desktop, laptop, mobile, tablet, etc. Websites also scale the content and the elements accordingly. The elements in an interface get enlarged or reduced according to the screen size of the device.



## Adaptive Web Design

Adaptive Web Design refers to designing different layouts of the website for different devices. Many different devices' screen size is taken into account and a different layout is designed for each of them. It is different from Responsive Web design as in Responsive Web design the same website interface adjusts itself depending on the device's screen size but Adaptive design changes the complete layout for that specific screen size. Since different layouts are created for different devices their User Experience might differ too. A different User Experience can be created for multiple devices.



## Interactive Web Design

Interactive Web Design is designing a website in which users can actively interact with the content and features of the website. These web pages include engaging elements such as buttons, forms, animations, videos etc. Users can perform various actions on an interactive webpage to trigger responses making it dynamic and engaging. An interactive web design aims to enhance user engagement and create a more enjoyable and user-friendly browsing experience.



# Principles of Design



Balance



Alignment



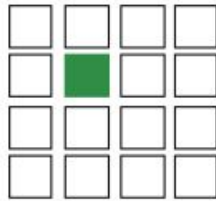
Repetition



Contrast



Hierarchy



Emphasis



White Spaces



Proportion

# Principles of Design?

The principles of design are basic rules that help make designs look good and work well. They guide how to arrange elements in a design to make it visually appealing and effective. Here are the main principles of design:

1. **Balance:** Distributing elements evenly to create a sense of stability.
2. **Contrast:** Using differences in color, shape, or size to make elements stand out.
3. **Emphasis:** Highlighting the most important parts of a design.
4. **Movement:** Guiding the viewer's eye through the design in a specific way.
5. **Pattern:** Repeating elements to create a sense of order and consistency.
6. **Rhythm:** Creating a sense of movement by repeating elements at regular intervals.
7. **Unity:** Making sure all parts of the design work together as a whole.

**Which of the following is the most important principle of effective web design?**

- (a) Adding maximum animations and effects
- (b) Using as many colors as possible
- (c) Maintaining consistency and simplicity in layout
- (d) Avoiding whitespace

**Which of the following is the most important principle of effective web design?**

- (a) Adding maximum animations and effects
- (b) Using as many colors as possible
- (c) Maintaining consistency and simplicity in layout
- (d) Avoiding whitespace

**Answer:** (c) Maintaining consistency and simplicity in layout

**Explanation:** A good web design emphasizes clarity, consistency, and simplicity to enhance usability and user experience. Excessive design elements distract users from the main content.



## What does “responsive web design” mean?

- (a) A website that automatically replies to user emails
- (b) A website that adjusts its layout based on the device and screen size
- (c) A website with advanced animations
- (d) A website developed only for desktop users

## What does “responsive web design” mean?

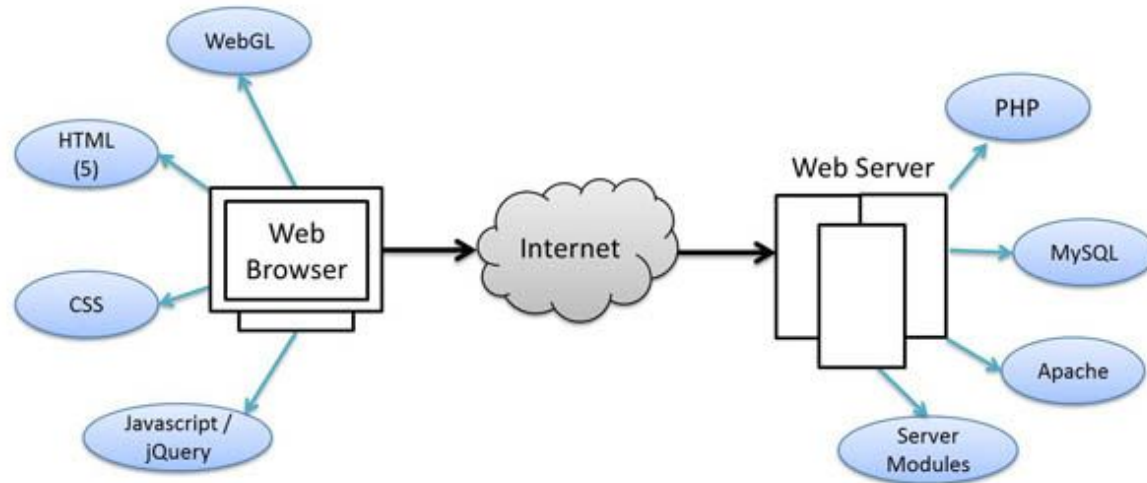
- (a) A website that automatically replies to user emails
- (b) A website that adjusts its layout based on the device and screen size
- (c) A website with advanced animations
- (d) A website developed only for desktop users

**Answer:** (b) A website that adjusts its layout based on the device and screen size

**Explanation:** Responsive web design ensures that a website adapts dynamically to different devices (mobile, tablet, desktop) using flexible grids and CSS media queries.

# Introduction to Web Technologies

Web Technology refers to the various tools and techniques that are utilized in the process of communication between different types of devices over the Internet. A web browser is used to access web pages. Web browsers can be defined as programs that display text, data, pictures, animation, and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers.



Web Technology can be classified into the following sections:

**World Wide Web (WWW):** The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML), and Hypertext Transfer Protocol (HTTP).

**Web Browser:** The web browser is an application software to explore www (World Wide Web). It provides an interface between the server and the client and requests to the server for web documents and services.

**Web Server:** Web server is a program which processes the network requests of the users and serves them with files that create web pages. This exchange takes place using Hypertext Transfer Protocol (HTTP).

**Web Pages:** A webpage is a digital document that is linked to the World Wide Web and viewable by anyone connected to the internet has a web browser.

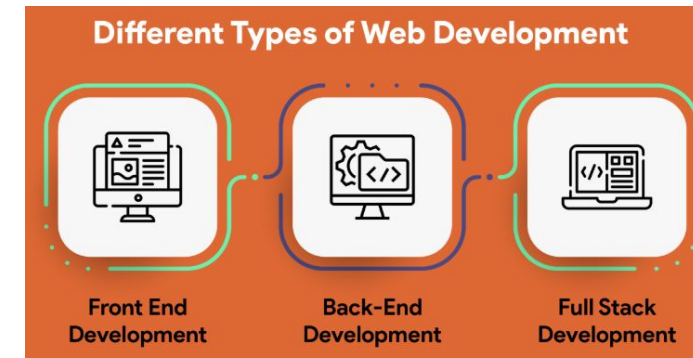
**Web Development:** Web development refers to the building, creating, and maintaining of websites. It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet i.e. websites.

Web Development can be classified into three ways:

**Frontend Development:** The part of a website that the user interacts directly is termed as front end. It is also referred to as the 'client side' of the application.

**Backend Development:** Backend is the server side of a website. It is the part of the website that users cannot see and interact. It is the portion of software that does not come in direct contact with the users. It is used to store and arrange data.

**Full Stack Development** refers to the end-to-end design, development, and deployment of web applications, encompassing both **front-end (client-side)** and **back-end (server-side)** technologies, as well as **database management**, **version control**, and **deployment practices**.



**Frontend Languages:** The front-end portion is built by using some languages which are discussed below:

**HTML:** HTML stands for Hypertext Markup Language. It is used to design the front-end portion of web pages using a markup language. HTML is the combination of Hypertext and Markup language. Hypertext defines the link between the web pages. The markup language is used to define the text documentation within the tag which defines the structure of web pages.

**CSS:** Cascading Style Sheets fondly referred to as CSS is a simply designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

**JavaScript:** JavaScript is a famous scripting language used to create magic on the sites to make the site interactive for the user. It is used to enhancing the functionality of a website to running cool games and web-based software.

**AJAX:** Ajax is an acronym for Asynchronous Javascript and XML. It is used to communicate with the server without refreshing the web page and thus increasing the user experience and better performance.

**Backend Languages:** The back end portion is built by using some languages which are discussed below:

**PHP:** PHP is a server-side scripting language designed specifically for web development. Since PHP code is executed on the server-side, so it is called a server-side scripting language.

**Node.js:** Node.js is an open-source and cross-platform runtime environment for executing JavaScript code outside a browser. You need to remember that NodeJS is not a framework, and it's not a programming language. Most people are confused and understand it's a framework or a programming language. We often use Node.js for building back-end services like APIs like Web App or Mobile App. It's used in production by large companies such as Paypal, Uber, Netflix, Walmart, and so on.

**Python:** Python is a programming language that lets you work quickly and integrate systems more efficiently.

# Databases

A database is where a website's data like user's data, product's data are stored and organized. It is part of the backend (server side) that manages and keeps this information safe. Websites use databases to save and access information like user details, content, and transactions. Some databases organize data in tables (called relational databases, like MySQL), while others store data in flexible formats (called NoSQL databases, like MongoDB).

There are basically two types of databases:

## 1. SQL/Relational Database

A relational database stores data in tables, similar to a spreadsheet, where each table has rows and columns. The rows hold individual records, and the columns define the data attributes. Tables can be linked to each other through special keys, allowing related data to be connected.



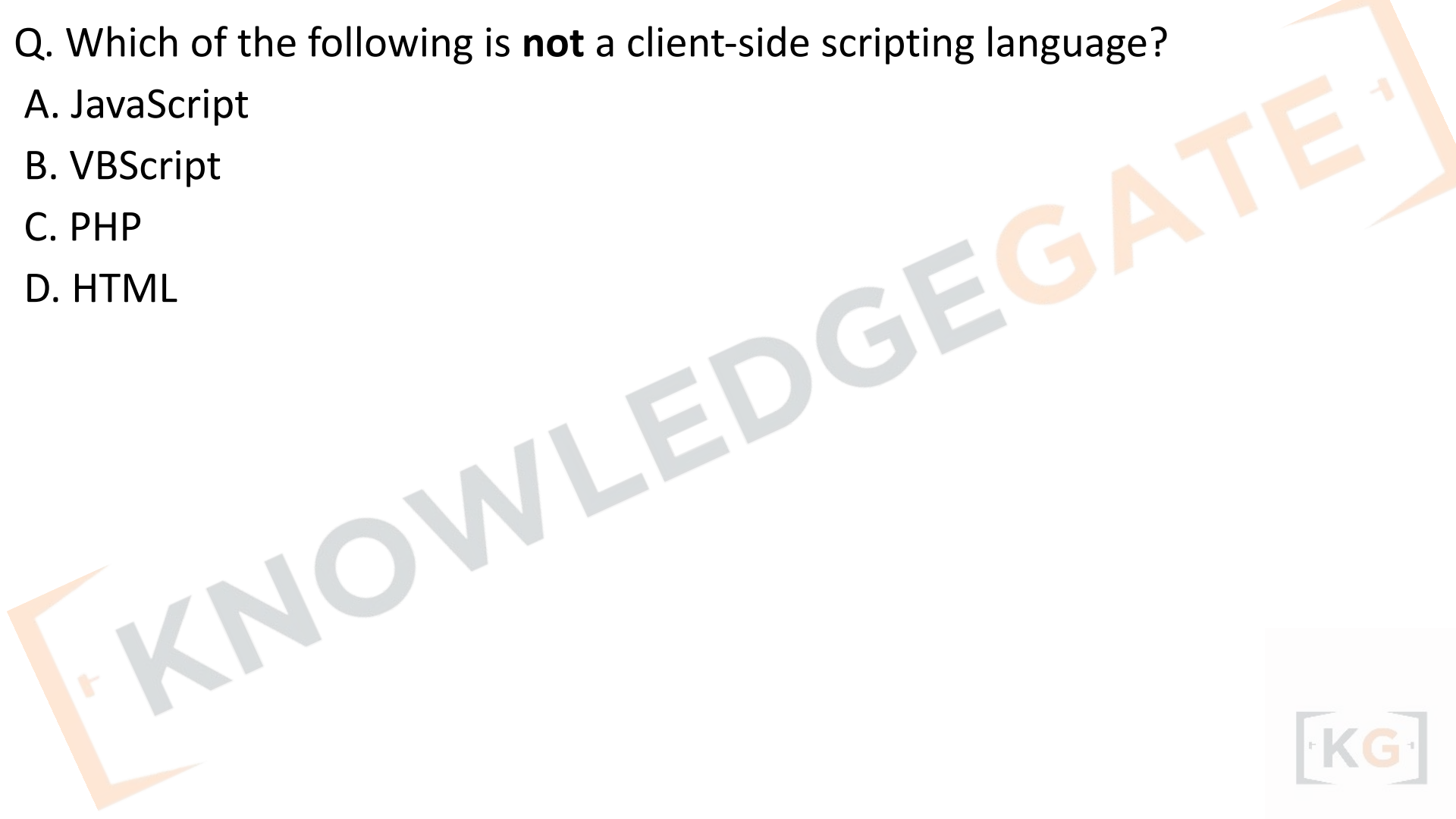
- **MYSQL :** MySQL is an open-source relational database management system that uses SQL for managing structured data. It's known for its reliability, ease of use, and performance, widely used in web applications.
- **PostgreSQL:** PostgreSQL is a powerful, open-source relational database that supports advanced SQL features and complex queries. It handles structured data, ensures ACID compliance, and is known for its reliability and extensibility.

## 2. NoSQL Databases

A NoSQL database stores data in a flexible, non-tabular format, unlike traditional relational databases. Instead of using tables with rows and columns, NoSQL databases might use documents, key-value pairs, wide-columns, or graphs to store data. This allows them to handle large amounts of unstructured or semi-structured data efficiently. They are designed to scale easily and manage big data applications.

- **MongoDB:** MongoDB is a NoSQL database storing data in JSON-like documents. It handles unstructured data, supports powerful queries, and scales easily across servers, making it popular for flexible, scalable applications.
- **Cassandra :** Apache Cassandra is an open-source NoSQL database that is used for handling big data. It has the capability to handle structure, semi-structured, and unstructured data.

- **Redis:** Redis is an in-memory NoSQL database known for its speed. It supports various data structures like strings, hashes, and lists, making it ideal for caching, real-time analytics, and messaging.



Q. Which of the following is **not** a client-side scripting language?

- A. JavaScript
- B. VBScript
- C. PHP
- D. HTML



Q. Which of the following is **not** a client-side scripting language?

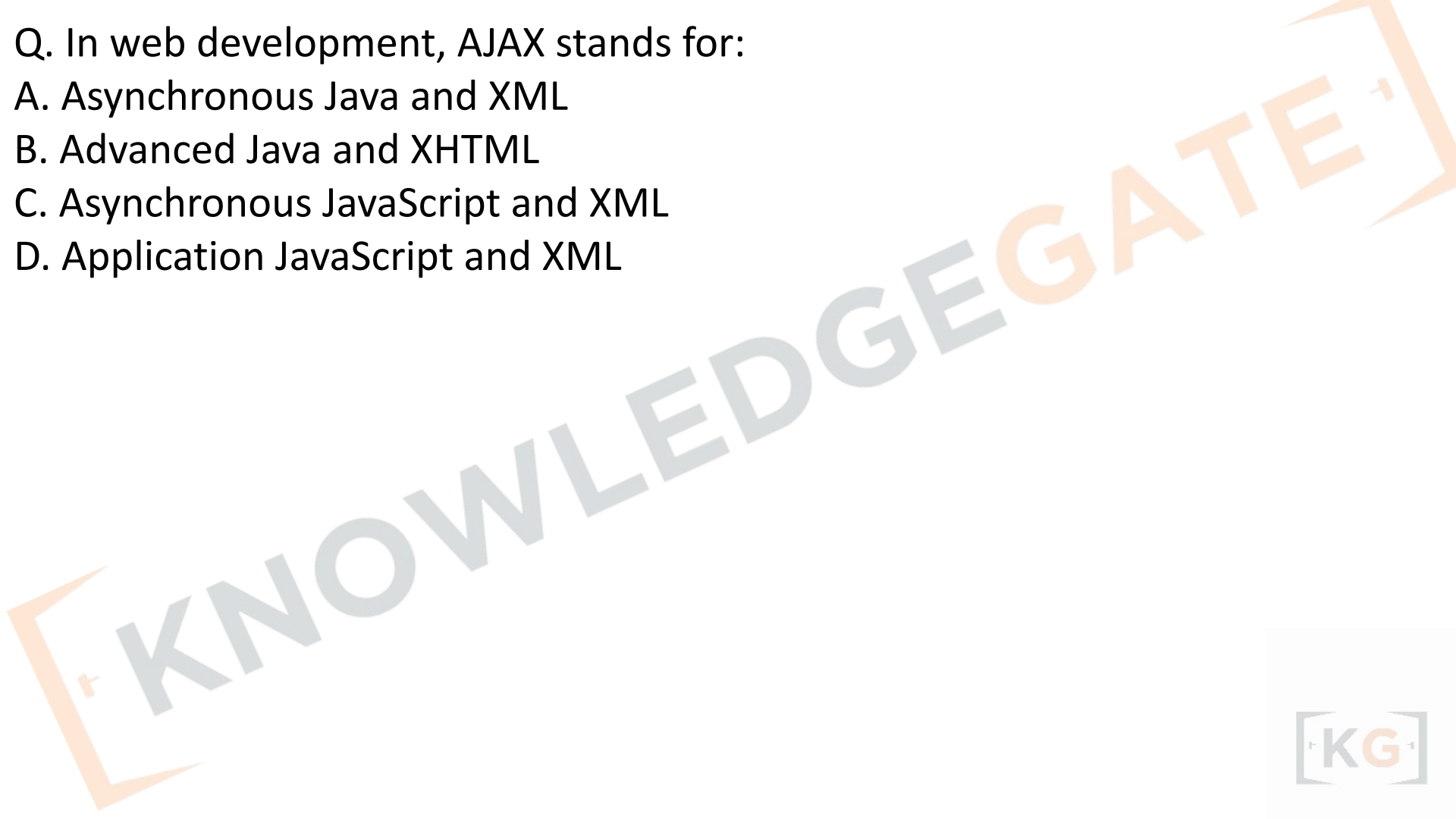
- A. JavaScript
- B. VBScript
- C. PHP
- D. HTML

**Answer:** C. PHP

**Explanation:** PHP is executed on the server before the page is sent to the client, whereas JavaScript and VBScript are executed on the client's browser.

Q. In web development, AJAX stands for:

- A. Asynchronous Java and XML
- B. Advanced Java and XHTML
- C. Asynchronous JavaScript and XML
- D. Application JavaScript and XML



Q. In web development, AJAX stands for:

- A. Asynchronous Java and XML
- B. Advanced Java and XHTML
- C. Asynchronous JavaScript and XML
- D. Application JavaScript and XML

**Answer:** C. Asynchronous JavaScript and XML

**Explanation:** AJAX allows web pages to be updated asynchronously by exchanging small amounts of data with the server behind the scenes.

# Chapter - 2

## Services on Internet, Electronic Mail



# INTERNET SERVICES

These are technologies that allow us to access huge amounts of information such as text, graphics, sound and software over the Internet.

Internet services include:

- THE World Wide Web
- E-mail
- FTP(File Transfer Protocol)
- Newsgroups
- Message boards
- Mailing lists
- Chatrooms
- Instant messaging
- Internet telephony
- Video conferencing

# INTERNET SERVICES

**1. World Wide Web(WWW)**- This is a system of interlinked hypertext documents accessed via the internet .

Retrieving documents on the World Wide Web.

- The URL (Uniform/ Universal Resource Locator ) specifies the internet address of a file stored on a host computer connected to the internet.
- Every file on the internet, no matter what its access protocol, has a unique URL.
- URLs are translated into numeric addresses using the Domain Name System (DNS)
- The DNS is a world wide system of servers that stores location pointers to web sites.

# Anatomy of a URL.

- URLs have the following format:

Protocol://www. Hostname.subdomain.top-level-domain.

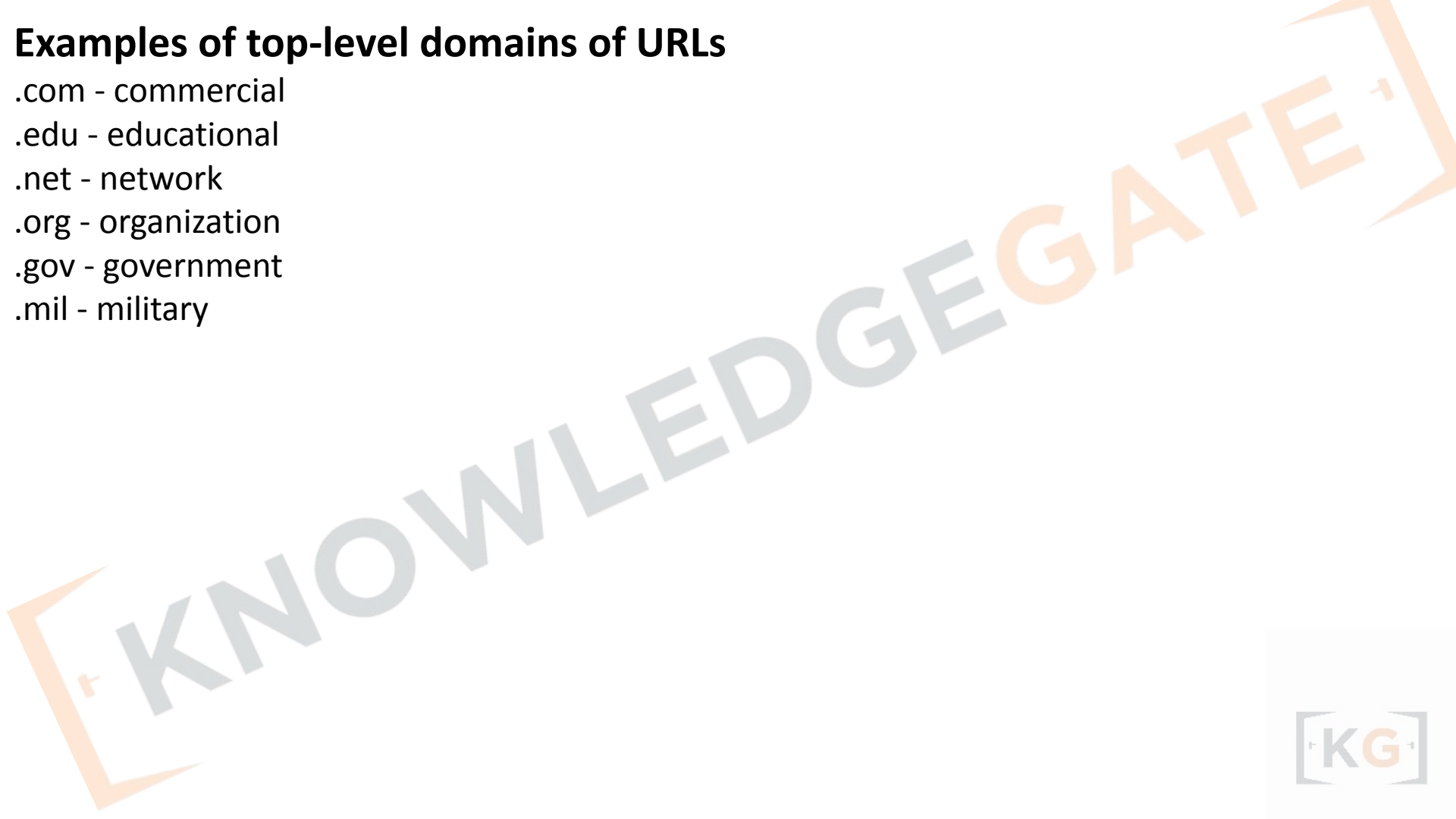
A school's URL, for example can be:

<http://www.kingscollegebudo.sc.ug> .

This naming structure will give you clues about the address. kingscollegebudo is the name of the host; The subdomain; sc. signifies that this is a school and .ug is the top-level-domain to signify Uganda.

# Examples of top-level domains of URLs

- .com - commercial
- .edu - educational
- .net - network
- .org - organization
- .gov - government
- .mil - military



## 2. ELECTRONIC MAIL (E-MAIL)

- Is the transmission of messages via a computer network such as Local Area Network or internet.
- The message can be simple text or can be include an attachment such as an Ms. Excel file, graphical image or video clip.
- E-mail allows computer users locally and world widely to exchange messages.
- Each user of an e-mail has a mail address to which messages are sent.

E.g

annet@gmail.com where annet is the user name, @- separator, gmail.com -domain name.

### Examples of E.mail software or programs:

- Eudora,
- Netscape,
- Messenger,
- Pegasus
- Outlook express.

These offer the ability to read files written in HTML.

## Examples internet mail service providers:

- Hot mail
- Yahoo
- Gmail

## Features of e-mail

- **To:** This feature requires the Email address of the person you are sending an e-mail message to.
- **C.C:** This is short for Carbon copy and this feature requires the email address of other recipients of the message and it indicates to each recipient who else has received a copy of the message.
- **BCC:** Blind Carbon Copy, makes a copy of an e-mail to other recipients, but all will not know that others have received a copy of the same e-mail message.
- **Subject:** This feature requires the title or heading of the message.
- **Compose** - Enables one to write a new e-mail message.
- **Inbox** - This feature(main folder) that holds list of all incoming e-mail messages.
- **Spam** – This feature that holds un solicited/ junk mails.

- **Attachments:** (Multimedia Internet Mail Extension)- MIME: These are files uploaded from a computer to an e-mail and sent to another email account. It can be a picture, Ms. Word Document etc.
- **Reply:** Is feature that enables one make a response to a received e-mail.
- **Forward:** This feature allows one send a received message to another or other recipient(s) without changing the message.

## **ADVANTAGES OF E-MAIL**

- It is a very fast means of Communication
- It is cheaper and the cost does not vary across the globe
- Highly private and confidential compared to other means of communications.
- e-mails are efficient as the mails reach the rightful person.
- In addition to text one can send graphics and sound information.
- It is not affected by geographical condition or weather.

## **DISADVANTAGES**

- When some one comes across a password he or she may have access to all your mails.
- e-mail can only be sent to people with these services.
- It only caters for computer literate people.



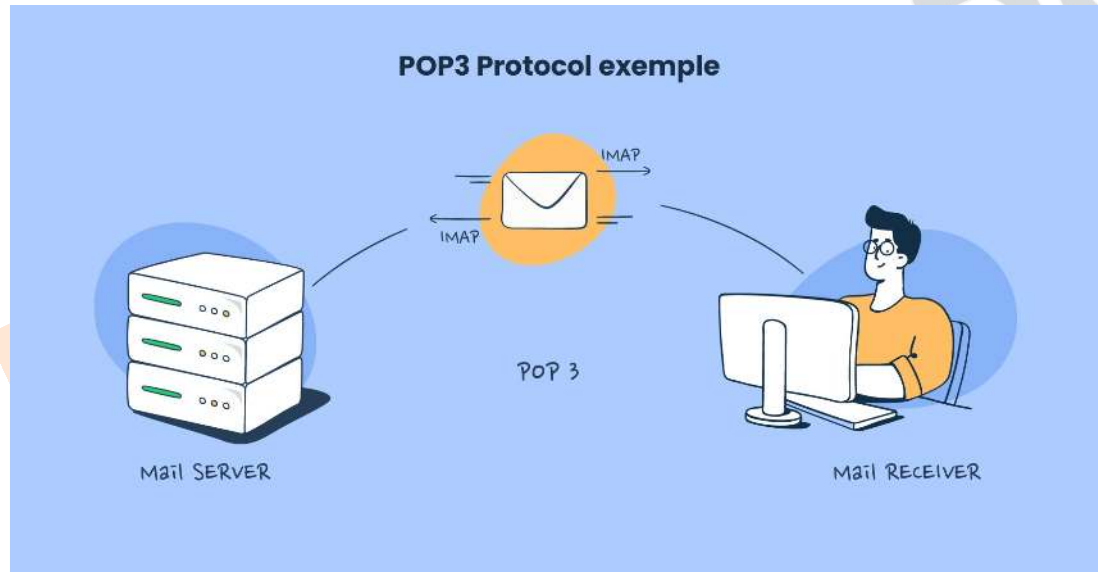
## Compare Email and Post Mail service

| Email                               | Post Mail                                            |
|-------------------------------------|------------------------------------------------------|
| Very fast                           | Slow on delivery                                     |
| Cost is uniform                     | Cost varies with distance and mail weight            |
| Provides automatic storage (Backup) | No Back up in case of future reference               |
| Can be accessed all over the world  | Can only be collected from a designated post office. |
| More secure                         | Easily accessed by illegal users                     |

## Protocols of E-mail:-

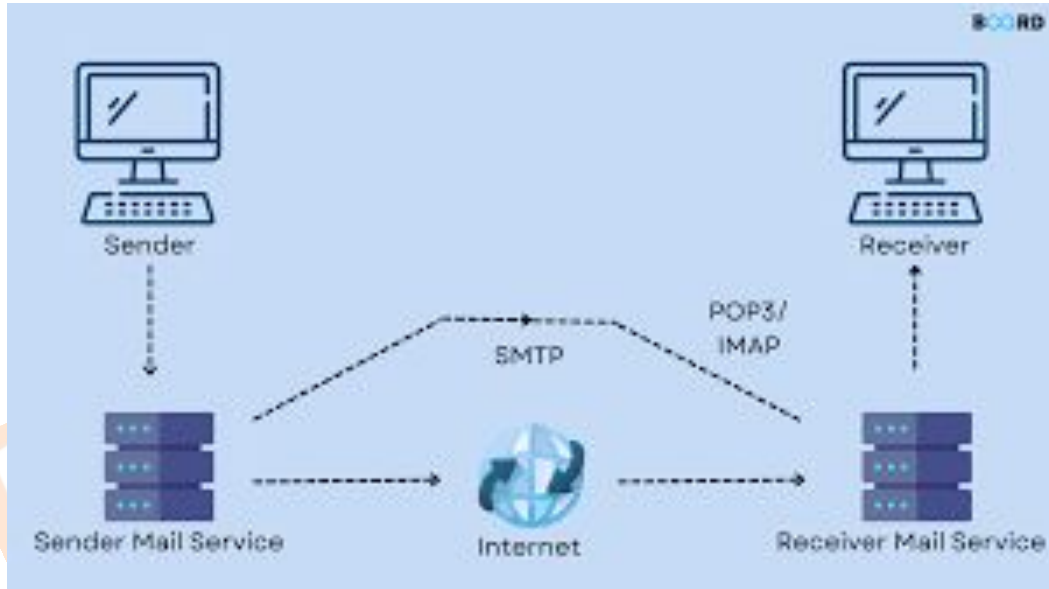
### 1- POP3 ( Post Office Protocol 3):-

POP3 is a protocol that define how e-mail messages are to be retrieved from a mail server. POP3 is one of the two protocol that a PC mail should be able to use if it is to send and receive e-mail over the internet. SMTP & POP3 can share information with each other. So we don't need to use the same e-mail client.



## 2- SMTP ( Simple Mail Transfer Protocol) :-

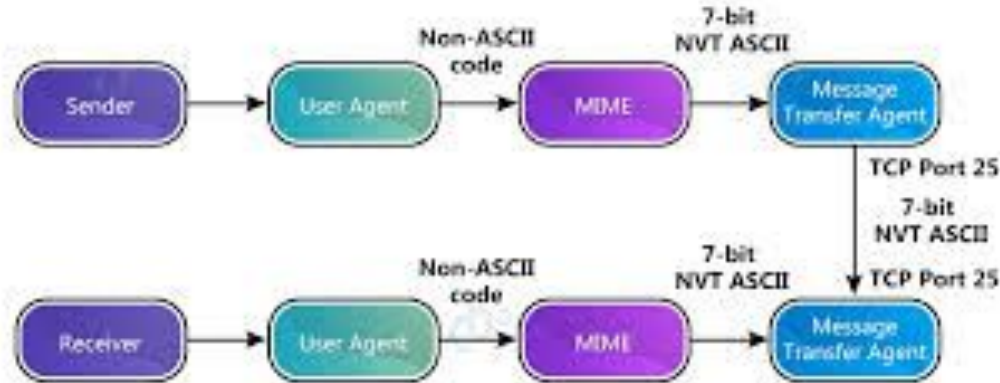
This is the method protocol computer use to send messages by message transfer agents (MTA) on the internet. MTA are client & server programs that perform e-mail services such as sending or receiving mail for a host computer.



## MIME ( Multipurpose Internet Mail Extension):-

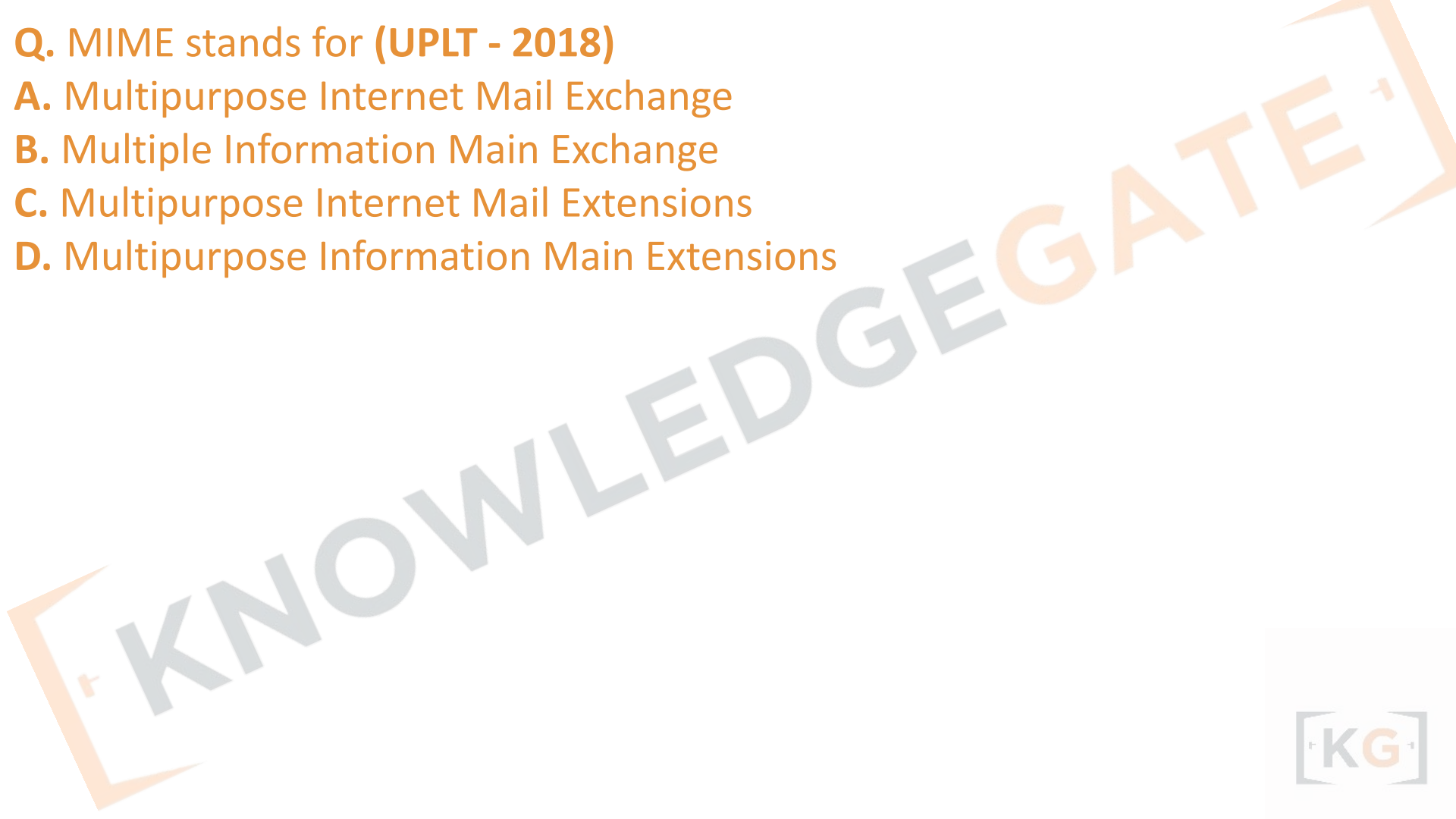
It represents a standard for describing data types. It gives information about data. It was designed as an extension to the simple mail transfer protocol allowing people to send binary data. A particular MIME type is a pair of elements delimited by “/”. The 1st element describes the type of data and 2nd element describe the format of data. Example:- Text/ HTML Image/JPEG Video/ MPEG.

### Working of MIME Protocol



**Q. MIME stands for (UPLT - 2018)**

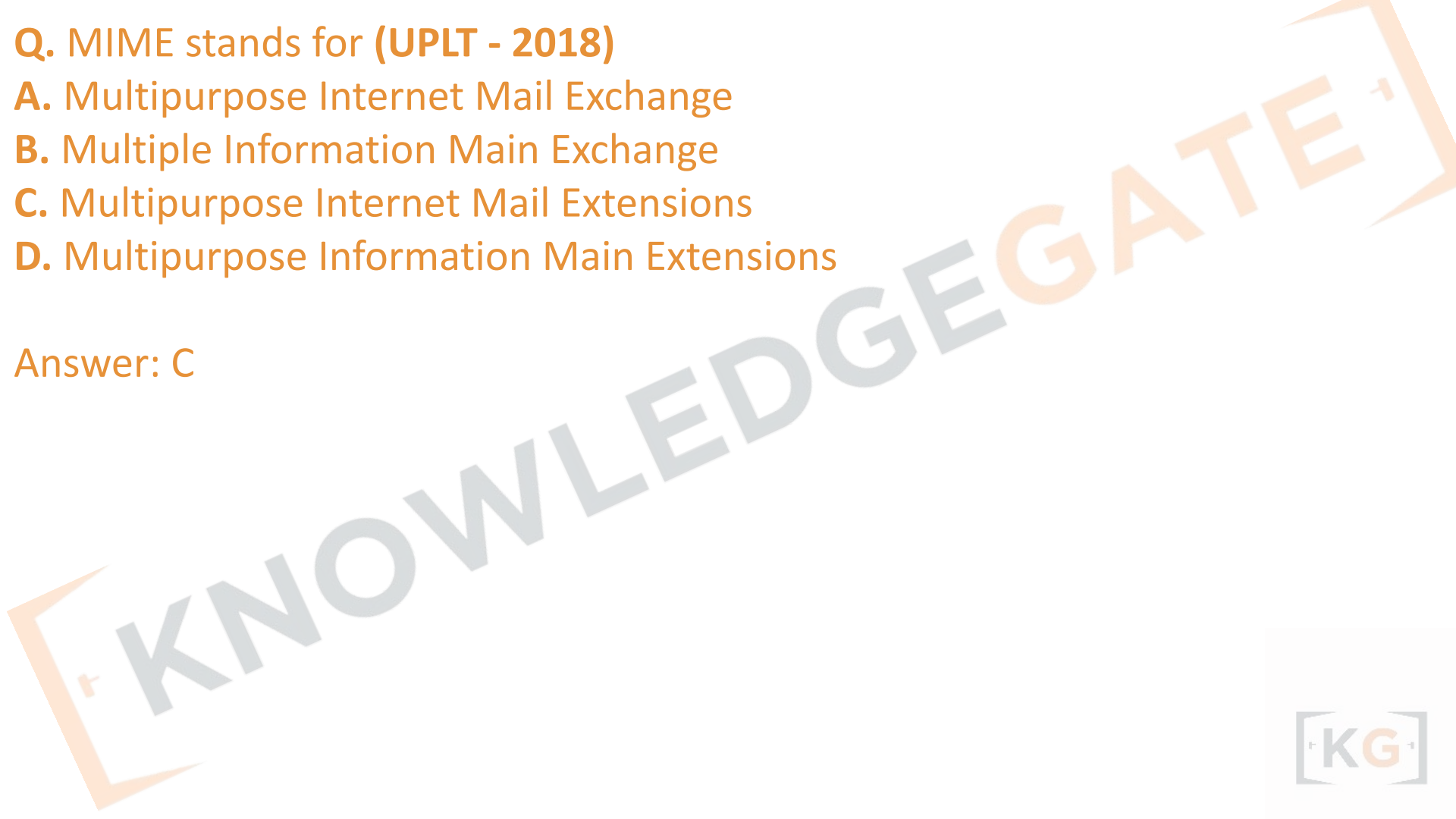
- A. Multipurpose Internet Mail Exchange**
- B. Multiple Information Main Exchange**
- C. Multipurpose Internet Mail Extensions**
- D. Multipurpose Information Main Extensions**



**Q. MIME stands for (UPLT - 2018)**

- A. Multipurpose Internet Mail Exchange**
- B. Multiple Information Main Exchange**
- C. Multipurpose Internet Mail Extensions**
- D. Multipurpose Information Main Extensions**

**Answer: C**



#### 4- IMAP ( Internet Message Access Protocol):-

It is a method of accessing e-mail or bulletin board messages that have kept on a mail server.

The latest version, IMAP4, is similar to POP3 but supports some additional features. For example, with IMAP4, you can search through your e-mail messages for keywords while the messages are still on mail server. In other words, it permits a client in a program to access remote message stores as if they were local.

Example:- E-mail stored in a IMAP server can be manipulated from a desktop computer at home, a work station at office and a notebook computer while travelling without the need to transfer messages or files back and forth between these computers.



**3. FTP(File Transfer Protocol)-** Is an internet standard that allows users to upload and download files with other computers.

**4. Newsgroups-** also called a discussion group, an online forum where users conduct written discussions about a particular subject.

- Message Boards- or discussion board is a web- based type of discussion group that does
- not require a news reader program.
- Mailing lists- A group of e-mail names and addresses given a single name for sending messages to multiple recipients .
- Chat rooms- A location on an internet server that permits users to chat with each other by typing lines of text on the computer.
- Instant Messaging- Is a areal time communication service that notifies a user when one or more people are online and allows the user to exchange messages or files with them.
- Internet telephony- Also called Voice over IP (VoIP), is a web-based telephone service that allows a user to talk to others for just the cost of the internet connection.
- Videoconferencing- A meeting between two or more geographically separated people who use a network or the internet to transmit audio and video data. Requirements A video conferencing software, microphone, speakers, webcam.



## Terms used in relation to internet.

- A Web Browser: Is a software application for retrieving, presenting, and traversing information resources on the World Wide Web.

Examples:

- Internet Explorer,
- Google Chrome,
- Mozilla Firefox,
- Opera etc

- **A Web Page:** It is a document commonly written in HTML code that contains text, pictures, sound, and video and links to other pages, files, or parts of the document

- **Home page:** This is the first page you see at a website, it acts like a title page of a book, the home page or welcome page identifies the web site and contains links to other pages.

- A Web server is software or hardware that uses HTTP (Hypertext Transfer Protocol) and other protocols to respond to client requests made over the World WideWeb (WWW).

- **Web publishing** : This is the designing and uploading of Web pages onto the www server to be shared by other people on the internet.
- **Hyperlink**: It is part of the text or graphic on a web page, that when clicked at will automatically take you to a different page/ website, download a file or launch an application, video, or sound.
- **Downloading**: Copying information from the internet to ones computer.
- **Uploading**: Posting information from a host computer to the internet e.g sending an e – mail attachment.
- **Browsing** : Searching for particular /specific items on the internet.
- **Surfing** : Means moving from one website to another for topics of interest, the term surfing is generally used to describe a rather undirected type of web browsing.

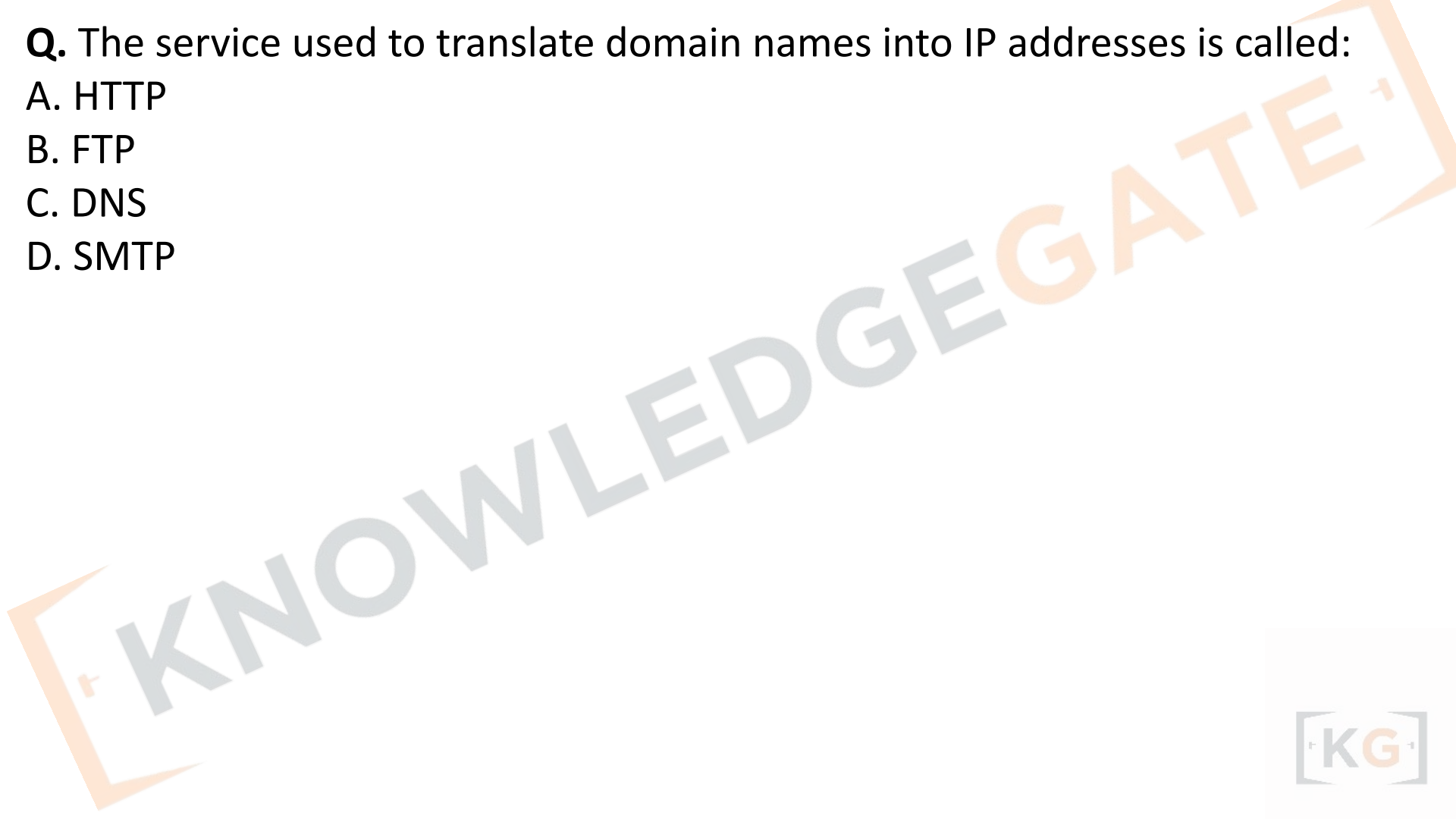
- **Bookmark:** Is a term used in Netscape to describe a Web site that is marked for later reference. Called a 'favourite' in Internet Explorer.
- HOST is any computer on a network that receives services from the server.
- A Webmaster is the individual responsible for developing Web pages and maintaining a Website.
- Search engine is a software program that can be used to find Websites, Web pages and files on the internet. Eg Yahoo, Google, Askme.com etc
- A **Website:** Is a location of data stored on the www server which can be freely accessed by people through surfing the internet using a domain name or url eg <https://www.newvision.co.ug/>

## Examples of websites:

- (i) **Rating site** Is a website designed for users to vote on or rate people, content, or other things
- (ii) **Blog (web log) Sites** generally used to post online diaries which may include discussion forums.
- (iii) A **web portal** is a specially designed website that brings information from diverse sources, like emails, online forums and search engines, together in a uniform way.

**Q.** The service used to translate domain names into IP addresses is called:

- A. HTTP
- B. FTP
- C. DNS
- D. SMTP



Q. The service used to translate domain names into IP addresses is called:

A. HTTP

B. FTP

C. DNS

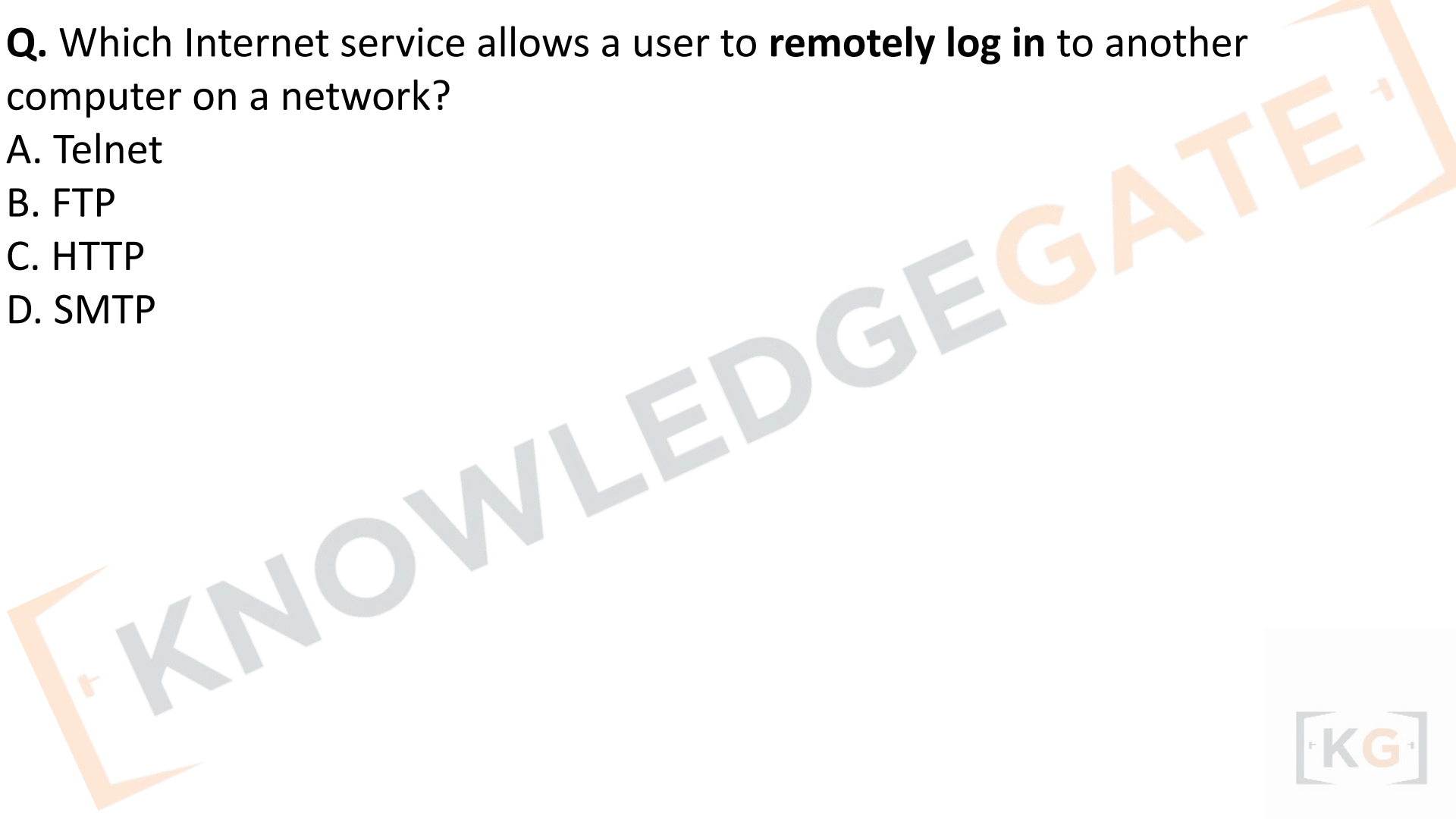
D. SMTP

**Answer:** C. DNS

**Explanation:** The Domain Name System (DNS) resolves human-readable domain names into machine-readable IP addresses.

**Q.** Which Internet service allows a user to **remotely log in** to another computer on a network?

- A. Telnet
- B. FTP
- C. HTTP
- D. SMTP



**Q.** Which Internet service allows a user to **remotely log in** to another computer on a network?

- A. Telnet
- B. FTP
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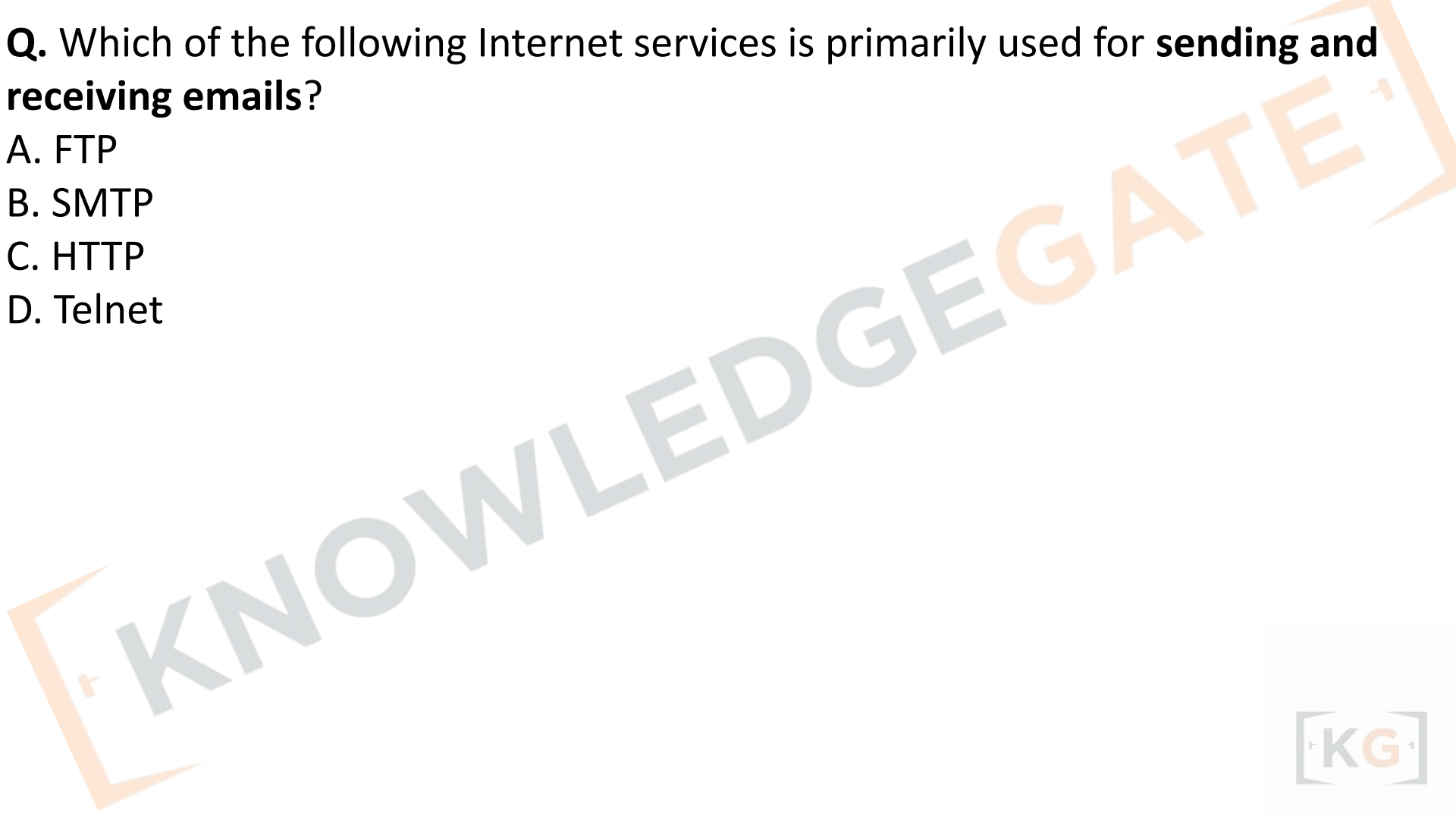
**Answer:** A. Telnet

**Explanation:** Telnet enables a user to remotely log into another system and execute commands as if they were local.



**Q. Which of the following Internet services is primarily used for **sending and receiving emails**?**

- A. FTP
- B. SMTP
- C. HTTP
- D. Telnet



**Q.** Which of the following Internet services is primarily used for **sending and receiving emails**?

- A. FTP
- B. SMTP
- C. HTTP
- D. Telnet

**Answer:** B. SMTP

**Explanation:** The Simple Mail Transfer Protocol (SMTP) handles email transmission between mail servers.

# Chapter - 3

## Current Trends on Internet

# Current Trends on Internet

**1. Introduction :** The Internet has become the backbone of modern communication, education, business, and entertainment. Current Trends on the Internet represent the technologies and applications that are shaping the way people interact, learn, and work. These trends determine how businesses grow, how people communicate, and how societies function in a digital-first world.

# Current Trends on Internet

**2. Internet Languages :** Internet languages are programming, markup, and scripting languages that make websites, apps, and online platforms function.

- Markup: HTML, XML
- Styling: CSS
- Scripting: JavaScript (with frameworks like React, Angular, Vue)
- Server-Side: PHP, Python, Java, Node.js
- Databases: SQL, NoSQL

Trends: Full-stack development with JavaScript, Python integration with AI, and multilingual support through Unicode.

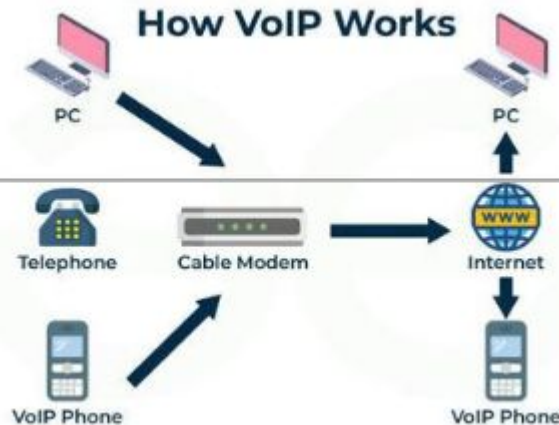
Trends: Full-stack development with JavaScript, Python integration with AI, and multilingual support through Unicode.

**3. Internet Phone (VoIP) :** VoIP (Voice over Internet Protocol) enables voice communication over IP networks instead of traditional telephone lines.

Examples: Skype, WhatsApp Calls, Google Voice, Zoom Audio.

Advantages: Low cost, multimedia integration, global connectivity.

Trends: Business VoIP call centers, cloud telephony, and mobile VoIP apps.



**4. Internet Video :** Internet Video refers to video content transmitted via the Internet, either in real-time or on-demand.

Examples: YouTube, Netflix, Amazon Prime, Zoom.

Technologies: Streaming protocols (RTP, HLS, MPEG-DASH), codecs (H.264, H.265, AV1).

Trends: Short-form videos (Reels, Shorts), live streaming (Twitch, webinars), and OTT platforms replacing traditional TV.



**5. Collaborative Computing :** Collaborative computing allows multiple people to work together in real-time through the Internet.

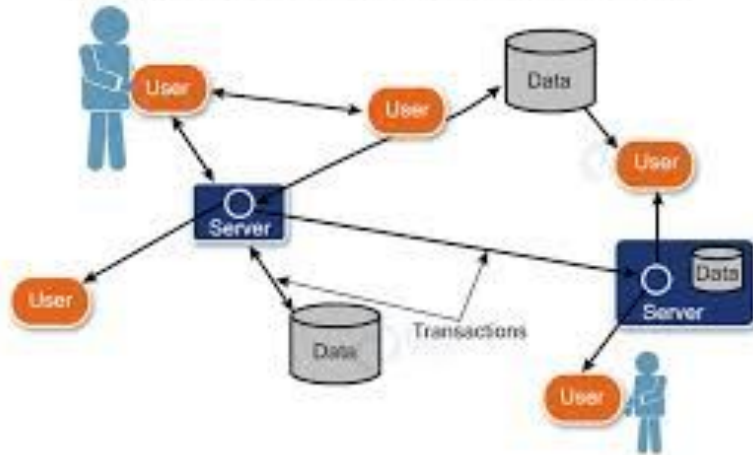
Examples: Google Docs, Microsoft Teams, Slack, GitHub.

Technologies: Cloud computing, APIs, WebRTC.

Trends: Remote work, online classrooms, virtual project management tools.

Advantages: Increased productivity, global collaboration, cost-effectiveness.

Collaborative Computing Architecture





**6. E-Commerce** : E-Commerce is the buying and selling of goods and services over the Internet.

Types: B2C (Amazon, Flipkart), B2B (Alibaba, IndiaMART), C2C (OLX, eBay), C2B (Freelancer platforms).

Technologies: Payment gateways, HTTPS/SSL, logistics integration.

Trends: Mobile commerce, social commerce (Instagram Shop), AI recommendations.

Challenges: Cyber fraud, privacy issues, logistics problems.



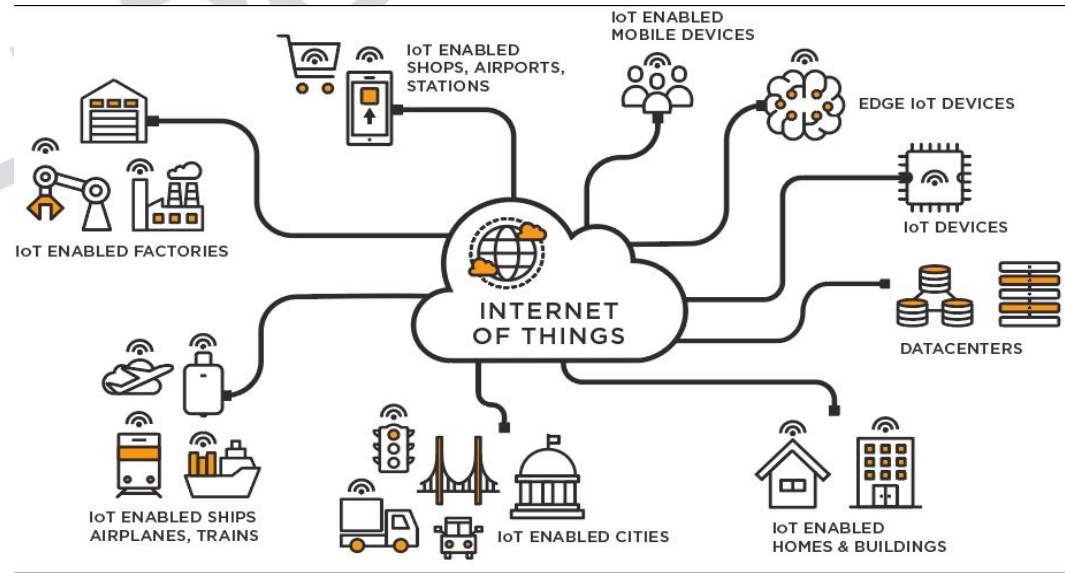
## 7. Other Current Trends

**7.1 Internet of Things (IoT) :** IoT refers to interconnected devices embedded with sensors and software that exchange data over the Internet.

Examples: Smart Homes (Alexa, Google Home), Smart Cities, Wearable healthcare devices.

Trends: AIoT (AI + IoT), Industrial IoT, smart agriculture.

Challenges: Security risks, privacy, lack of standards.



## 7. Other Current Trends

**7.2 Artificial Intelligence (AI) :** AI powers search engines, recommendation systems, chatbots, and personalization on the Internet.

Applications: Google Search, Netflix recommendations, Siri, ChatGPT, adaptive learning.

Trends: NLP for communication, AI in e-commerce, AI in cybersecurity.

Challenges: Bias, ethical issues, automation impacts on jobs.



## 7.3 Cybersecurity & Privacy : Cybersecurity protects data and networks from attacks, while privacy ensures responsible use of personal data.

Threats: Phishing, ransomware, identity theft.

Trends: Zero Trust architecture, stronger encryption, global cyber laws.

Importance: Protecting sensitive information and building user trust.

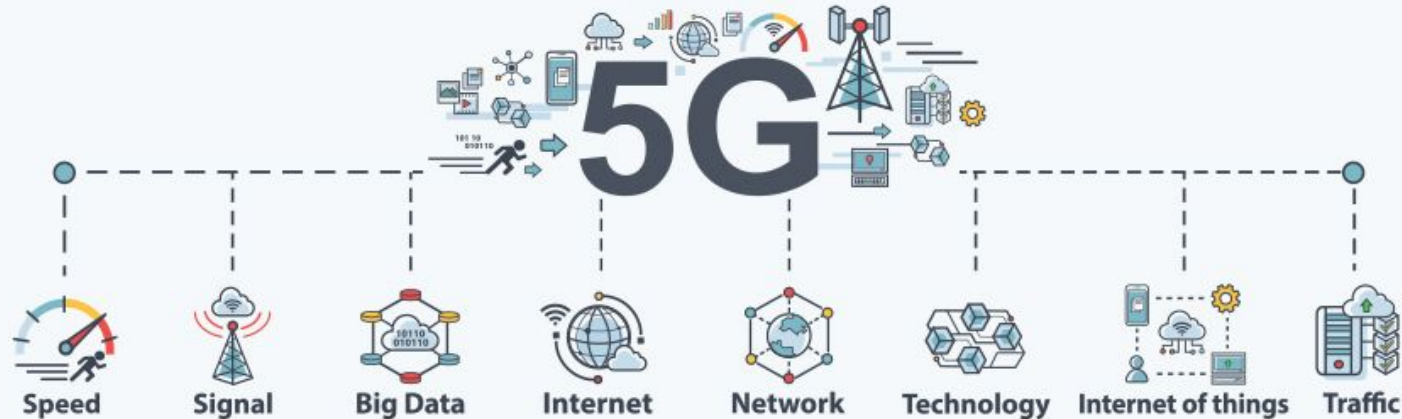


## 7.4 5G and Future Internet : 5G offers high-speed, low-latency communication enabling IoT and advanced applications.

Applications: AR/VR, autonomous vehicles, telemedicine.

Future: Research on 6G, edge computing for faster services.

Challenges: High infrastructure cost, security risks.



**7.5 Web3 and Blockchain :** Web3 is the decentralized phase of the Internet using blockchain. Blockchain operates as a decentralized distributed [database](#), with data stored across multiple computers, making it resistant to tampering. Transactions are validated through a consensus mechanism, ensuring agreement across the network.

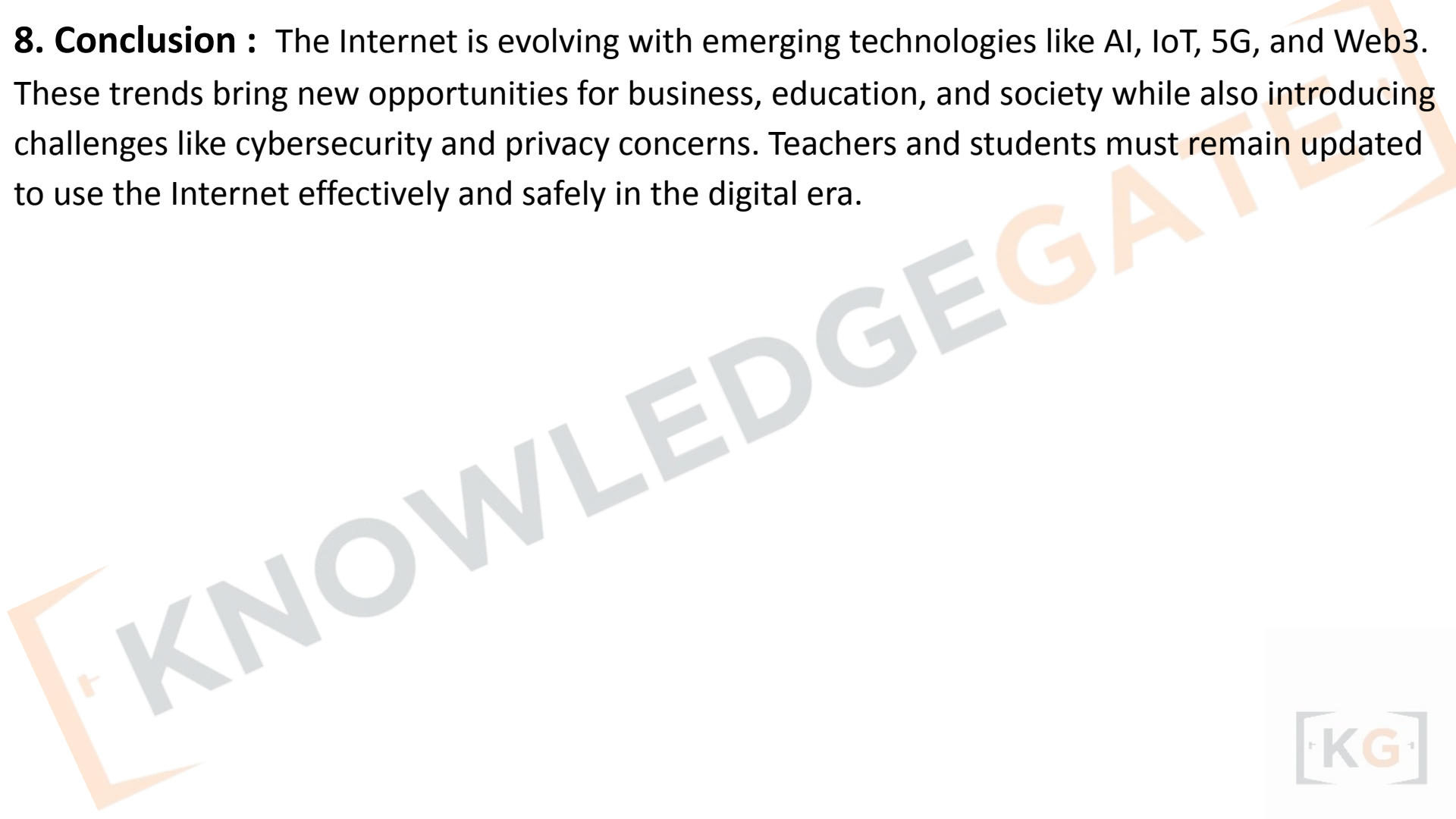
Applications: Cryptocurrencies, NFTs, decentralized apps (DApps), DeFi.

Trends: NFT marketplaces, smart contracts, decentralized storage.

Challenges: Regulations, energy use, fraud risks.

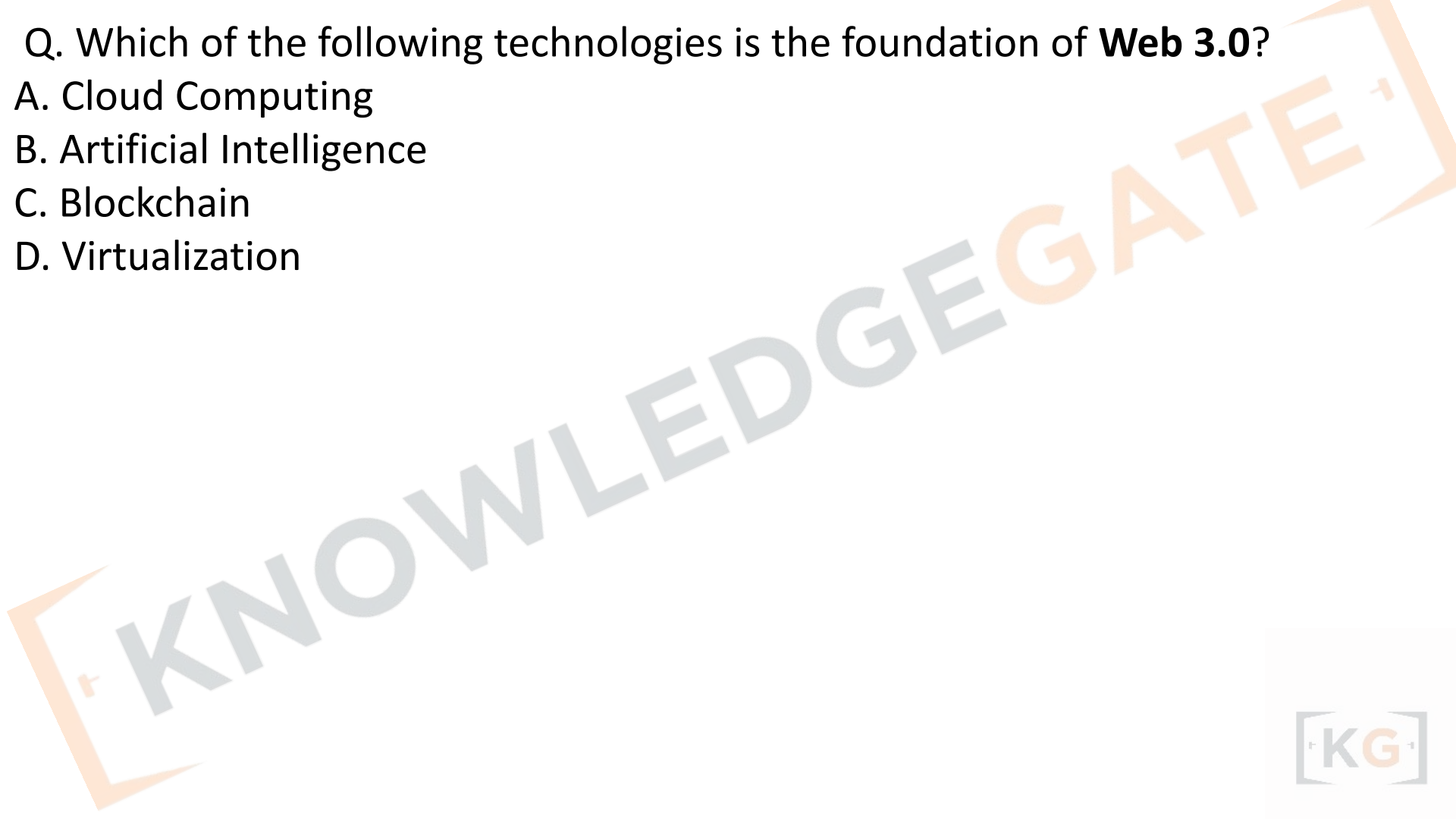


**8. Conclusion :** The Internet is evolving with emerging technologies like AI, IoT, 5G, and Web3. These trends bring new opportunities for business, education, and society while also introducing challenges like cybersecurity and privacy concerns. Teachers and students must remain updated to use the Internet effectively and safely in the digital era.



Q. Which of the following technologies is the foundation of **Web 3.0**?

- A. Cloud Computing
- B. Artificial Intelligence
- C. Blockchain
- D. Virtualization





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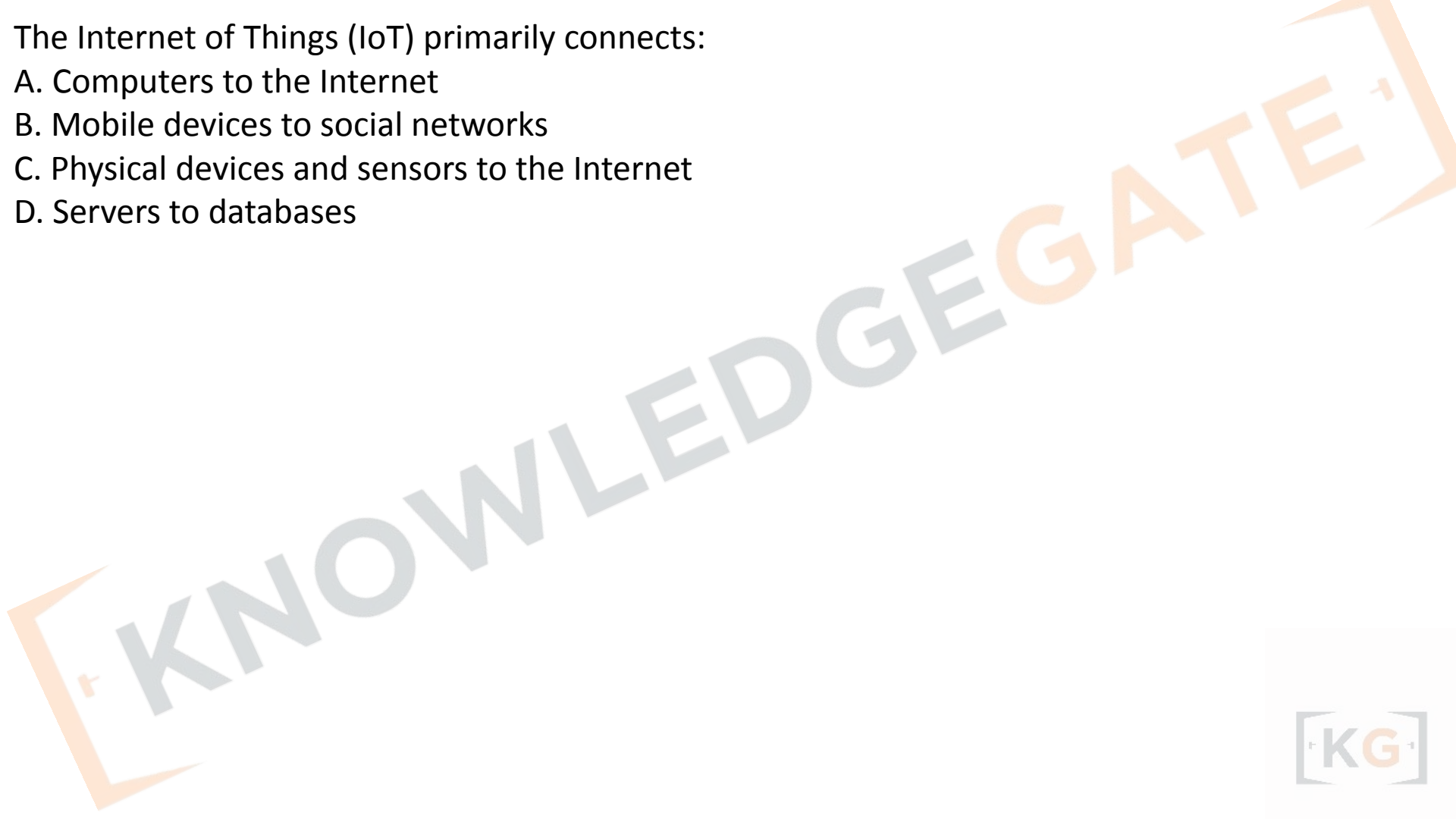
**Answer:** C. Blockchain

**Explanation:** Web 3.0 emphasizes decentralization, transparency, and user control, which are made possible through blockchain technology.



The Internet of Things (IoT) primarily connects:

- A. Computers to the Internet
- B. Mobile devices to social networks
- C. Physical devices and sensors to the Internet
- D. Servers to databases



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- A. Computers to the Internet
- B. Mobile devices to social networks
- C. Physical devices and sensors to the Internet
- D. Servers to databases

**Answer:** C. Physical devices and sensors to the Internet

**Explanation:** IoT enables communication among smart devices using embedded sensors and network connectivity.

Which of the following is an example of **AI-driven Internet service**?

- A. Google Maps route prediction
- B. FTP file transfer
- C. Telnet remote login
- D. DNS name resolution

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- D. DNS name resolution

**Answer:** A. Google Maps route prediction

**Explanation:** AI algorithms analyze real-time traffic data to optimize routes and provide intelligent navigation.

# Chapter - 4

## Web Publishing and Browsing

# Web Publishing and Browsing

## Overview:

Web publishing is the process of publishing original content on the Internet.

The process includes building and uploading websites, updating the associated webpages, and posting content to these webpages online. Web publishing comprises of personal, business, and community websites in addition to e-books and blogs.

The content meant for web publishing can include text, videos, digital images, artwork, and other forms of media. Publishers must possess a web server, a web publishing software, and an Internet connection to carry out web publishing.

Web publishing is also known as online publishing

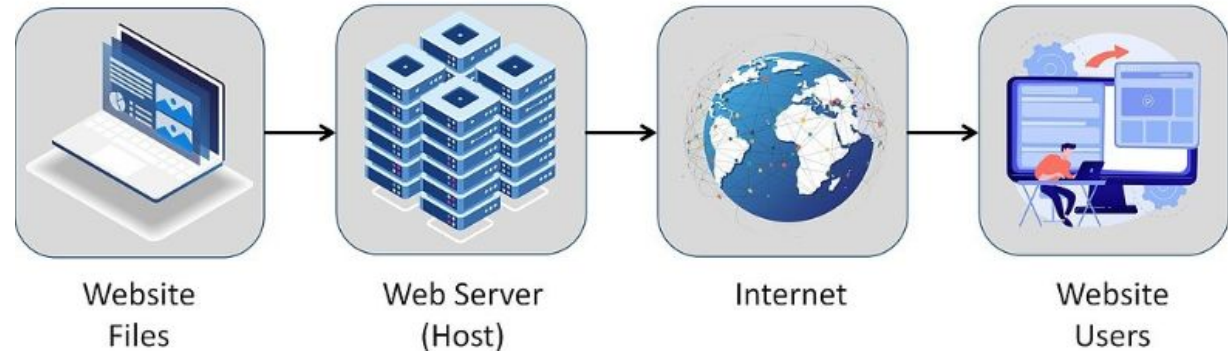
# Web hosting Basics:

Web hosting is a service of providing online space for storage of web pages. These web pages are made available via World Wide Web. The companies which offer website hosting are known as Web hosts.

The servers on which web site is hosted remain switched on 24 x7. These servers are run by web hosting companies. Each server has its own IP address. Since IP addresses are difficult to remember therefore, webmaster points their domain name to the IP address of the server their website is stored on.

It is not possible to host your website on your local computer, to do so you would have to leave your computer on 24 hours a day. This is not practical and cheaper as well. This is where web hosting companies comes in.

## WEB HOSTING





# Hosting Types:

## Different Types of Web Hosting Services



Self  
Hosting



Shared  
Hosting



Virtual  
Private Server



Dedicated  
Hosting



Reseller  
Hosting



Managed  
WordPress Hosting



Colocation  
Hosting



Cloud  
Hosting

# Hosting Types:

Following are most widely used hosting types –

**Free Hosting:** Free web hosting is best suited for small sites with low traffic, like family sites or sites about hobbies. There are many service providers who will give you free space on their web server with a condition that you will allow them to run their advertisement at your web pages. So if you are OK with this option, then you have nothing to pay for a space. There are some websites like **geocities.com**, **lycos.com**, **myspace.com**, etc. that give you space to build your web pages.



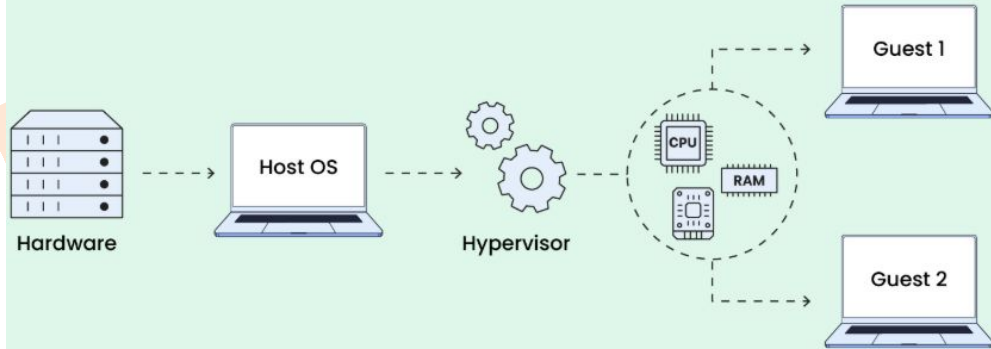
# Hosting Types:

**Shared Hosting:** With shared hosting, your website is hosted on a powerful server along with other websites. On a shared host, you will have your own user ID and password to login to the shared host and you will be allowed to work in your work area. You would not be able to touch any file or directory belonging to other host partner. Even you would not know how many sites are hosted on your shared host. This type of hosting is very cost effective and good for small websites where your space and speed are not very important. Here traffic on one site will affect the speed of all other hosted sites.



**Virtual Dedicated Hosting :** This type of hosting is better for medium size business. With virtual dedicated hosting, you will have a dedicated bandwidth and dedicated RAM for your site. You will be given a root ID and password to maintain your Web server. You will be the complete owner of your virtual dedicated server and will be able to install or uninstall any software. This type of hosting is created on a single server, but it is managed in such a way that every user will have dedicated speed and bandwidth. This is bit more expensive but really good one for medium size business.

### How Virtual Private Server (VPS) Works



**Dedicated Hosting** : This type of hosting is very similar to virtual dedicated hosting, but here, a complete machine will be allotted for you. They are more expensive than virtual dedicated hosting and should be considered when you have a very high traffic requirement.

## Dedicated Web Hosting



**Collocated Hosting :** It is very difficult to set dedicated resources such as high-security against fire and vandalism, regulated backup power, dedicated Internet connections and more. Collocated hosting, or "colo," is a service where a business rents space in a third-party data center to house its own servers and hardware. The data center provides the physical infrastructure, such as power, cooling, security, and network connectivity, while the business retains full ownership and control over its equipment. This setup allows companies to leverage the benefits of a professional data center without the cost of building one themselves.

### Colocation vs Dedicated Hosting

## Which is Right for Your Business



**Colocation**

VS



**Dedicated**



## **Dedicated Hosting**



Rent Servers



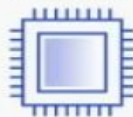
Customers rent servers as well as the infrastructure needed to house and maintain them



## **Colocation Hosting**



Own Hardware



Companies own their own hardware, but are looking to augment its usefulness by **renting server space in a data center**

## **Benefits of Colocation Hosting**



Improved Internet Bandwidth



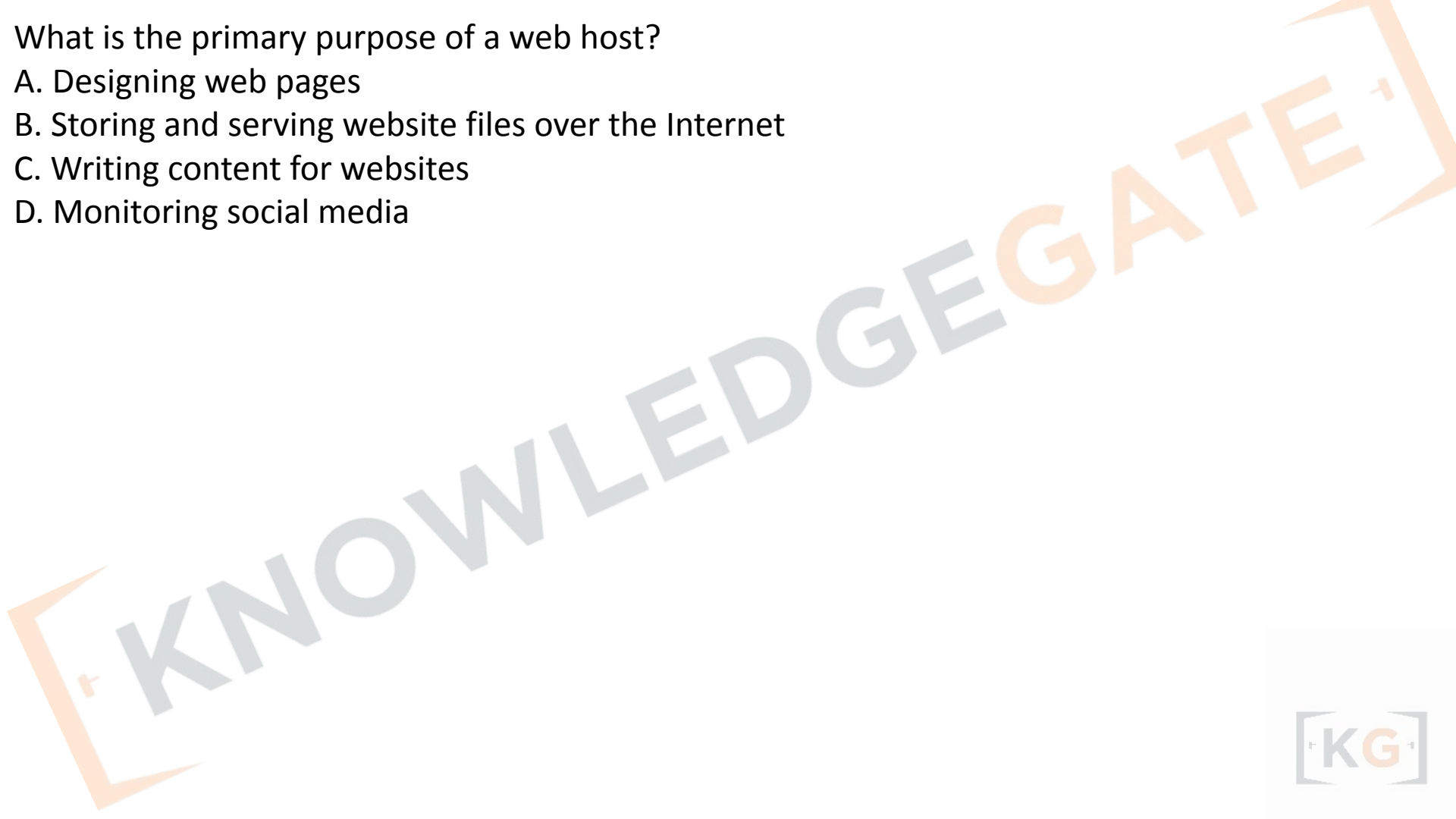
Server Cooling Systems and Climate Control



Reliable Power

What is the primary purpose of a web host?

- A. Designing web pages
- B. Storing and serving website files over the Internet
- C. Writing content for websites
- D. Monitoring social media





What is the primary purpose of a web host?

- A. Designing web pages
- B. Storing and serving website files over the Internet
- C. Writing content for websites
- D. Monitoring social media

**Answer:** B. Storing and serving website files over the Internet

**Explanation:** Web hosting services provide server space, bandwidth, and other resources needed for websites to be accessible online.

Which type of web hosting allows multiple websites to share the same server resources?

- A. Dedicated Hosting
- B. Shared Hosting
- C. Cloud Hosting
- D. VPS Hosting



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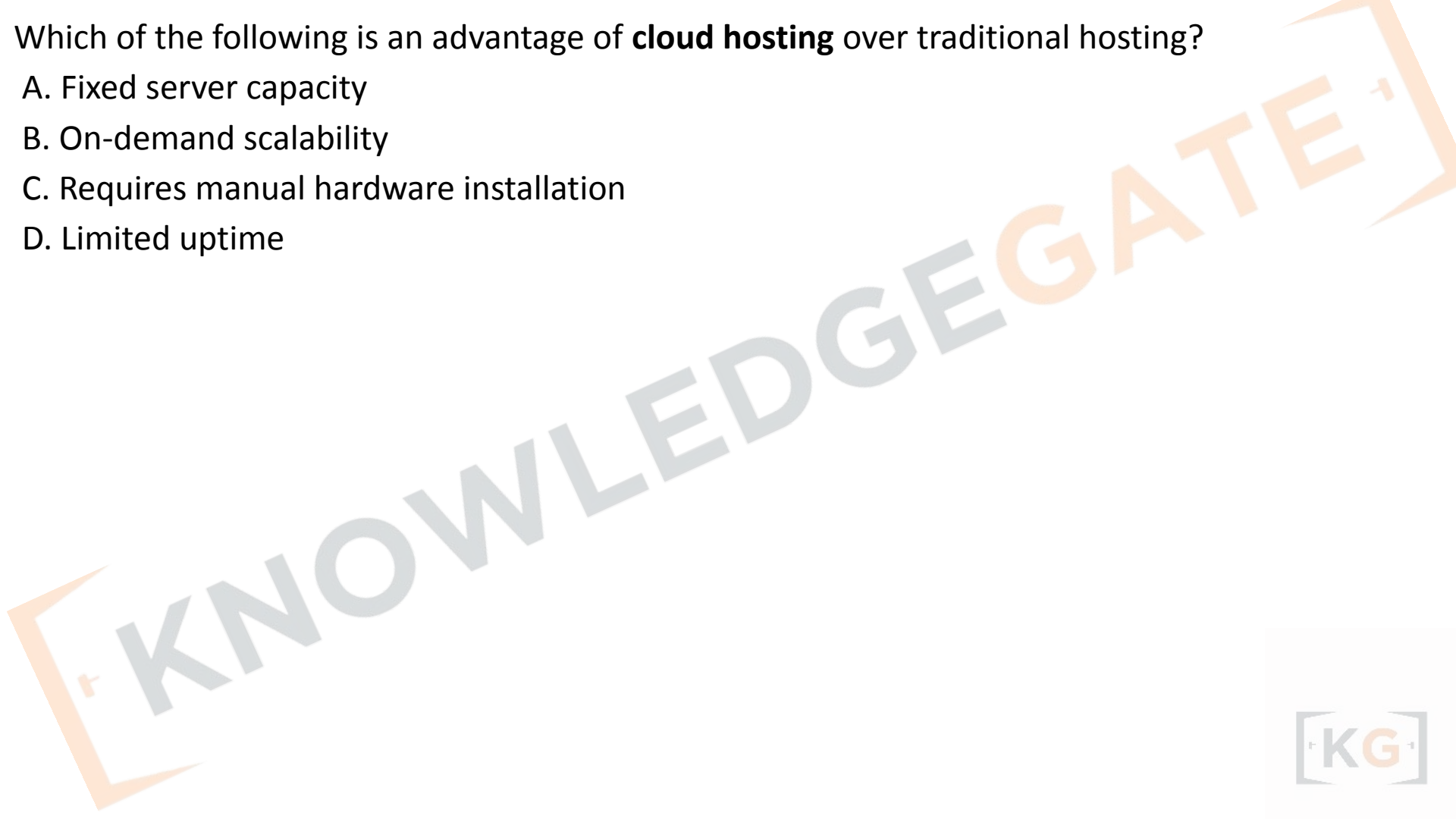
**Answer:** B. Shared Hosting

**Explanation:** Shared hosting is cost-effective as multiple users share CPU, memory, and bandwidth on the same server.



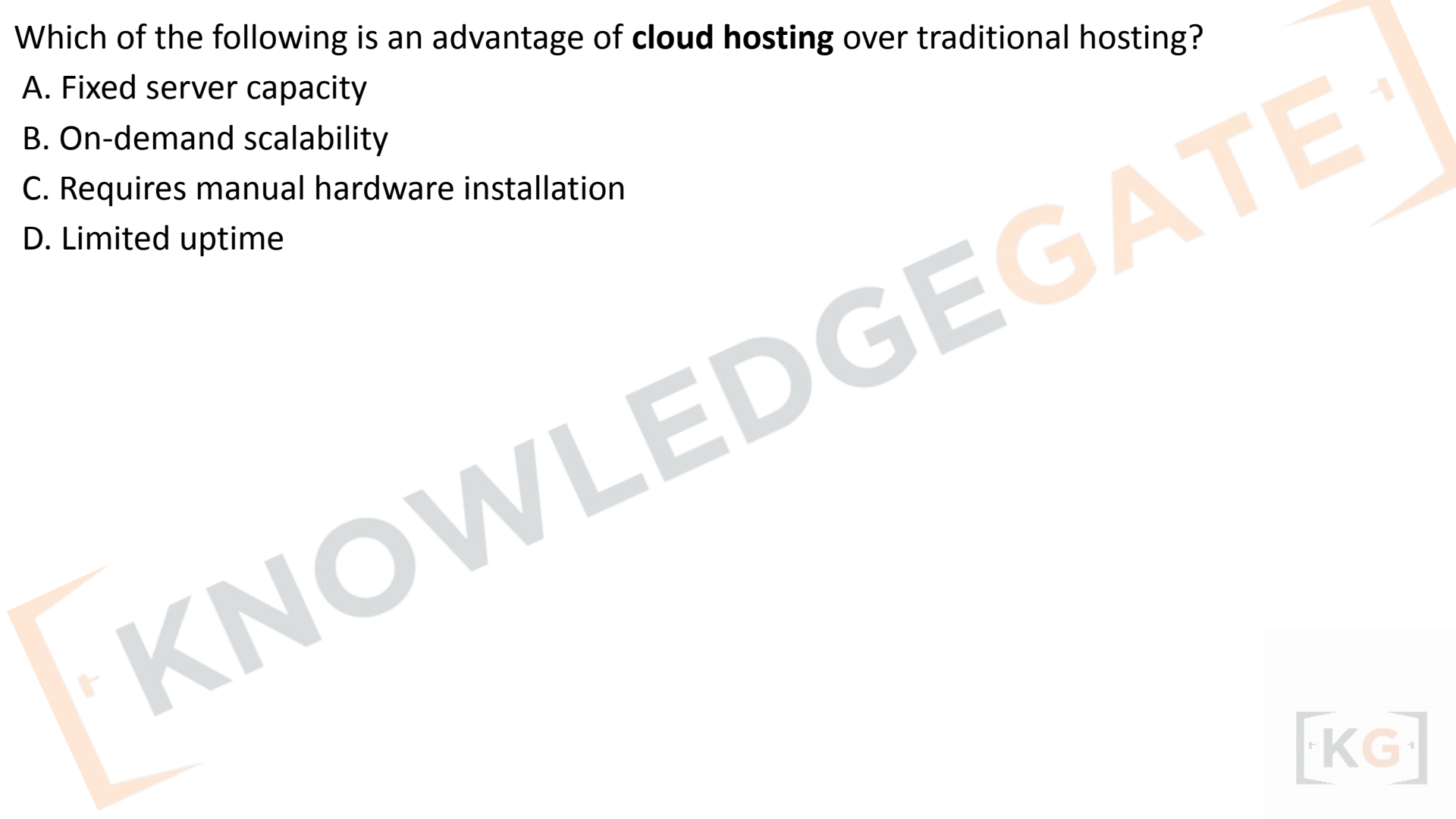
Which of the following is an advantage of **cloud hosting** over traditional hosting?

- A. Fixed server capacity
- B. On-demand scalability
- C. Requires manual hardware installation
- D. Limited uptime



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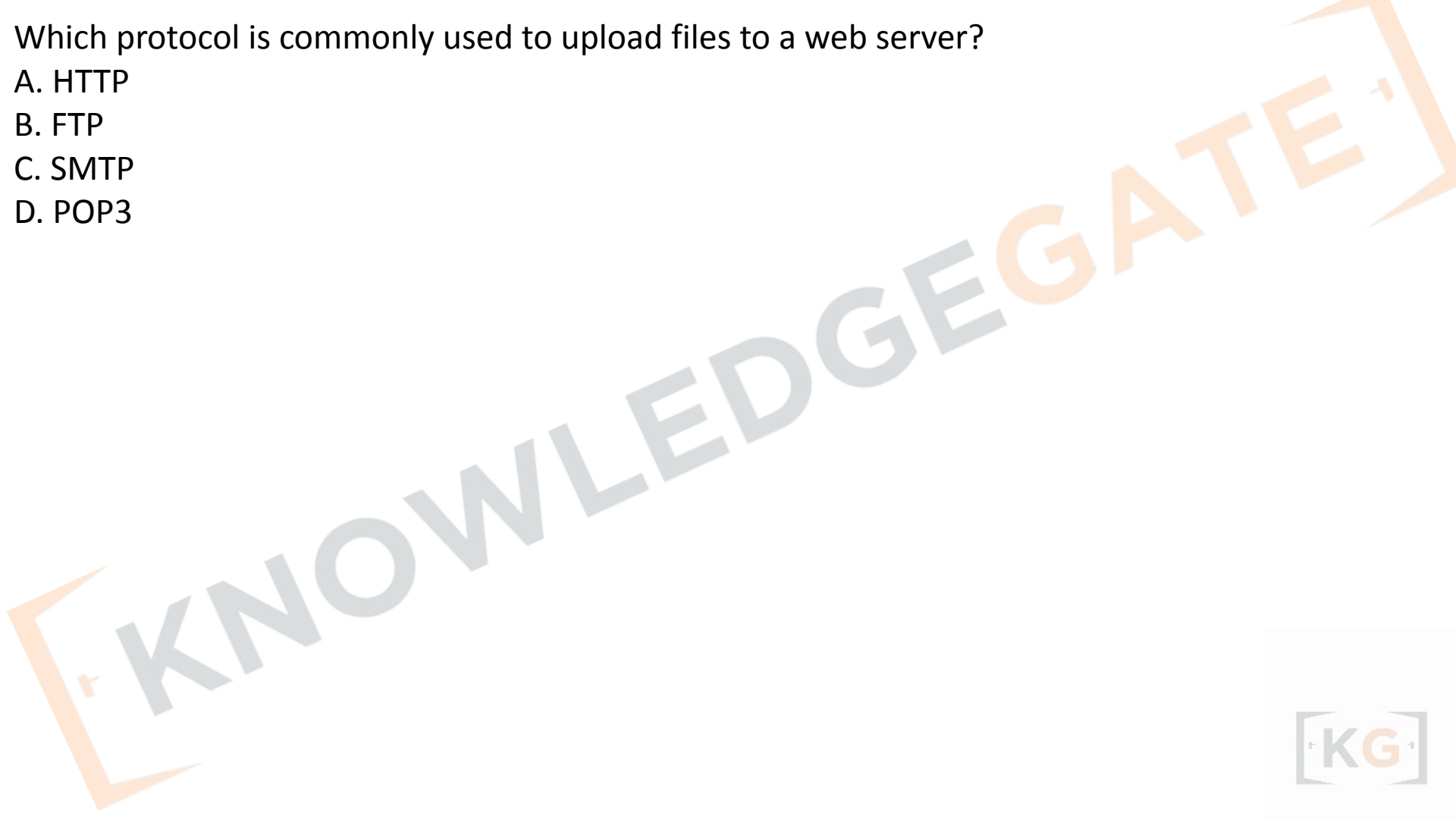
- A. Fixed server capacity
- B. On-demand scalability
- C. Requires manual hardware installation
- D. Limited uptime

**Answer:** B. On-demand scalability

**Explanation:** Cloud hosting allows resources to scale dynamically based on website traffic and demand.

Which protocol is commonly used to upload files to a web server?

- A. HTTP
- B. FTP
- C. SMTP
- D. POP3



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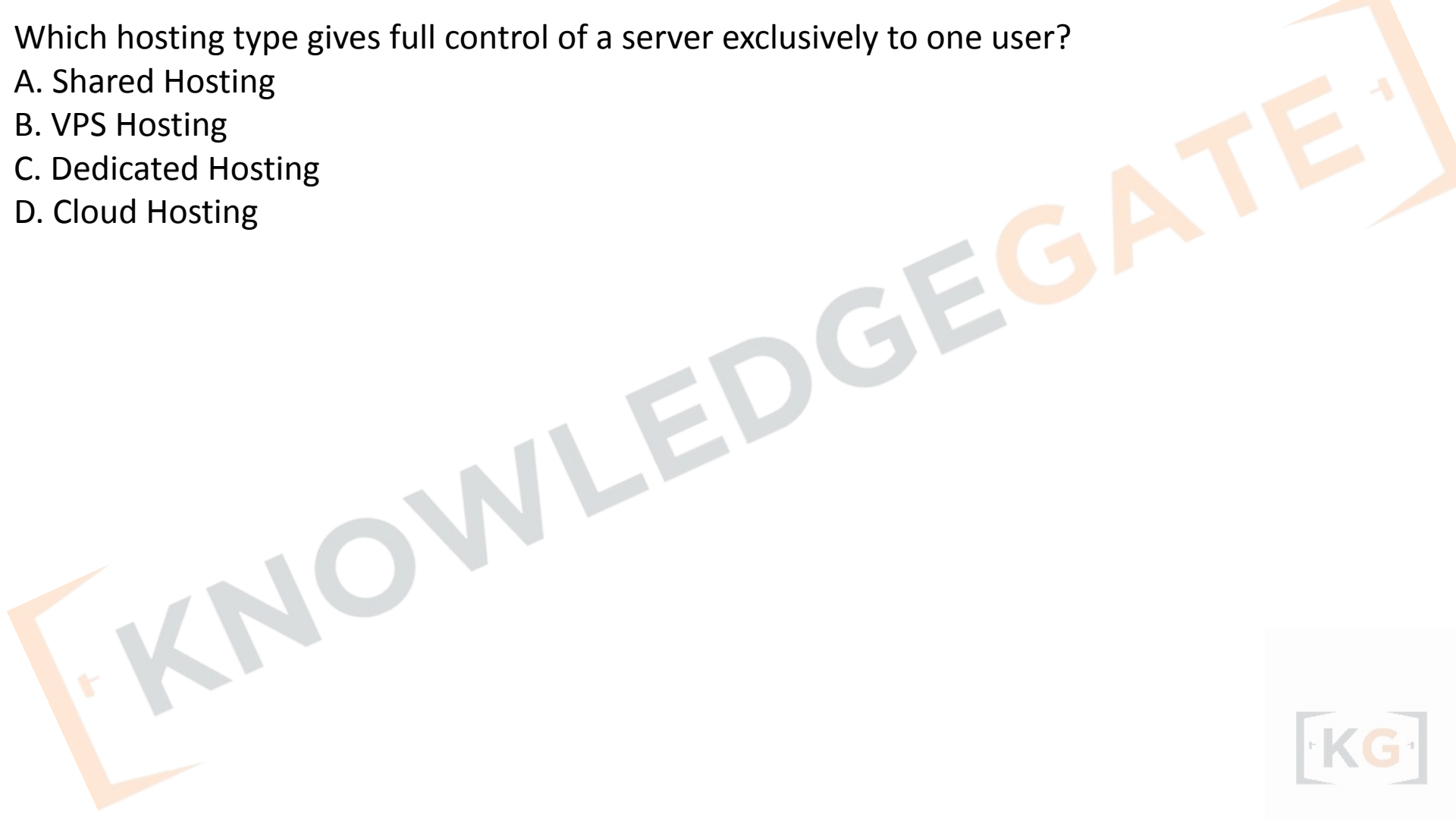
**Answer:** B. FTP

**Explanation:** File Transfer Protocol (FTP) allows transferring files from local systems to remote web servers.



Which hosting type gives full control of a server exclusively to one user?

- A. Shared Hosting
- B. VPS Hosting
- C. Dedicated Hosting
- D. Cloud Hosting



Which hosting type gives full control of a server exclusively to one user?

- A. Shared Hosting
- B. VPS Hosting
- C. Dedicated Hosting
- D. Cloud Hosting

**Answer:** C. Dedicated Hosting

**Explanation:** Dedicated hosting provides complete server resources to a single client, allowing full customization and high performance.

# Components of web publishing

Web publishing is a large process and identical to book publishing. It starts right from writing the first word of the first page to the moment it gets published. Likewise, Web publishing includes different stages till it becomes available for the premiere. The process includes deciding a name of domain, registering it with the competent registrar, starts writing contents for the site, designing pages for them, uploading the site with a host. Every stage is crucial and one has to be careful while choosing perfect software for designing and uploading and deciding about the most appropriate host in terms of facilities offered and investment as well.

There are basically five phases in web publishing as follows-

- Domain name planning and registration
- Web hosting
- Web design and development
- Promotion
- Maintenance

## **Domain Name Planning and Registration**

Domain name is an address where Internet users can find one on the web. The first phase of Web Publishing is domain name planning and registration. There are certain things one must consider while one is going to get a domain name registered.

### **Getting a Domain Name**

The first thing one needs to do before anything else is to get a domain name for the proposed site. A domain name is the name one want to give to one's website. For example, blogspot.com is a domain name. To get a domain name, you have to pay an annual fee to a registrar for the right to use that name. Getting a name does not get you a website or anything like that. It's just a name. It's sort of like registering a business name with some competent government to avoid duplication of the name.

### **Choosing a Domain Name**

Before one rush out and choose your domain name or name your website, you would like to consider the following points

## **Your Domain Name should be Your Website Name**

Domain name that matches one's brand name is very important. The very name that one uses to advertise one's product is the name that one will want for one's domain, because that is the first thing that people will try in their browser.

## **Your domain name should not be too long**

Domain names can be of any length up to 67 characters. one doesn't have to settle for an obscure domain name like bgo.com when one mean's blogspot.com

If one manages to get a short domain name though, the key is to make sure it's meaningful combination of characters

## **Your domain name should be hyphens free**

Domain name with hyphenation has got some advantage as well as some disadvantages. Search engines can distinguish your keyword better if you have domain name with hyphen. Disadvantage is that is it easy to forget hyphens when typing a name. Moreover, when somebody recommends one's site to his./her friends verbally having hyphens in your domain name leads to more potential errors that when the name does not contain hyphens. Therefore, it is advisable to avoid domain name with hyphens

## **Your domain name should not contain plural words**

Plural form of the domain name always creates loss, since the chance of the visitor failing to type the "s" in the name is very great. For example, somebody wants to log on to websites.com but mistakenly leaves website.com may lead to a loss of clients if the products are the same on the both domains

## **Your top level domain name should portray the nature of your site**

If your site is of a charitable institution or a voluntary organization, never get a domain name with .com Neither forget to choose country specific top level domain if one is working for the clients in a particular country only. If one is doing business for profit .com is the best top level domain. And if one is doing business in India, select in as country specific top level domain.

# **World Wide Web (known as "WWW", "Web" or "W3"):**

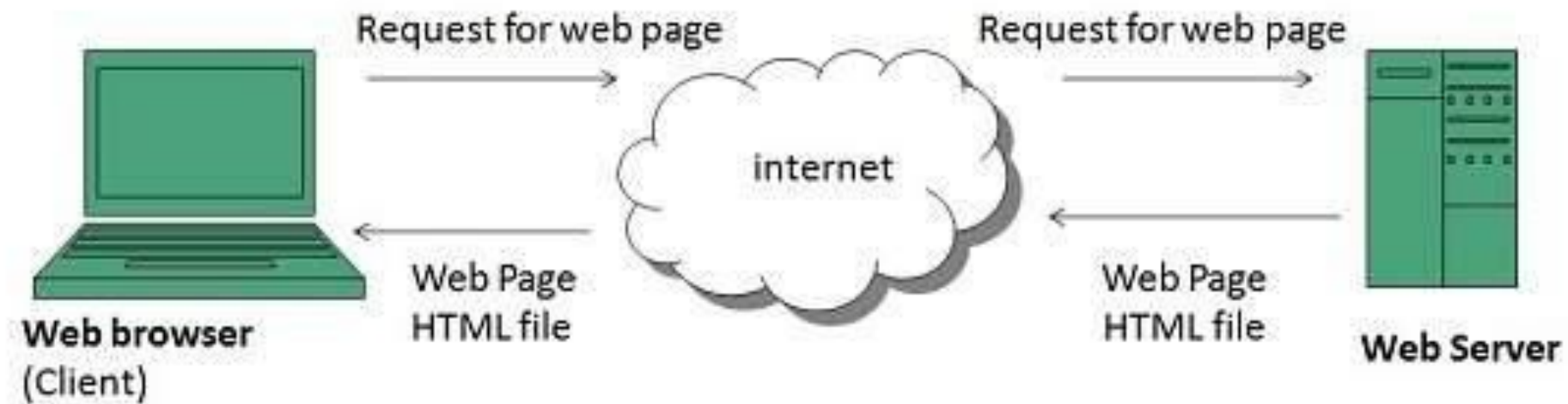
Tim Berners-Lee invented the World Wide Web in 1989 at CERN in Geneva, The World Wide Web is not synonymous with the Internet.

The World Wide Web is the universe of network-accessible information, an embodiment of human knowledge.

World Wide Web (abbreviated as WWW or W3, commonly known as the web) is a system of interlinked documents that are accessed via the internet.

In contrast, the World Wide Web is a global collection of documents and other resources, linked by hyperlinks and uniform resource identifier.

**www operation diagram:**





## Web or Internet :

1. It is important to understand that the Internet and the World Wide Web are not the same thing.
2. The internet is a collection of computers and other devices connected by equipment that allows them to communicate with each other.
3. The web is a collection of software and protocols that has been installed on most, if not all, of the computers on the internet.
4. The internet was quite useful before the web was developed and it is still useful without it.
5. However, it is now the case that most users of the internet use it through the web.
6. The internet is a global data communications system including hardware and software infrastructure that provides connectivity between computers. In contrast, the web is one of the services provided via the internet.
7. Web is a collection of interconnected documents and other resources, linked by hyperlinks and URLs. Thus in short the web is an application running on the internet.

## **Basic Elements of World Wide Web (WWW) :**

The web is based on a Client/Server System. We use a web client to communicate with the web server.

A web server is a computer connected to the internet that runs a program (which is also called web server) that takes responsibility for sorting, retrieving and distributing some of the web files.

A web client or web browser is a computer that request files from the web.

URLs are used by web browsers to locate internet resources. A URL specifies the protocol to be used in accessing the resource, the name of the server on which the resource resides and optionally the path to a resource. For example, <http://www.nielit.in>

# **Search and Meta Search Engine:**

## **Search Engine:**

**A search engine is a web-based tool that enables users to locate information on the World Wide Web. Popular examples of search engines are Google, Yahoo!, and MSN Search.**

Search engines utilize automated software applications (referred to as robots, bots, or spiders) that travel along the Web, following links from page to page, site to site. The information gathered by the spiders is used to create a searchable index of the Web.

A search engine is an online tool that searches for results in its database based on the search query (keyword) submitted by the internet user. The results are usually websites that semantically match with the search query.

## **The popular search engine are-**

### **Google**

Google Search Engine is the best search engine in the world and it is also one of most popular products from Google. It is used very frequently. Its popularity is assumed in the way like most of people have set this as its home page and default search engine for the browser. It is a web portal contains number of services like Mail, Books, Translate, MAP, Google+, YouTube and so on.

## **Yahoo**

Yahoo! Search is a web search engine owned by Yahoo, headquartered in Sunnyvale, California. As of October 2018, it is the fourth largest search engine worldwide across all platforms.

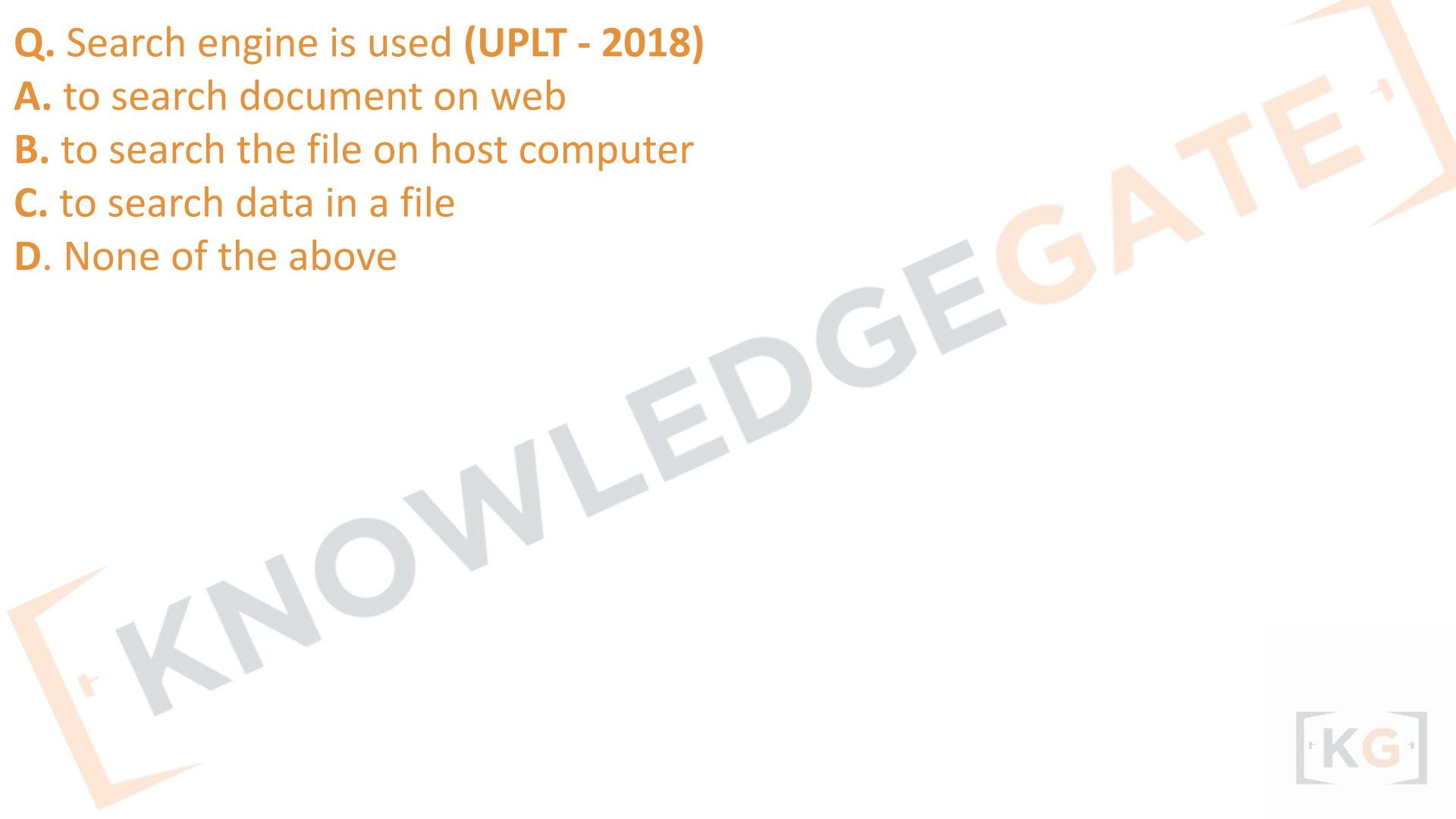
## **Bing**

Bing is the newer search engine among mentioned search engines. It is developed by Microsoft Corporation. It has replaced mostly MSN(Microsoft Network ) search engines of its own developer team. This is very interactive user friendly search Engine.



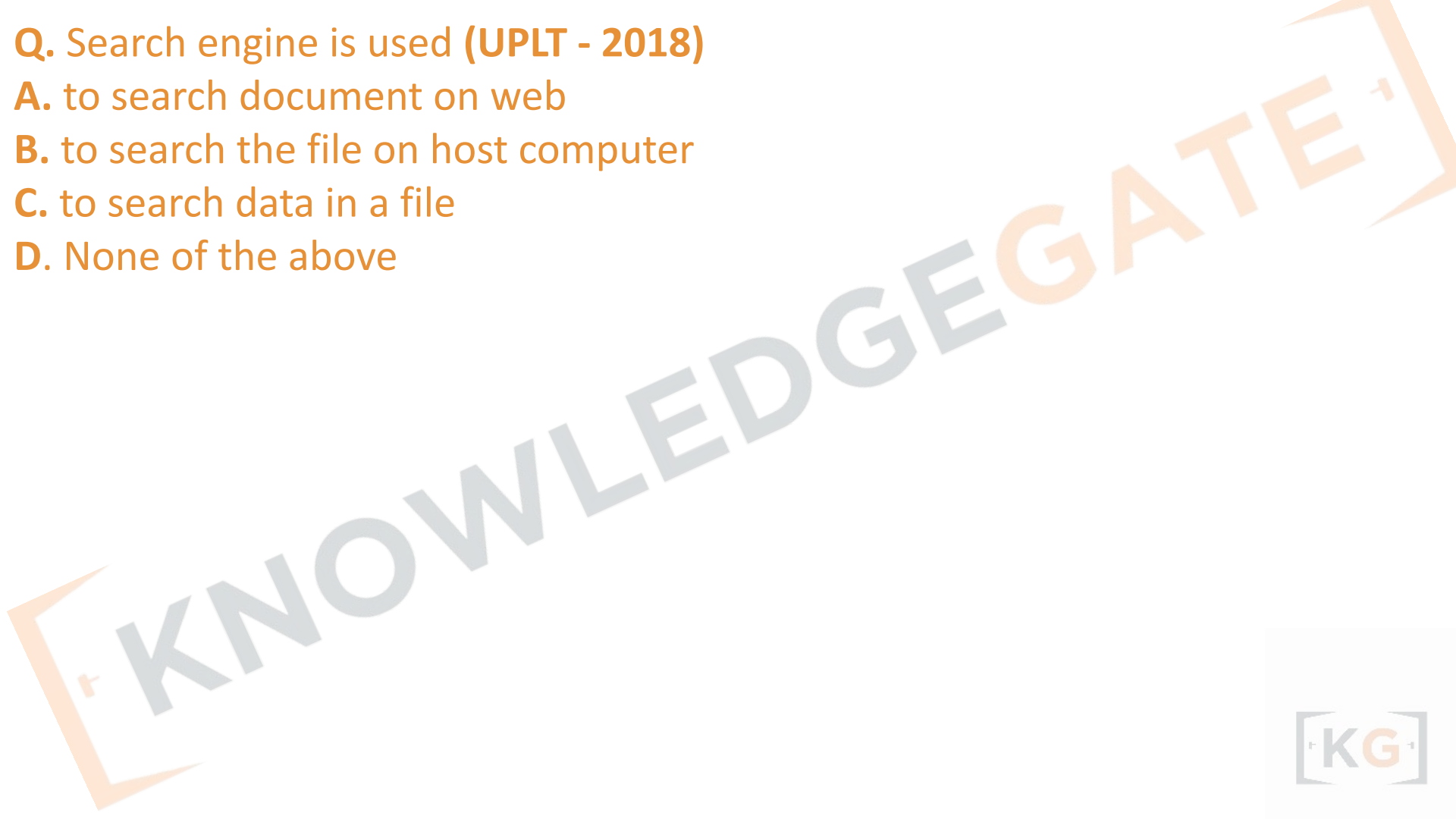
**Q. Search engine is used (UPLT - 2018)**

- A. to search document on web**
- B. to search the file on host computer**
- C. to search data in a file**
- D. None of the above**



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## **Meta Search Engine:**

**The Metasearch Engine is a search engine that combines the results of various search engines into one and gives one result. It can also be stated as an online information retrieval tool.**

The metasearch engines run the queries on most other search engines and in turn reflect the result in the form of the summarization of such sites.

HumHailIndia.com which was India's first meta search engine that was developed by Sumeet Lamba in the year 2000. It was later called Tazaa.com

## **Need for Metasearch Engines**

The Metasearch Engines were developed to cover the entire web, unlike most search engines. Individual search Engines try to spam people to enhance their page rankings. This is an illegitimate way of promoting. The individual Search engines are unable to find results from other search engines. This is when the Metasearch Engine comes handy

## List of popular meta search engine:

**Dogpile** is a metasearch engine developed by InfoSpace LLC. It writes back results from the individual search engines of Google and Yahoo. It combines the search results of text, images, new, etc.

**Sputtr** is comparably one of the best meta search engines that combines the results from various popular search engines like Google, Yahoo, Bing, Ask.com, Twitter, Facebook, YouTube, LinkedIn, IMDb, Slashdot, Dictionary.com, About.com, New York Times, etc.

**Trivago**, a hotel booking site that mentions this concept in its advertisements. It gives back results from over 100 other hotel booking sites including Booking.com, Airbnb, Expedia, Trip.com, and Agoda. Trivago can be visited at [www.trivago.com](http://www.trivago.com).

**WebCrawler**(<https://www.webcrawler.com/>) is a search engine, and is the oldest surviving search engine on the web today. For many years, it operated as a metasearch engine. it is Created by: Brian Pinkerton.

**MetaCrawler** is a search engine. It is a registered trademark of InfoSpace and was created by Erik Selberg. It was originally a metasearch engine. It was launched on 7 July 1995.



# Advantages of Metasearch Engines

1. The meta search engine searches for all the search engines at once hence individual search is unnecessary.
2. The metasearch engine comes free of cost. Its services are similar to those offered by other search engines.
3. The users can customize the metasearch engine according to their preferences.
4. The results offer coverage data that include more information than results by individual sites.
5. It reduces the workload of the users by allowing comprehensive results in very little time skipping the time and hard labour of searching for results in the individual sites.
6. Metasearching is a better alternative to searching if getting an overview or quick results is the motive.
7. The Metasearch engines provide a means of hiding the IP addresses from the searches. This provides a lot of security to the user.

## **Disadvantages of Metasearch Engines**

1. The meta search engine is not capable of fully translating the query syntax.
2. It is not as popular as other searches like Google and Yahoo search engines. The results usually incorporate up to 10 links from a particular site leaving out important links from major sites.

# Web Browser

**Web Browser is an application software that allows us to view and explore information on the web. User can request for any web page by just entering a URL into address bar.**

Web browsers are used on a range of devices, including desktops, laptops, tablets, and smartphones

Web browser can show text, audio, video, animation and more. It is the responsibility of a web browser to interpret text and commands contained in the web page.

Earlier the web browsers were text-based while now a days graphical-based or voice-based web browsers are also available.

Following are the most common web browser available today:

| <b><u>Browser Name</u></b> | <b><u>Description</u></b>                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Internet Explorer</b>   | Internet Explorer (IE) is a product from software giant Microsoft. This is the most commonly used browser in the universe. This was introduced in 1995.                                 |
| <b>Microsoft Edge</b>      | Microsoft Edge is a web browser developed by Microsoft, released in July 2015. Microsoft Edge is the default browser in Windows 10. Edge cannot be used on previous version of Windows. |
| <b>Google Chrome</b>       | Google Chrome is a web browser developed by Google in 2008.                                                                                                                             |
| <b>Mozilla Firefox</b>     | Mozilla Firefox is an open-source web browser developed by Mozilla.                                                                                                                     |
| <b>Netscape Navigator</b>  | Netscape Communications Corp.                                                                                                                                                           |
| <b>Opera</b>               | Opera is a web browser developed by Opera Software.                                                                                                                                     |
| <b>Safari</b>              | Safari is a web browser developed by Apple Inc. and included in Mac OS X. It was first released as a public beta in January 2003.                                                       |

# **Publishing Tools :**

## **Hardware**

- The most important item is the computer, which must have a modem or network connection to allow access to the Internet.
- The type of computer will obviously be dependent on user's budget.
- The power of the computer (how fast it will run) depends on processor speed and RAM (Random Access Memory). Most computers in schools use either the PC (Windows) or the Macintosh operating system.

## **Software**

- User will need the following software before they begin creating the web pages.
- A text editing package. For example, Notepad (Windows) or SimpleText (Mac).
- Graphics processing software. For example, PaintShop Pro.

## **A web browser**

The two most commonly used today are Microsoft Internet Explorer and Netscape Navigator. It may be a good idea to have both these browsers on the computer and to test the web pages on each.

# Other Useful Tools :

## Hardware

- A scanner, handy for including project work and/or photographs on your web site. There are two main types: hand-held or flat-bed.
- Digital stills camera, simplifies the process of including high-quality photos on the web site.
- Video camera, can be analogue or digital.

## Software

- HTML editor
- Image mapping software
- Animation software

# Chapter - 5

## HTML Programming Basics

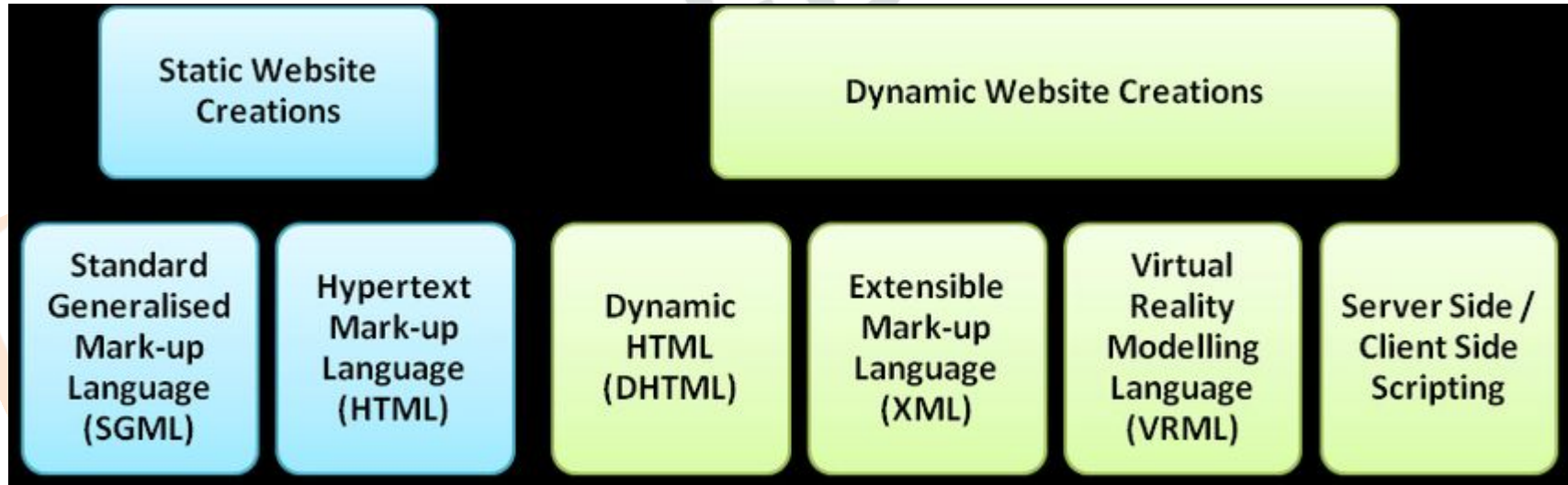


# SGML (Standard Generalized Markup Language)

SGML (Standard Generalized Markup Language) is a standard for how to specify a document markup language or tag set. Such a specification is itself a document type definition (DTD).

SGML is not in itself a document language, but a description of how to specify one. It is metadata.

SGML is based on the idea that documents have structural and other semantic elements that can be described without reference to how such elements should be displayed. The actual display of such a document may vary, depending on the output medium and style preferences.

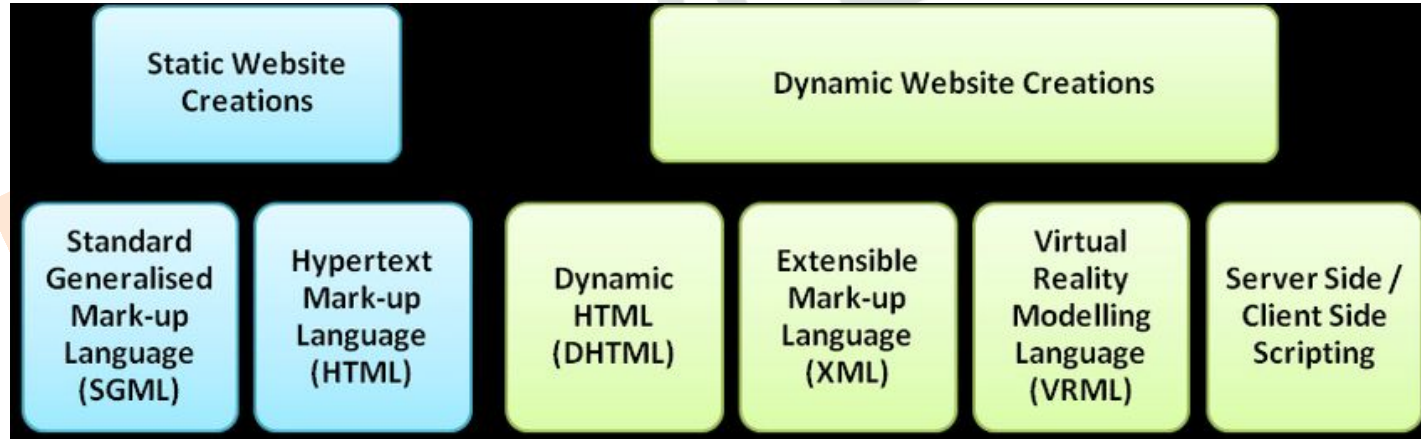




# SGML (Standard Generalized Markup Language)

**Some advantages of documents based on SGML are:**

1. They can be created by thinking in terms of document structure rather than appearance characteristics (which may change over time).
2. They will be more portable because an SGML compiler can interpret any document by reference to its document type definition (DTD).
3. Documents originally intended for the print medium can easily be re-adapted for other media, such as the computer display screen.



The language that Web browser uses, Hypertext Markup Language (HTML), is an example of an SGML-based language. There is a document type definition for

HTML (and reading the HTML specification is effectively reading an expanded version of the document type definition). In today's distributed networking environment, many documents are being described with the Extensible Markup Language (XML) which is a data description language (and a document can be viewed as a collection of data) that uses SGML principles.

SGML is based somewhat on earlier generalized markup languages developed at IBM, including General Markup Language (GML) and ISIL.

# Introduction to HTML

HTML stands for Hypertext Markup Language, and it is the most widely used language to write Web Pages.

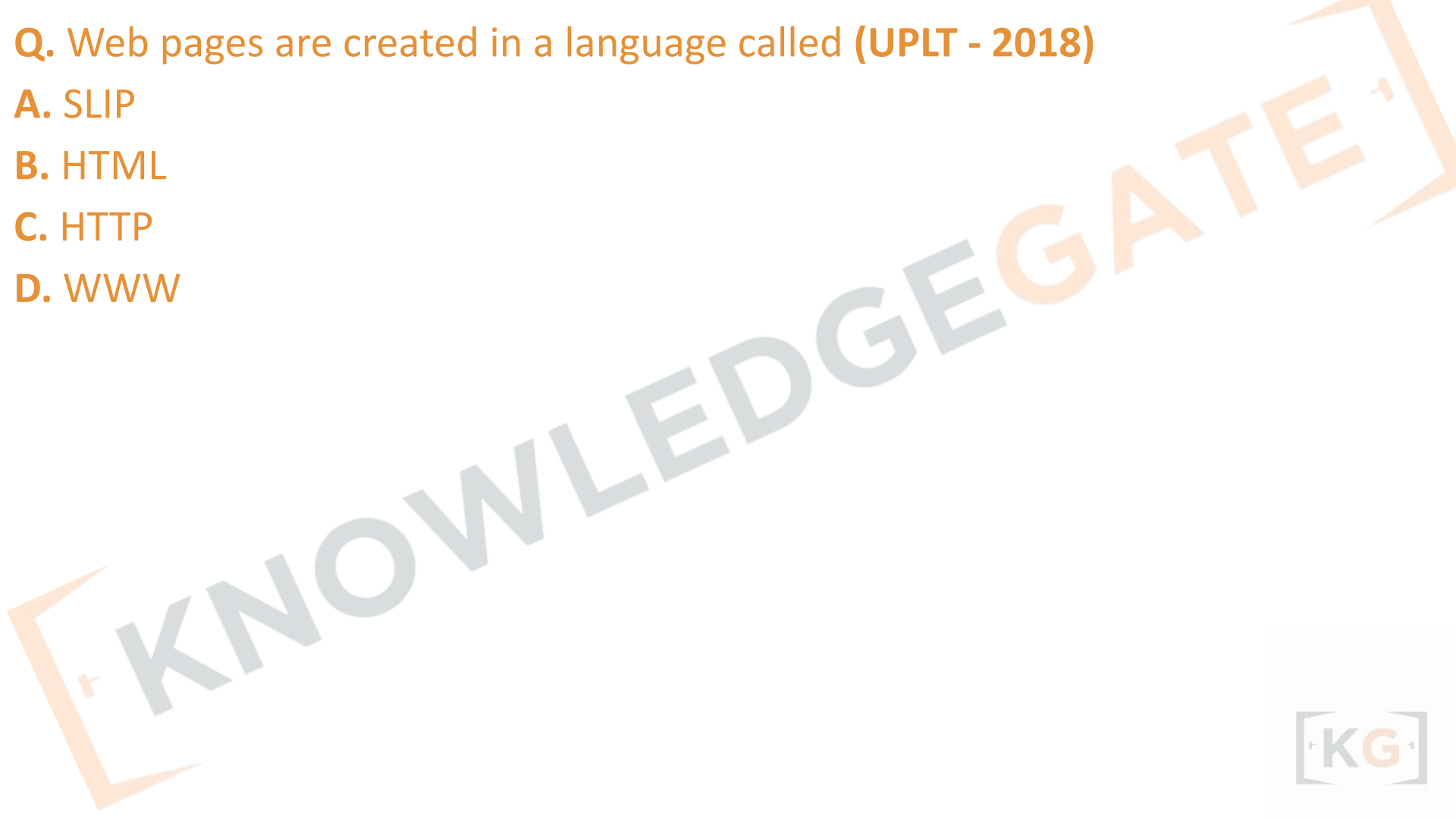
- Hypertext refers to the way in which Web pages (HTML documents) are linked together. Thus, the link available on a webpage is called Hypertext.
- As its name suggests, HTML is a Markup Language which means We use HTML to simply "markup" a text document with tags that tell a Web browser how to structure it to display.

## HTML Tags

As told earlier, HTML is a markup language and makes use of various tags to format the content. These tags are enclosed within angle braces <Tag Name>. Except few tags, most of the tags have their corresponding closing tags. For example, <html> has its closing tag </html> and <body> tag has its closing tag </body> tag etc.

Q. Web pages are created in a language called (UPLT - 2018)

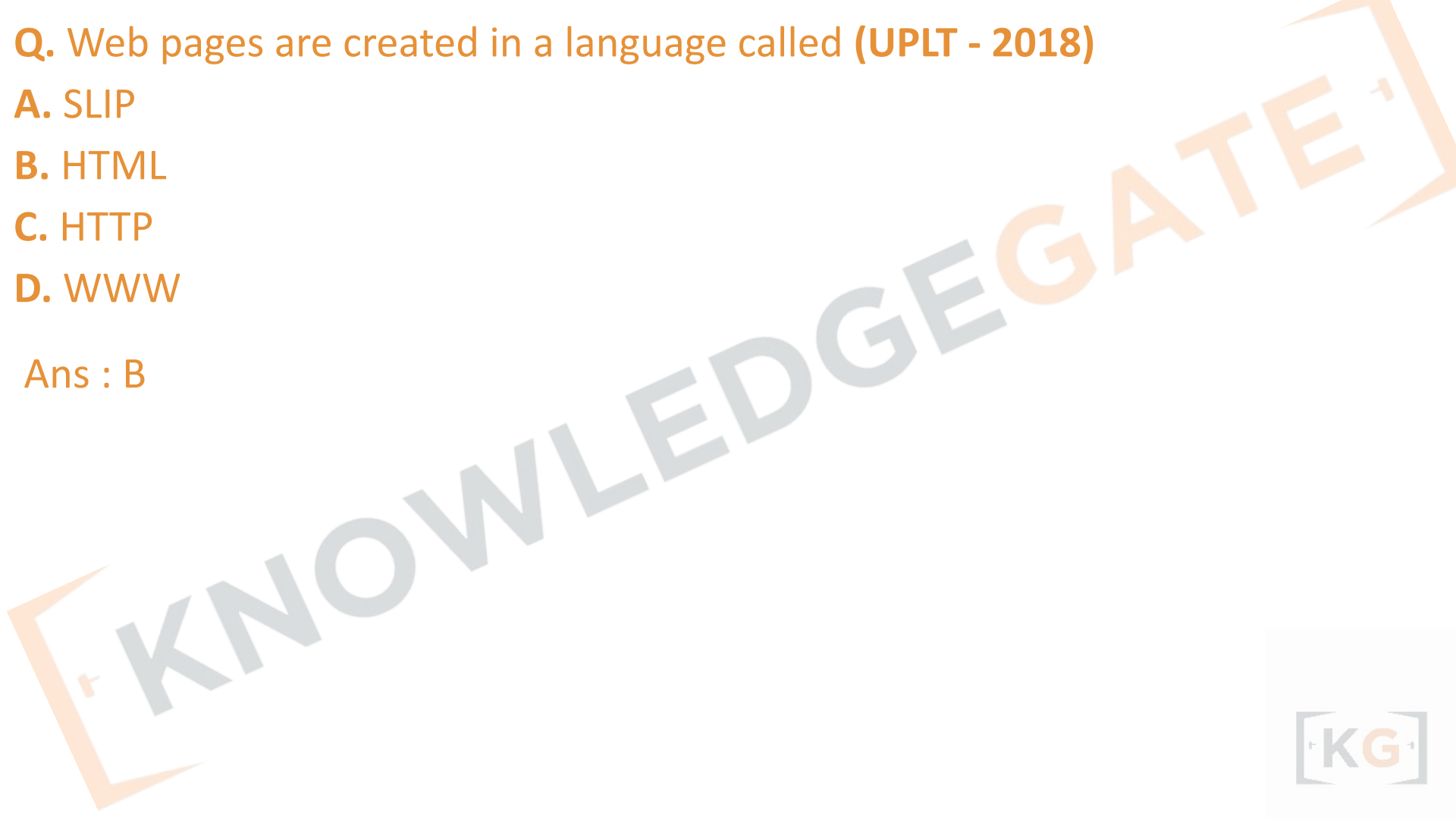
- A. SLIP
- B. HTML
- C. HTTP
- D. WWW



Q. Web pages are created in a language called (UPLT - 2018)

- A. SLIP
- B. HTML
- C. HTTP
- D. WWW

Ans : B



# HTML Basic Tags

## Heading Tags

Any document starts with a heading. We can use different sizes for our headings. HTML also has six levels of headings, which use the elements `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`. While displaying any heading, browser adds one line before and one line after that Heading.

## Paragraph Tag

The `<p>` tag offers a way to structure our text into different paragraphs. Each paragraph of text should go in between an opening `<p>` and a closing `</p>` tag.

## Line Break Tag

Whenever We use the `<br />` element, anything following it starts from the next line. This tag is an example of an empty element, where We do not need opening and closing tags, as there is nothing to go in between them.

The `<br />` tag has a space between the characters `br` and the forward slash. If We omit this space, older browsers will have trouble rendering the line break, while if We miss the forward slash character and just use `<br>` it is not valid in XHTML.

## Centering Content

We can use `<center>` tag to put any content in the center of the page or any table cell.

## Horizontal Lines

Horizontal lines are used to visually breakup sections of a document. The `<hr>` tag creates a line from the current position in the document to the right margin and breaks the line accordingly.

## Preserve Formatting

Sometimes, We want our text to follow the exact format of how it is written in the HTML document. In these cases, We can use the preformatted tag `<pre>`.

Any text between the opening `<pre>` tag and the closing `</pre>` tag will preserve the formatting of the source document.

# HTML Attributes

**An attribute is used to define the characteristics of an HTML element and is placed inside the element's opening tag. All attributes are made up of two parts – a name and a value**

- The name is the property We want to set. For example, the paragraph <p> element in the example carries an attribute whose name is align, which We can use to indicate the alignment of paragraph on the page.
- The value is what We want the value of the property to be set and always put within quotations. The below example shows three possible values of align attribute: left, center and right.

## Core Attributes

The four core attributes that can be used on the majority of HTML elements although not all) are

- Id
- Title
- Class
- Style



## The Id Attribute

The id attribute of an HTML tag can be used to uniquely identify any element within an HTML page. There are two primary reasons that We might want to use an id attribute on an element –

- If an element carries an id attribute as a unique identifier, it is possible to identify just that element and its content.
- If We have two elements of the same name within a Web page (or style sheet), We can use the id attribute to distinguish between elements that have the same name.

We will discuss style sheet in separate tutorial. For now, let's use the id attribute to distinguish between two paragraph elements as shown below.

### Example

```
<p id = "html">This para explains what is HTML</p>
```

```
<p id = "css">This para explains what is Cascading Style Sheet</p>
```

## **The title Attribute**

The title attribute gives a suggested title for the element. The syntax for the title attribute is similar as explained for id attribute –

The behavior of this attribute will depend upon the element that carries it, although it is often displayed as a tooltip when cursor comes over the element or while the element is loading.

## **The class Attribute**

The class attribute is used to associate an element with a style sheet, and specifies the class of element. We will learn more about the use of the class attribute when we will learn Cascading Style Sheet (CSS). So for now we can avoid it.

The value of the attribute may also be a space-separated list of class names. For example – `class = "className1 className2 className3"`

# The style Attribute

The style attribute allows We to specify Cascading Style Sheet (CSS) rules within the element.

```
<!DOCTYPE html>
<html>
<head>
<title>The style Attribute</title>
</head>
<body>
<p style = "fontfamily:
arial; color:#FF0000;">Some text...</p>
</body>
</html>
```

# HTML Formatting

If We use a word processor, We must be familiar with the ability to make text bold, italicized, or underlined; these are just three of the ten options available to indicate how text can appear in HTML and XHTML.

## Bold Text

Anything that appears within `<b>...</b>` element, is displayed in bold as shown below –

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Bold Text Example</title>
```

```
</head>
```

```
<body>
```

```
<p>The following word uses a <b>bold</b> typeface.</p>
```

```
</body>
```

```
</html>
```

# Italic Text

Anything that appears within `<i>...</i>` element is displayed in italicized as shown below –

## Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Italic Text Example</title>
```

```
</head>
```

```
<body>
```

```
<p>The following word uses an italicized typeface.</p>
```

```
</body>
```

```
</html>
```

## Underlined Text

Anything that appears within `<u>...</u>` element, is displayed with underline as shown below –

### Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Underlined Text Example</title>
```

```
</head>
```

```
<body>
```

```
<p>The following word uses an <u>underlined</u> typeface.</p>
```

```
</body>
```

```
</html>
```

## Strike Text

Anything that appears within `<strike>...</strike>` element is displayed with strikethrough, which is a thin line through the text as shown below –

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Strike Text Example</title>
```

```
</head>
```

```
<body>
```

```
<p>The following word uses a <strike>strikethrough</strike> typeface.</p> </
```

```
body>
```

```
</html>
```

# Monospaced Font

The content of a `<tt>...</tt>` element is written in monospaced font. Most of the fonts are known as variable-width fonts because different letters are of different widths (for example, the letter 'm' is wider than the letter 'i'). In a monospaced font, however, each letter has the same width.

## Example

```
<!DOCTYPE html>
<html>
<head>
<title>Monospaced Font Example</title>
</head>
<body>
<p>The following word uses a <tt>monospaced</tt> typeface.</p>
</body>
</html>
```



## Superscript Text

The content of a `<sup>...</sup>` element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

### Example

```
<!DOCTYPE html>
<html>
<head>
<title>Superscript Text Example</title>
</head>
<body>
<p>The following word uses a <sup>superscript</sup> typeface.</p>
</body>
</html>
```

## Subscript Text

The content of a `<sub>...</sub>` element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

### Example

```
<!DOCTYPE html>
<html>
<head>
<title>Subscript Text Example</title>
</head>
<body>
<p>The following word uses a <sub>subscript</sub> typeface.</p>
</body>
</html>
```

# Inserted Text

Anything that appears within `<ins>...</ins>` element is displayed as inserted text.

## Example

```
<!DOCTYPE html>
<html>
<head>
<title>Inserted Text Example</title>
</head>
<body>
<p>I want to drink <del>cola</del> <ins>wine</ins></p>
</body>
</html>
```

# Deleted Text

Anything that appears within `<del>...</del>` element, is displayed as deleted text.

## Example

```
<!DOCTYPE html>
<html>
<head>
<title>Deleted Text Example</title>
</head>
<body>
<p>I want to drink <del>cola</del> <ins>wine</ins></p>
</body>
</html>
```

## Larger Text

The content of the `<big>...</big>` element is displayed one font size larger than the rest of the text surrounding it as shown below –

### Example

```
<!DOCTYPE html>
<html>
<head>
<title>Larger Text Example</title>
</head>
<body>
<p>The following word uses a <big>big</big> typeface.</p>
</body>
</html>
```

## Smaller Text

The content of the `<small>...</small>` element is displayed one font size smaller than the rest of the text surrounding it as shown below –

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Smaller Text Example</title>
```

```
</head>
```

```
<body>
```

```
<p>The following word uses a <small>small</small> typeface.</p>
```

```
</body>
```

```
</html>
```

## Grouping Content

The `<div>` and `<span>` elements allow us to group together several elements to create sections or subsections of a page. For example, we might want to put all of the footnotes on a page within a `<div>` element to indicate that all of the elements within that `<div>` element relate to the footnotes. We might then attach a style to this `<div>` element so that they appear using a special set of style rules.

## HTML Text

### Links

A webpage can contain various links that take us directly to other pages and even specific parts of a given page. These links are known as hyperlinks. Hyperlinks allow visitors to navigate between web sites by clicking on words, phrases, and images. Thus we can create hyperlinks using text or images available on a webpage.

## Linking Documents

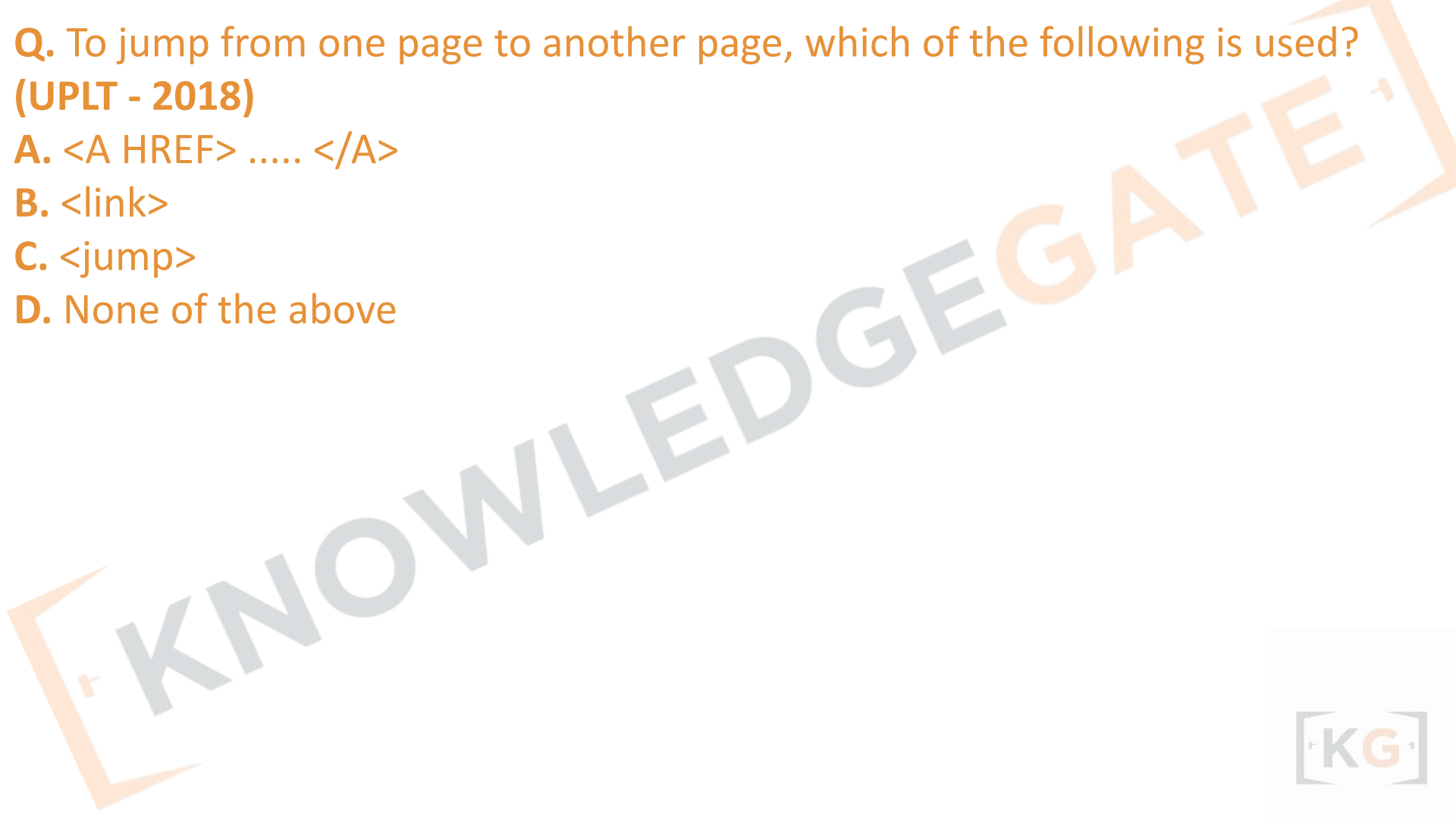
A link is specified using HTML tag `<a>`. This tag is called anchor tag and anything between the opening `<a>` tag and the closing `</a>` tag becomes part of the link and a user can click that part to reach to the linked document. Following is the simple syntax to use `<a>` tag.

`<a href = "Document URL" ... attributeslist> Link Text</a>`



**Q. To jump from one page to another page, which of the following is used?  
(UPLT - 2018)**

- A. <A HREF> ..... </A>**
- B. <link>**
- C. <jump>**
- D. None of the above**



**Q. To jump from one page to another page, which of the following is used? (UPLT - 2018)**

- A. <A HREF> ..... </A>**
- B. <link>**
- C. <jump>**
- D. None of the above**

**Answer: A**

## HTML - Image Links

We have seen how to create hypertext link using text and we also learnt How to use images in our webpages. Now, we will learn how to use images to create hyperlinks.

```
<img src = "/images/logo.png" alt = "" border = "0"/>
```

## HTML Images

Images are very important to beautify as well as to depict many complex concepts in simple way on our web page. This tutorial will take We through simple steps to use images in our web page . Insert Image We can insert any image in our web page by using <img> tag. Following is the simple syntax to use this tag.

```
<img src = "Image URL" ... attributeslist/>
```

The <img> tag is an empty tag, which means that, it can contain only list of attributes and it has no closing tag.

## HTML Tables

The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells. The HTML tables are created using the `<table>` tag in which the `<tr>` tag is used to create table rows and `<td>` tag is used to create data cells. The elements under `<td>` are regular and left aligned by default

## Example

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Tables</title>
</head>
<body>
<table border = "1">
<tr>
<td>Row 1, Column 1</td> <td>Row 1, Column 2</td>
</tr>
<tr>
<td>Row 2, Column 1</td> <td>Row 2, Column 2</td>
</tr>
</table>
</body>
</html>
```

## **Cellpadding and Cellspacing Attributes**

There are two attributes called cellpadding and cellspacing which We will use to adjust the white space in our table cells. The cellspacing attribute defines space between table cells, while cellpadding represents the distance between cell borders and the content within a Cell.

## **Colspan and Rowspan Attributes**

We will use colspan attribute if We want to merge two or more columns into a single column. Similar way We will use rowspan if We want to merge two or more rows

## HTML Lists

HTML offers web authors three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain –

- `<ul>` – An unordered list. This will list items using plain bullets.
- `<ol>` – An ordered list. This will use different schemes of numbers to list our items.
- `<dl>` – A definition list. This arranges our items in the same way as they are arranged in a dictionary.

## HTML Unordered Lists

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML <ul> tag. Each item in the list is marked with a bullet.

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>HTML Unordered List</title>
```

```
</head>
```

```
<body>
```

```
<ul>
```

```
<li>Beetroot</li>
```

```
<li>Ginger</li>
```

```
<li>Potato</li>
```

```
<li>Radish</li>
```

```
</ul>
```

```
</body>
```

```
</html>
```



**HTML Ordered Lists :** If We are required to put our items in a numbered list instead of bulleted, then HTML ordered list will be used. This list is created by using <ol> tag. The numbering starts at one and is incremented by one for each successive ordered list element tagged with <li>.

**Example**

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Ordered List</title>
</head>
<body>
<ol>
<li>Beetroot</li>
<li>Ginger</li>
<li>Potato</li>
<li>Radish</li>
</ol>
</body>
</html>
```

# Chapter - 6

**Interactivity Tools, Internet Security Management Concepts, Information Privacy and Copyright Issues**



# Interactivity Tools in Internet

## 1. Introduction

The Internet has evolved from Web 1.0 (static, read-only) to Web 2.0 (interactive, read-write) and now towards Web 3.0 (intelligent and decentralized). **Interactivity tools are at the core of this transformation, allowing two-way communication between users and websites.** They make the web dynamic, participatory, and engaging. Without these tools, the Internet would remain a one-way medium like television. Today, they are essential for education, business, governance, healthcare, and entertainment.

## 2. Types of Interactivity Tools

### (a) Communication Tools

These tools allow real-time communication between users. Examples include chat rooms, instant messaging apps (WhatsApp, Messenger, Telegram), and VoIP applications (Skype, Zoom, Google Meet). Modern email services like Gmail and Outlook also support interactivity through calendars, video calls, and file sharing.

### **(b) Social Media Tools**

Platforms like Facebook, Instagram, Twitter (X), and LinkedIn enable users to create, share, and interact with content. They allow commenting, liking, sharing, and live streaming. These tools are vital for digital marketing, community building, and even political communication.

### **(c) Collaboration Tools**

Collaboration tools support teamwork and productivity. Examples include Google Docs, Sheets, and Slides for real-time editing, Microsoft Teams, Slack for messaging and video calls, Trello and Asana for project management, and GitHub for collaborative coding.

### **(d) Web-Based Interactive Tools**

These tools are embedded within web pages to engage users. Examples include polls, surveys, and quizzes (Google Forms, Kahoot, Mentimeter), discussion forums (Quora, Reddit, Stack Overflow), and interactive animations created with HTML5 and JavaScript.

### **(e) E-Learning Tools**

E-learning platforms like Moodle, Blackboard, Google Classroom, and Coursera use interactivity to improve learning. They offer interactive videos, virtual labs, gamified quizzes, and instant feedback. These tools encourage active participation and personalized education.

## **(f) Web Design / Scripting Tools**

Technical tools like JavaScript frameworks (React, Angular, Vue), AJAX, and HTML5 APIs enable interactive websites. They allow live updates without refreshing (Twitter feeds), video playback, geolocation, and online gaming applications.

## **3. Importance of Interactivity Tools**

Interactivity tools are crucial because they:

- Engage users and keep them active.
- Enable collaboration across locations.
- Personalize user experiences (e.g., Amazon recommendations).
- Enhance education through interactive simulations and quizzes.
- Improve business services via chatbots and live support.
- Boost productivity by replacing physical meetings with online collaboration.

## **4. Current Trends in Interactivity Tools**

Some emerging trends include:

- AI Chatbots: ChatGPT, Alexa, Siri for automated conversations.
- Live Streaming + Chat: YouTube Live, Twitch for interactive broadcasts.
- Gamification: Adding game elements to education and marketing.
- Augmented Reality (AR) & Virtual Reality (VR): Immersive interactive experiences.
- Collaborative Whiteboards: Tools like Miro, Jamboard for brainstorming.
- Instant Polls & Surveys: Used in webinars, meetings, and classrooms.

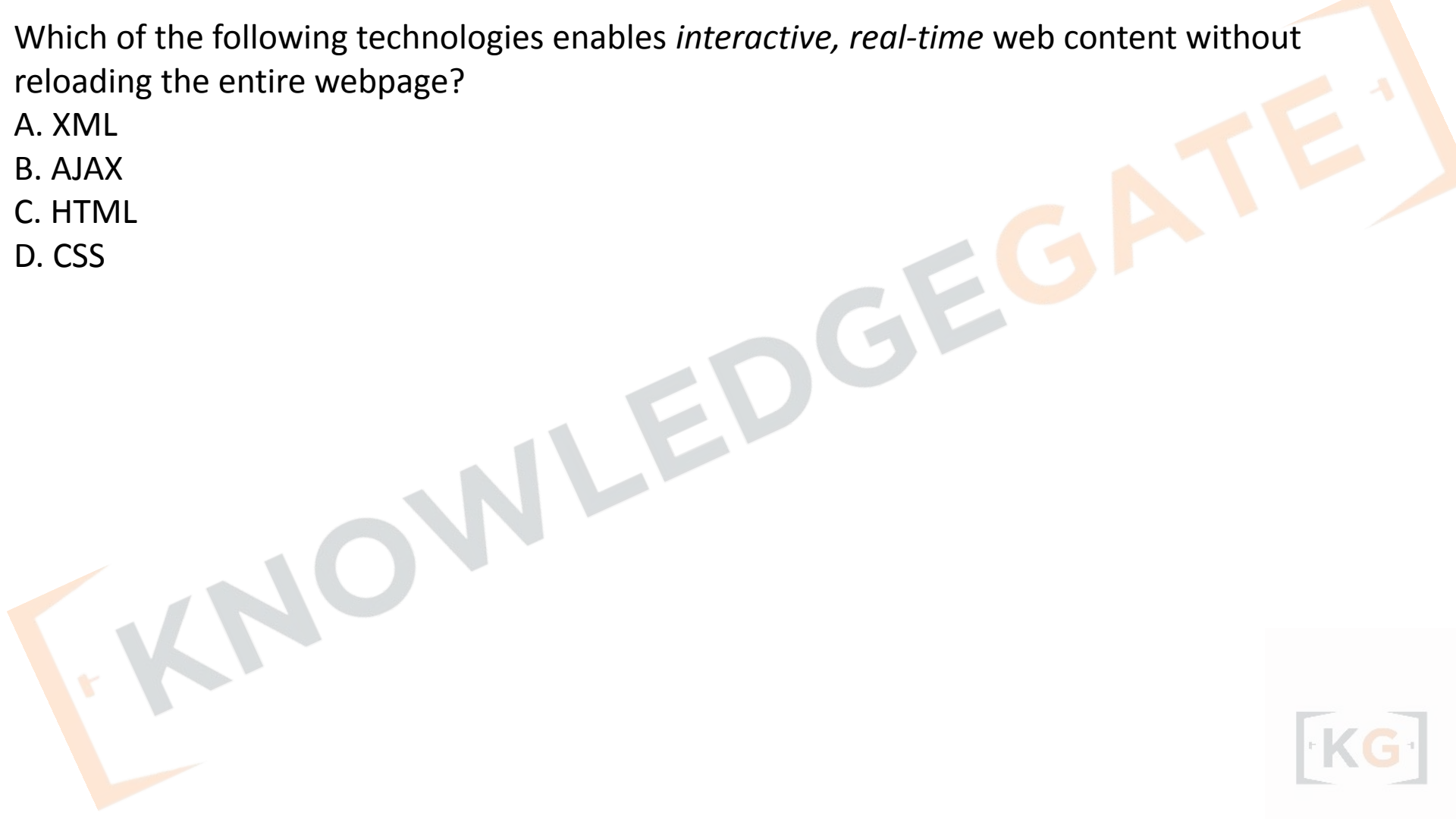
## 5. Challenges of Interactivity Tools

Despite their benefits, interactivity tools face challenges:

- Security Risks: Vulnerability to hacking and data theft.
- Privacy Concerns: Collection of personal data for personalization and ads.
- Bandwidth Needs: High-speed internet required for video conferencing and AR/VR.
- Digital Divide: Unequal access in rural or underdeveloped regions.
- Distractions: Overuse of interactive apps may reduce productivity and attention spans.

Which of the following technologies enables *interactive, real-time* web content without reloading the entire webpage?

- A. XML
- B. AJAX
- C. HTML
- D. CSS



Which of the following technologies enables *interactive, real-time* web content without reloading the entire webpage?

- A. XML
- B. AJAX
- C. HTML
- D. CSS

**Answer:** B. AJAX

**Explanation:** AJAX (Asynchronous JavaScript and XML) allows asynchronous data exchange between browser and server, updating parts of a page dynamically without full reload.



# Internet security and Management concepts

- In internet, as you know that we are basically connecting our computer system to another computer system.
- There is every likelihood that while doing this, some unauthorized person can also get into it and access data.
- Manipulating security becomes more important when we use computer as commercial transaction.
- User need to make sure that message are private and financial transactions are secure.



## Authentication

- **It is the process of determining identify of a user who is attempting to access a system.**
- Also for security purpose, the verification and identify of a person or process is necessary.
- Thus, it is the verification of the user's claim of identity by passwords, personal challenge, response calculation or random password generator.



## Authentication methods

- There are two primary models for token based user authentication systems.

**One time password systems:** The user token displays a “current password” which the user enters in any context that would require a conventional password.

The server system can calculate the password on demand and verify the authentication request.

**Challenge/ Response systems:** The user first receives a challenge generated by the authentication server. The challenge is then copied by the user into his token.

The token then calculate and displays a response which depends on the challenge. The user then copies this response back into the application prompting for authentication information.

- The server system calculates the expected response and compares with the response submitted by the user.
- Since challenges are generated random by the server, attacks associated with replaying authentication information are an issue for well-designed system.
- As the result, the server does not need to retain the information regarding which “passwords” have been used.

Some of the authentication methods are given in the following subsections.

## Passwords

- Passwords are the most common form of computer security.
- Some networks require multiple level of passwords to gain access to various servers or databases.

## Callback

- Callback is a security feature that works like when a user dials into a communication server and enters the user name and password.
- The communication server then hangs up the modem connection, searches its database to authentication the user and then calls the user back at a pre-defined number.



## Packet Filters

- Packet filters allow network administrators to limit a user's access to specific services on the network.
- For example, a user may allowed to send electronic mail but not copy data files from the network.

## Authentication servers

- They can be set up in variety of ways, depending on the security scheme of the network they are serving.
- The basic process of authenticating a user includes the following steps:

A user dials into a network through a **communication server or Network Access Server (NAS)**. The NAS forwards the user identification and password to the Authentication server. Then the authentication server validate the user and provides access privileges to the network.

## RADIUS (Remote Authentication Dial-up User Service)

- RADIUS is a system of distributed security that solves the problems associated with meeting the security requirements of remote computing.
- The solution eliminates the need of hardware and provides access to a variety of state of the art security solutions.
- Distributed security separates user authenticated data and authorization from the communications process and creates a single central location for user authentication data.
- RADIUS is freely available distributed security system developed by Lucent Technologies.
- RADIUS provides an open and scalable client-server security system.



## Authorization

- **Authorization is the process of determining how an authenticated user is permitted to use specific resources.**
- An authorization mechanism automatically enforces a management policy regarding resources object use.
- For example, in a computer system, resources typically includes data files, operator commands, transaction I/O devices and program process.
- The specific rules for authorizing access to data objects usually enforce confidentiality and integrity by either granting or denying access to read, modify or create data records and by controlling the creation or deletion of data objects.
- While authentication control who can access network resources, authorization says what they can do once they have access the resources.
- Authorization grants privileges to processes and users.
- Authorization lets a security administrators control part of a network, for example, directories and files on servers.

## **Auditing and Accountability**

- Auditing is the process of data collection and analysis that allows administrators and others, such as IT auditors, to verify that the users and authorization rules are producing the intended results are defined in a company's business and security policy.
- Individual accountability for attempts to violate the intended policy depends on monitoring relevant security events, which initiates the auditing feedback reporting loop.
- The monitoring process can be implemented as a continuous automatic function, as an occasional verification that proper procedures are being followed.
- The auditing information may be used by security administrators, internal audit personnel, external auditors, government regulatory officials and in legal proceedings.
- To effectively analyze the security of a network and to respond to security incidents, procedures should be established for collecting network activity data.



# What is information security?

Information security is defined as the preservation of confidentiality, integrity and availability of information.

It typically involves preventing or at least reducing the probability of unauthorized/inappropriate access, use, disclosure, disruption, deletion/destruction, corruption, modification, inspection, recording or devaluation, although it may also involve reducing the adverse impacts of incidents. Information may take any form, e.g., electronic or physical, tangible (e.g. paperwork) or intangible (e.g. knowledge).



# What is information security?

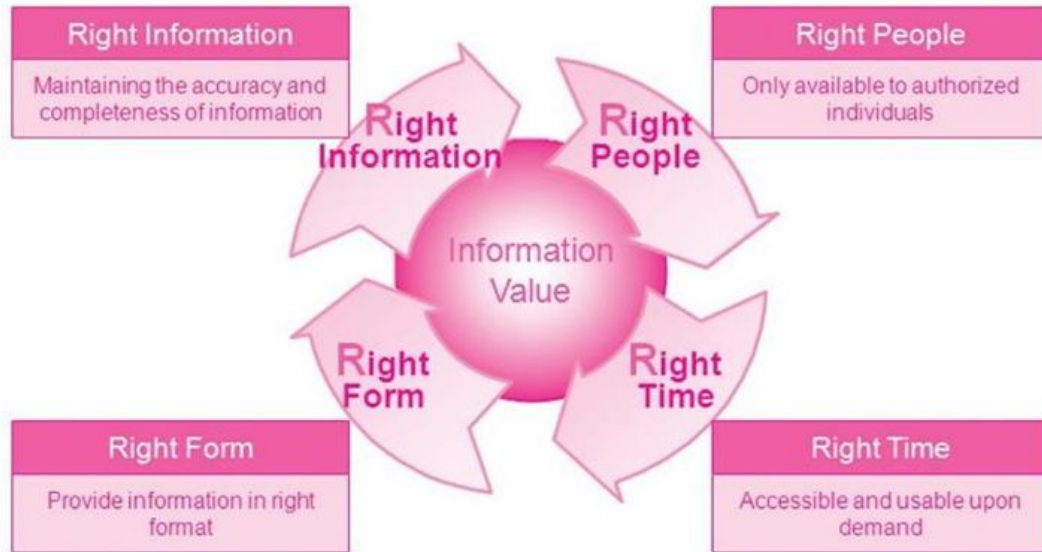
Information security's primary focus is the balanced protection of the confidentiality, integrity and availability of data (also known as the CIA triad) while maintaining a focus on efficient policy implementation, all without hampering organization productivity.



## 4Rs of information security

The 4Rs of information security are **Right Information, Right People, Right Time and Right Form**. Control over the 4Rs is the most efficient way to maintain and control the value of information.

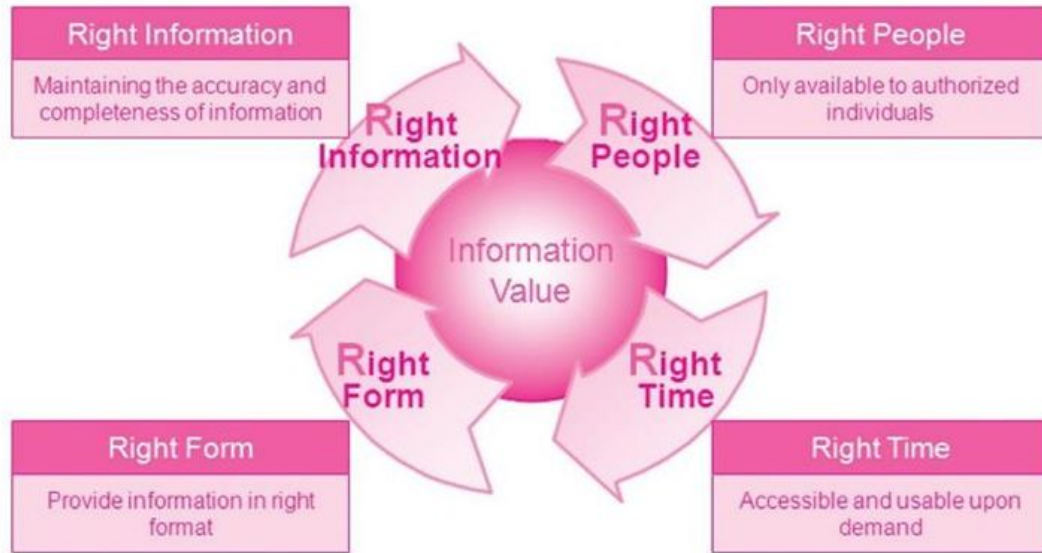
4Rs of Information Security “Right Information” refers to the accuracy and completeness of information, which guarantees the integrity of information.



“Right People” means that information is available only to authorized individuals, which guarantees confidentiality.

“Right Time” refers to the accessibility of information and its usability upon demand by an authorized entity. This guarantees availability.

“Right Form” refers to providing information in the right format.



# Copyright Issues

Copyright issues involve the unauthorized use of protected material, with common problems including plagiarism, content theft, and infringement of exclusive rights like reproduction, distribution, and public performance. These issues often arise from using copyrighted works without permission, and disputes can be resolved through negotiation, legal action, or the "notice and take down" process.

## Common copyright issues

- **Copyright Infringement:** Using a work that is protected by copyright without the holder's permission. This can include copying, sharing, or publicly performing the work.
- **Plagiarism :** Presenting someone else's work or ideas as your own. While not always a copyright violation if credit is given, it can be a legal issue.
- **Content Stealing:** Taking content directly from another website or source and using it on your own.
- **Unauthorized Use:** Even using a work found online for free, or attributing it to the original creator, can be considered infringement if permission was not granted.

# Chapter - 7

**WebTechnology: protocols, development strategies, applications, Web project and team**

# Protocols Governing Web

**A protocol is a set of rules that is used to communicate applications to each other.**

OR

**A protocol is the interface required for communicating the different applications. Protocols which are a set of rules that help in governing the way a particular technology will function for communication.** In other words, it can be said that the protocols are digital languages implemented in the form of networking algorithms. There are different networks and network protocols, user's use while surfing.

## **List of Internet Protocols:**

1. HTTP(Hyper Text Transfer Protocol)
2. HTTPS(Hyper Text Transfer Protocol Secure)
3. TCP/IP(Transmission Control Protocol/Internet Protocol)
4. FTP(File Transfer Protocol)
5. PPP(Point to Point Protocol)
6. SMTP(Simple Mail Transfer Protocol)
7. TELNET(TErминаL NETwork or Teletype Network)
8. POP3(Post Office Protocol 3)



## 1. HTTP:

It is the primary protocol used to distribute information on the web. This protocol is used to access, send and receive Hypertext markup language (HTML) files on the Internet. Initial HTTP 0.9 does not allow for content typing and does not have provisions for supplying meta-information.

**Content Typing:** To identify the type of data being transferred.

**Meta Information:** It is supplemental data, such as environment variables that identify the client's computer. Current version is HTTP 1.0

**2. HTTPS:** It is an extension of the Hypertext Transfer Protocol (HTTP). It is used for secure communication over a computer network with the SSL/TLS protocol for encryption and authentication. So, generally, a website has an HTTP protocol but if the website is such that it receives some sensitive information such as credit card details, debit card details, OTP, etc then it requires an SSL certificate installed to make the website more secure. So, before entering any sensitive information on a website, we should check if the link is HTTPS or not. If it is not HTTPS then it may not be secure enough to enter sensitive information.

**3. TCP/IP:** It is a set of rules that an application can use to package its information for sending across the networks of networks.

**TCP:** This protocol ensures the delivery of information packets across network.

**IP:** This protocol is responsible logical addressing called IP address to route information between networks.

**4. FTP:** This protocol is used for transferring files from one system to the other. This works on a client-server model. When a machine requests for file transfer from another machine, the FTO sets up a connection between the two and authenticates each other using their ID and Password. And, the desired file transfer takes place between the machines.

**5. PPP:** It is a communication protocol that is used to create a direct connection between two communicating devices. This protocol defines the rules using which two devices will authenticate with each other and exchange information with each other. For example, A user connects his PC to the server of an Internet Service Provider also uses PPP. Similarly, for connecting two routers for direct communication it uses PPP.

**6. Simple Mail Transfer Protocol(SMTP):** These protocols are important for sending and distributing outgoing emails. This protocol uses the header of the mail to get the email id of the receiver and enters the mail into the queue of outgoing mails. And as soon as, it delivers the mail to the receiving email id, it removes the email from the outgoing list.

**7. Telnet:** Telnet lets you remotely log into another system and browse files and directories on that remote system.

**8. POP3:** POP3 stands for Post Office Protocol version 3. It has two Message Access Agents (MAAs) where one is client MAA (Message Access Agent) and another is server MAA(Message Access Agent) for accessing the messages from the mailbox. This protocol helps us to retrieve and manage emails from the mailbox on the receiver mail server to the receiver's computer.

**9. Gopher:** Gopher is a collection of rules implemented for searching, retrieving as well as displaying documents from isolated sites. Gopher also works on the client/server principle.

# Introduction and Web Development Strategies



# Introduction and Web Development Strategies

1. **Identify target user:** Identify the user of the website by doing market research.
2. **Make our design portable:** To be successful, website design should be portable and accessible across different browsers, operating systems, and computer platforms. Designers should test the website in a different environment whether they look same to all their users.
3. **Design for low bandwidth:** Web pages in websites should be accessible at any connection speeds. If a page is downloaded slowly then users will leave the website before they see the content.
4. **Plan for clear presentation and easy access to information:** Presentation of the information on the website must be clear and easily accessible to the user.
5. **Create smooth transitions:** Plan to create a unified look among the sections and pages of site. Reinforce the identifying elements of the site and create smooth transitions from one page to another. Any software development project, a methodology should be followed to ensure project consistency and completeness.

**Web:**

The World Wide Web ("WWW" or "The Web") is the part of the Internet that contains websites and web pages.

**Web Page:**

“A hypertext document on the World Wide Web. OR “A document which can be displayed in a web browser such as Firefox, Google Chrome, Opera, Microsoft Internet Explorer or Edge, or Apple's Safari.”

**Home Page:**

The introductory page or first page of a website, typically serving as a table of contents for the site is known as Home Page. OR Home page is a web page set as the default or start-up page on a browser.

## **Website:**

A website is simply a collection of interlinked web pages.

Classification of Website:

- a) Corporate Website
- b) Individual website

### **Corporate Website:**

- i. In this, there is certain no. of persons, who develop their website for a particular organization.
- ii. The corporate website are formed when group of people have common interest and objective.
- iii. The purpose of this website is to convey the information of organization to all over the world.

### **Individual Website:**

It is just like profile management system. In this type of website an individual wants to develop website for h-projection, career growth etc.

## **Web Applications:**

Simple office software (word processors, online spread sheets, and presentation tools). More advanced applications such as project management, computer-aided design, video editing and point-of-sale.

## **Web Project:**

A web project is the process of developing and creating a website, activities in a network. The goal of web project is the transfer of static and dynamic content.



## Writing web Projects and Target Users:

1. **Write a project mission statement:** Write the Specific mission statement that you want to do.
2. **Identify Objectives:**
  - i. Specific
  - ii. Measurable
  - iii. Attainable
  - iv. Realistic
  - v. Time limited

**3. Identify your target users:** The matter of a website will be determined by the Users whom you want to visit the site. This is totally depending upon:

- i. Market research
- ii. Focus group
- iii. Understanding intranet audiences

**4. Determine the scope:** By supporting documents and client's approval.

**5. Budget:**

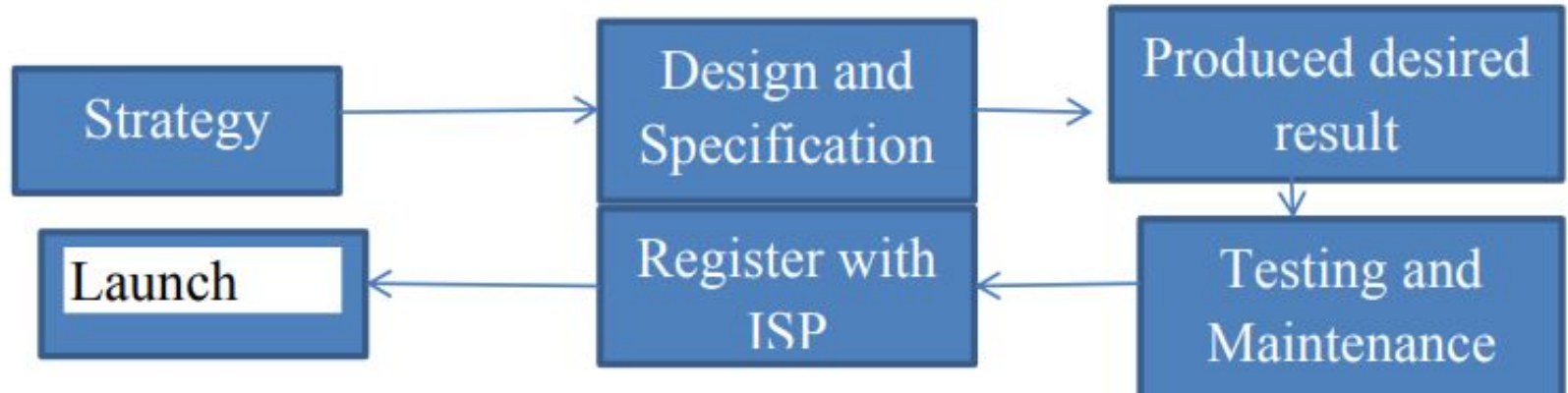
- i. Assumption for budgets.
- ii. Budget categories.
- iii. Determine hidden costs and tools.

**6. Planning issues:**

- i. Discuss client's existing information system.
- ii. Project team and developing infrastructure.
- iii. Where the website will place.

## Identification of objects:

- a) Object identification: All the components which are visible in website are objects or in other words, we can say that all visible components in the web browser are defined as objects. Ex. Text box, command button etc.
- b) Web development process:



## **I. Strategy:**

- Goals and objectives
- Team building
- Research and review
- Project proposal

## **II. Design and Specification:**

- Developing concept
- Content planning
- Rough design
- Final design

## **III. Produced desired Result:**

- Build prototype
- Prototype testing
- Original design
- Satisfy the clients need

#### **IV. Testing and maintenance:**

- Test the code
- Maintain the web server.

#### **V. Register with ISP:**

- Register domain name

#### **VI. Get web space Launch:**

- Connect domain name with web server
- Finally host the web accordingly

## **Web Team:**

Web team is a group of various technical experts in a developing site from coding the page to maintain the web server.

### **Types of Web team:**

- 1. Server Side:** hired by a company to develop a website.
- 2. Client Side:** part of the company putting together the website.

### **Assessment techniques used to comprise a web team:**

**a) Deciding roles and responsibilities:** The composition of team varies to depend on the audience, scope and complexity level of the web. There are key roles on each virtual project. One should always decide for core, extended and special team members in a web and shares responsibilities accordingly.

**b) Common Team Compositions:** It is possible to acknowledge specific type of teams and determine to be based on the kind of project, who is likely to be part of the team though these are all type of web project. As for example the team composition might be account manager, creative lead, project manager, designer etc.

c) **Putting together with right team:** It meets out the needs with low price, more effectively in reasonable time. So it is the team that fulfils the website requirements successfully.

d) **Identifying Necessary Skills:** The skill set in the base of web team. It must have in care to plan, design, build and deploy a website.

## Classification of Web Team:



# **1. Core Team member:**

## **i. Project Manager:**

- Specify the work.
- Developing the project plan.
- Scheduling.
- Allocation resources.
- Budgeting and managing the team.

## **ii. Technical lead:**

- Managing programmers.
- Chooses specialized team such as security expert, database Programmers.

## **iii. Web Production specialist:**

- Integrate the site using html or java script.



## 2. Extended Team Member:

1. **Account Manager:** It interacts with the client, project manager and creative lead.
2. **Programmer:** develops applications for the web projects.
3. **Network Engineer:** configuring a web server.
4. **Information architects:** It understands how to display information visually to users and how to interact with the website.
5. **Content Writer:** write contents for the website.
6. **Tester:** It tests the web project based on the team plan that QA lead writes.

## 3. Special Team Member:

1. Security Experts: security handling and encryption techniques.
2. Audio, Video Engineer
3. 3-D Modeller
4. Web Cast Specialist
5. Media Buyer
6. Strategic Planner

**Which of the following best defines a Web Development Strategy?**

- A) A marketing plan for promoting a website
- B) A structured approach for designing, developing, and maintaining a web application
- C) A user manual for website navigation
- D) A database backup policy

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- A) A marketing plan for promoting a website
- B) A structured approach for designing, developing, and maintaining a web application
- C) A user manual for website navigation
- D) A database backup policy

**Answer:** B

**Explanation:** A web development strategy is a systematic plan defining goals, architecture, technologies, design, testing, and deployment methods to ensure efficient web project delivery.

**The role of a Web Project Manager primarily involves:**

- A) Writing code for web pages
- B) Managing budget, schedule, and team coordination
- C) Creating multimedia content
- D) Maintaining server hardware

**The role of a Web Project Manager primarily involves:**

- A) Writing code for web pages
- B) Managing budget, schedule, and team coordination
- C) Creating multimedia content
- D) Maintaining server hardware

**Answer: B**

**Explanation:** The project manager oversees the entire web project, ensuring resources, timelines, and quality standards are maintained according to client requirements.

# Chapter - 8

**Web Page Designing, Scripting, Server Site Programming**

# Introduction to script

A scripting or script language is a programming language for a special run-time environment that automates the execution of tasks, the tasks could alternatively be executed one-by-one by a human operator. Scripting languages are often interpreted, rather than compiled.

## JavaScript

JavaScript (js) is a light-weight object-oriented programming language which is used by several websites for scripting the webpages. It is an interpreted, full-fledged programming language that enables dynamic interactivity on websites when applied to an HTML document.

# Advantages of JavaScript:

1. JavaScript is a client side language.
2. JavaScript is an easy language to learn.
3. JavaScript is comparatively fast for the end user.
4. Extended functionality to web pages
5. No compilation needed
6. Easy to debug and test
7. Platform independent
8. Event-Based Programming language
9. Procedural programming capabilities



# Client-side scripting

Client-side scripting (embedded scripts) is code that exists inside the client's HTML page. This code will be processed on the client machine and the HTML page will NOT perform a Post Back to the web-server. Traditionally, **client-side scripting is used for page navigation, data validation and formatting**. The language used in this scripting is JavaScript. JavaScript is compatible and is able to run on any internet browser.

## Advantages of Client-side scripting

- More interactive since it responds immediately to the user action.
- Quick Execution because they don't require a trip to the server.
- improve the user experience for the user whose browser support script.
- An alternative option is available for the user whose browser didn't support script.
- Reusable and obtainable from many resources.

# Disadvantages of client-side scripting.

- It is not supported by all browsers. if no alternative is given for the script than the user might get an error.
- Need more testing. because of the different browser and its version support script differently.
- If the Script is not available through other resources than more development time and effort required.
- Developers have more control over the look and behaviour of their Web widgets; however, usability issues can arise if a Web widget seems like a standard control but behaves different way or vice-versa.

# JavaScript Variables

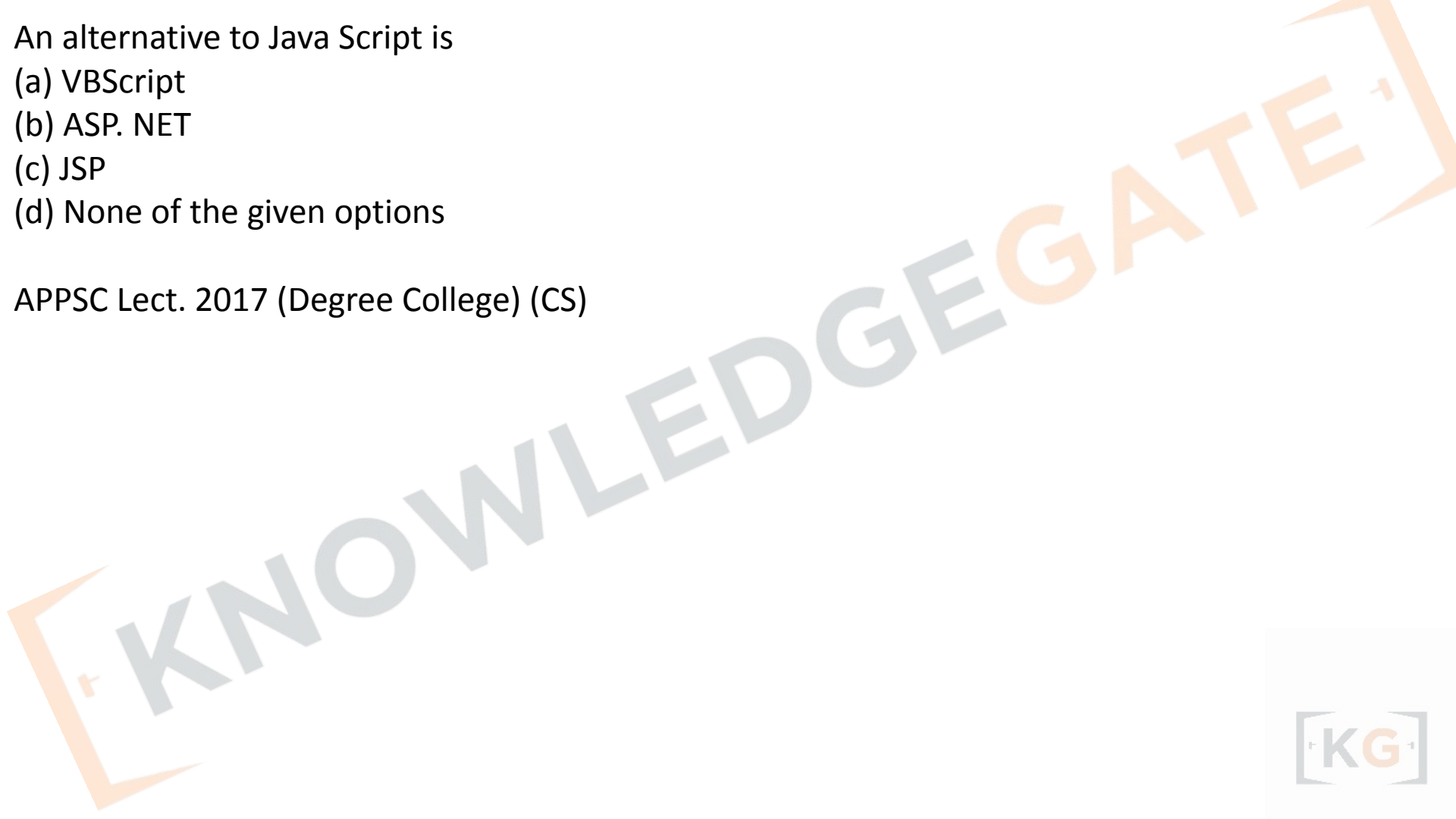
Like many other programming languages, JavaScript has variables. Variables can be thought of as named containers. You can place data into these containers and then refer to the data simply by naming the container. Before you use a variable in a JavaScript program, you must declare it. Variables are declared with the var keyword as follows.

```
<script type="text/javascript">  
<!--  
var money; var name;  
//-->  
</script>
```

An alternative to Java Script is

- (a) VBScript
- (b) ASP. NET
- (c) JSP
- (d) None of the given options

APPSC Lect. 2017 (Degree College) (CS)



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APPSC Lect. 2017 (Degree College) (CS)

Ans. (a)

HTML (Hypertext Markup Language) has language elements which permit certain actions other than describing the structure of the web document. Which one of the following actions is NOT supported by pure HTML (Without any server or client side scripting) pages?

- (a) Embed web objects from different sites into the same page
- (b) Refresh the page automatically after a specified interval
- (c) Automatically redirect to another page upon download
- (d) Display the client time as part of the page

. NIELIT Scientists-B 22.07.2017 (IT)

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. NIELIT Scientists-B 22.07.2017 (IT) Ans. (d)

JavaScript numbers are stored as \_\_\_\_\_

- (a) integers
- (b) double precision floating point
- (c) double
- (d) floating point

GPSC Asstt. Prof. 30.06.2016



JavaScript numbers are stored as \_\_\_\_\_

- (a) integers
- (b) double precision floating point
- (c) double
- (d) floating point

GPSC Asstt. Prof. 30.06.2016 Ans. (b) :

# Common Built-in Functions

- JavaScript provides number of built-in functions.

| Functions              | Description                                                                                                                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| isNaN()                | <b>Returns true</b> , if the object is Not a Number.<br><b>Returns false</b> , if the object is a number.                                                                                                                                                                 |
| parseFloat<br>(string) | If the string begins with a number, the function reads through the string until it finds the end of the number; cuts off the remainder of the string and returns the result.<br>If the string does not begin with a number, the function returns NaN.                     |
| parseInt<br>(string)   | If the string begins with an integer, the function reads through the string until it finds the end of the integer, cuts off the remainder of the string and returns the result.<br>If the string does not begin with an integer, the function returns NaN (Not a Number). |
| String (object)        | Converts the object into a string.                                                                                                                                                                                                                                        |
| eval()                 | Returns the result of evaluating an arithmetic expression.                                                                                                                                                                                                                |

In Javascript, what does is NaN function do?

- (a) Return true, if the argument is not a number
- (b) Return false, if the argument is not a number
- (c) Return true, if the argument is a number
- (d) None of the given options

APPSC Lect. Degree College 07.06.2017 (CS)

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- (b) Return false, if the argument is not a number
- (c) Return true, if the argument is a number
- (d) None of the given options

APPSC Lect. Degree College 07.06.2017 (CS) Ans. (a)



# JavaScript Array

JavaScript array is an object that represents a collection of similar type of elements.

**The syntax of creating array directly is given below:**

```
var arrayname=new Array();
```

Here, new keyword is used to create instance of array.

## Javascript Operators

JavaScript includes operators as in other languages. An operator performs some operation on single or multiple operands (data value) and produces a result.

JavaScript includes following categories of operators.

1. Arithmetic Operators
2. Comparison Operators
3. Logical Operators
4. Assignment Operators
5. Conditional Operators

The correct syntax to write "Hi There" in JavaScript is

- (a) jscript. write (Hi There")
- (b) response. write ("Hi There")
- (c) Print ("Hi There")
- (d) print jscript ("Hi there")

ISRO Scientist/Engineer 2015

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- (c) Print ("Hi There")
- (d) print jscript ("Hi there")

ISRO Scientist/Engineer 2015 Ans. (b)

# Arithmetic Operators

Arithmetic operators are used to perform mathematical operations between numeric operands.

| Operator | Description                                          |
|----------|------------------------------------------------------|
| +        | Adds two numeric operands.                           |
| -        | Subtract right operand from left operand             |
| *        | Multiply two numeric operands.                       |
| /        | Divide left operand by right operand.                |
| %        | Modulus operator. Returns remainder of two operands. |
| ++       | Increment operator. Increase operand value by one.   |
| --       | Decrement operator. Decrease value by one.           |



# Comparison Operators

JavaScript language includes operators that compare two operands and return Boolean value true or false.

| Operator<br>s | Description                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------|
| ==            | Compares the equality of two operands without considering type.                                                  |
| ===           | Compares equality of two operands with type.                                                                     |
| !=            | Compares inequality of two operands.                                                                             |
| >             | Checks whether left side value is greater than right side value. If yes then returns true otherwise false.       |
| <             | Checks whether left operand is less than right operand. If yes then returns true otherwise false.                |
| >=            | Checks whether left operand is greater than or equal to right operand. If yes then returns true otherwise false. |
| <=            | Checks whether left operand is less than or equal to right operand. If yes then returns true otherwise false.    |

# Logical Operators

Logical operators are used to combine two or more conditions. JavaScript includes following logical operators.

| Operator | Description                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| &&       | && is known as AND operator. It checks whether two operands are non-zero (0, false, undefined, null or "" are considered as zero), if yes then returns 1 otherwise 0. |
|          | is known as OR operator. It checks whether any one of the two operands is non-zero (0, false, undefined, null or "" is considered as zero).                           |
| !        | ! is known as NOT operator. It reverses the boolean result of the operand (or condition)                                                                              |

# Assignment Operators

JavaScript includes assignment operators to assign values to variables with less key strokes.

| Assignment operators | Description                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------|
| =                    | Assigns right operand value to left operand.                                                             |
| +=                   | Sums up left and right operand values and assign the result to the left operand.                         |
| -=                   | Subtract right operand value from left operand value and assign the result to the left operand.          |
| *=                   | Multiply left and right operand values and assign the result to the left operand.                        |
| /=                   | Divide left operand value by right operand value and assign the result to the left operand.              |
| %=                   | Get the modulus of left operand divide by right operand and assign resulted modulus to the left operand. |

# Ternary Operator

JavaScript includes special operator called ternary operator :? that assigns a value to a variable based on some condition. This is like short form of if-else condition.

Syntax:

```
<condition> ? <value1> : <value2>;
```

Ternary operator starts with conditional expression followed by ? operator. Second part ( after ? and before : operator) will be executed if condition turns out to be true. If condition becomes false then third part (after :) will be executed.

## Conditional statements

In JavaScript we have the following conditional statements:

- Use if to specify a block of code to be executed, if a specified condition is true.
- Use else to specify a block of code to be executed, if the same condition is false
- Use else if to specify a new condition to test, if the first condition is false
- Use switch to specify many alternative blocks of code to be executed



## The if Statement

Use the if statement to specify a block of JavaScript code to be executed if a condition is true.

Syntax

```
if (condition) {  
  // block of code to be executed if the condition is true  
}
```

## The else Statement

Use the else statement to specify a block of code to be executed if the condition is false. if (condition) {

```
  // block of code to be executed if the condition is true  
} else {  
  // block of code to be executed if the condition is false  
}
```

## The else if Statement

Use the else if statement to specify a new condition if the first condition is false.

Syntax

```
if (condition1) {  
    // block of code to be executed if condition1 is true  
} else if (condition2) {  
    // block of code to be executed if the condition1 is false and condition2 is true  
} else {  
    // block of code to be executed if the condition1 is false and condition2 is false  
}
```

consider the following script: What will be the output of the above script?

```
<html>
<head><title>JavaScript</title></head>
<body>
<script language= "JavaScript">
var a = 80
var b = (a ==80 ? "pass": "fail");
document. write(b)
</script>
</body>
</html>
```

- (a) pass
- (b) fail
- (c) 80
- (d) Error at line = 6.



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</html>
```

- (a) pass
- (b) fail
- (c) 80
- (d) Error at line = 6.



# The JavaScript Switch Statement

Use the switch statement to select one of many code blocks to be executed.

Syntax

```
switch(expression) { case x:  
  // code block break;  
  case y:  
    // code block break; default:  
    // code block  
}
```

# JavaScript Loops

Loops are handy, if you want to run the same code over and over again, each time with a different value.

The statements for loops provided in JavaScript are:

- for statement
- do...while statement
- while statement

## for statement

A for loop repeats until a specified condition evaluates to false. The JavaScript for loop is similar to the Java and C for loop.

A for statement looks as follows:

for ([initialExpression]; [conditionExpression]; [incrementExpression]) statement

## **do...while statement**

The do...while statement repeats until a specified condition evaluates to false.

A do...while statement looks as follows:

```
do  
statement  
while (condition);
```

## **while statement**

A while statement executes its statements as long as a specified condition evaluates to true. A while statement looks as follows:

```
while (condition)  
statement
```

# Document Object Model

Every web page resides inside a browser window which can be considered as an object. A Document object represents the HTML document that is displayed in that window. The Document object has various properties that refer to other objects which allow access to and modification of document content.

The way a document content is accessed and modified is called the Document Object Model, or DOM. The Objects are organized in a hierarchy. This hierarchical structure applies to the organization of objects in a Web document.

- Window object – Top of the hierarchy. It is the outmost element of the object hierarchy.
- Document object – Each HTML document that gets loaded into a window becomes a document object. The document contains the contents of the page.
- Form object – Everything enclosed in the `<form>...</form>` tags sets the form object.
- Form control elements – The form object contains all the elements defined for that object such as text fields, buttons, radio buttons, and checkboxes.

# DOM compatibility

If you want to write a script with the flexibility to use either W3C DOM or IE 4 DOM depending on their availability, then you can use a capability-testing approach that first checks for the existence of a method or property to determine whether the browser has the capability you desire.

**For example** – if (document.getElementById) {

// If the W3C method exists, use it

} else if (document.all) {

// If the all[] array exists, use it

} else {

// Otherwise use the legacy DOM

}

In Java Script \_\_\_\_\_ is an object of the target language data type that encloses an object of the source language.

- (a) a wrapper
- (b) a link
- (c) a cursor
- (d) a form

RPSC VPITI 2018 (IT)

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- (b) a link
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- (d) a form

RPSC VPITI 2018 (IT) Ans. (a)

In Java Script, if you add a leading zero like “0127” the number system followed is

- (a) Decimal
  - (b) Hexadecimal
  - (c) Binary
  - (d) Octal
- RPSC VPITI 2018 (IT)



In Java Script, if you add a leading zero like “0127” the number system followed is

- (a) Decimal
  - (b) Hexadecimal
  - (c) Binary
  - (d) Octal
- RPSC VPITI 2018 (IT)

Ans. (d)

# Creating Functions,

A JavaScript function is defined with the function keyword, followed by a name, followed by parentheses (). Function names can contain letters, digits, underscores, and dollar signs (same rules as variables).

The parentheses may include parameter names separated by commas:  
(parameter1, parameter2, ...)

The code to be executed, by the function, is placed inside curly brackets:

```
{ } function name(parameter1, parameter2, parameter3) {  
// code to be executed  
}
```

# Event handling in JavaScript

## Introduction to Event Handling

- Event Handling is a software routine that processes actions, such as keystrokes and mouse movements.
- It is the receipt of an event at some event handler from an event producer and subsequent processes.
- Functions of Event Handling
- Event Handling identifies where an event should be forwarded.
- It makes the forward event.
- It receives the forwarded event.
- It takes some kind of appropriate action in response, such as writing to a log, sending an error or recovery routine or sending a message.
- The event handler may ultimately forward the event to an event consumer.

# Event Handlers

| Event Handler | Description                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| onAbort       | It executes when the user aborts loading an image.                                                                                                                                  |
| onBlur        | It executes when the input focus leaves the field of a text, textarea or a select option.                                                                                           |
| onChange      | It executes when the input focus exits the field after the user modifies its text.                                                                                                  |
| onClick       | In this, a function is called when an object in a button is clicked, a link is pushed, a checkbox is checked or an image map is selected. It can return false to cancel the action. |
| onError       | It executes when an error occurs while loading a document or an image.                                                                                                              |
| onFocus       | It executes when input focus enters the field by tabbing in or by clicking but not selecting input from the field.                                                                  |
| onLoad        | It executes when a window or image finishes loading.                                                                                                                                |
| onMouseOver   | The JavaScript code is called when the mouse is placed over a specific link or an object.                                                                                           |
| onMouseOut    | The JavaScript code is called when the mouse leaves a specific link or an object.                                                                                                   |

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| onReset  | It executes when the user resets a form by clicking on the reset button.            |
| onSelect | It executes when the user selects some of the text within a text or textarea field. |
| onSubmit | It calls when the form is submitted.                                                |
| onUnload | It calls when a document is exited.                                                 |

#### Example : Simple Program on onload() Event handler

```
<html>
<head>
<script
type="text/javascript">
function time()
{
    var d = new Date();
    var ty = d.getHours() +
    ":"+d.getMinutes()+":"+d.getSeconds();
    document.frmty.timeetxt.value=ty;
    setInterval("time()",1000)
}
</script>
</head>
<body onload="time()">
<center><h2>Displaying Time</h2>
<form name="frmty">
    <input type="text" name="timeetxt" size="8">
</form>
</center>
</body>
</html>
```

# Embedding JavaScript in HTML

Client-side JavaScript code is embedded within HTML documents in a number of ways:

- Between a pair of `<script>` and `</script>` tags
- From an external file specified by the `src` attribute of a `<script>` tag
- In an event handler, specified as the value of an HTML attribute such as `onclick` or `onmouseover`
- As the body of a URL that uses the special `javascript:` protocol

**The `<script>` Tag:** Client-side JavaScript scripts are part of an HTML file and are coded within `<script>` and `</script>` tags.

You may place any number of JavaScript statements between these tags; they are executed in order of appearance, as part of the document loading process. `<script>` tags may appear in either the `<head>` or `<body>` of an HTML document.

**Including JavaScript Files:** As of JavaScript 1.1, the `<script>` tag supports a `src` attribute. The value of this attribute specifies the URL of a file containing JavaScript code. It is used like this:

```
<script src="../../javascript/util.js"></script>
```

A JavaScript file typically has a `.js` extension and contains pure JavaScript, without `<script>` tags or any other HTML.

**Event Handlers :** JavaScript code in a script is executed once, when the HTML file that contains it is read into the web browser. A program that uses only this sort of static script cannot dynamically respond to the user. More dynamic programs define event handlers that are automatically invoked by the web browser when certain events occur. `<input type="checkbox" name="opts" value="ignore-case"`

```
onclick="ignore_case = this.checked;">
```



Java script is contained inside the\_\_\_\_\_ tags.

- (a) <font>.....</font>
- (b) <script>.....</script>
- (c) <head>.....</head>
- (d) <body>.....</body>

TRB Poly. Lect. 2017

Java script is contained inside the\_\_\_\_\_ tags.

- (a) <font>.....</font>
- (b) <script>.....</script>
- (c) <head>.....</head>
- (d) <body>.....</body>

TRB Poly. Lect. 2017 Ans. (b)

# Cookies

- Cookies are data, stored in small text files, on your computer.
- When a web server has sent a web page to a browser, the connection is shut down, and the server forgets everything about the user.
- Cookies were invented to solve the problem "how to remember information about the user":
- When a user visits a web page, his/her name can be stored in a cookie.
- Next time the user visits the page, the cookie "remembers" his/her name.

## Create a Cookie with JavaScript

JavaScript can create, read, and delete cookies with the `document.cookie` property.

With JavaScript, a cookie can be created like this:

```
document.cookie = "username=John Doe";
```

Client-side technology, such as \_\_\_\_\_ is used for validating user inputs.

- (a) JavaScript
- (b) ASP.NET
- (c) PHP
- (d) None

RPSC VPITI 2018 (IT)

Client-side technology, such as \_\_\_\_\_ is used for validating user inputs.

- (a) JavaScript
- (b) ASP.NET
- (c) PHP
- (d) None

RPSC VPITI 2018 (IT)

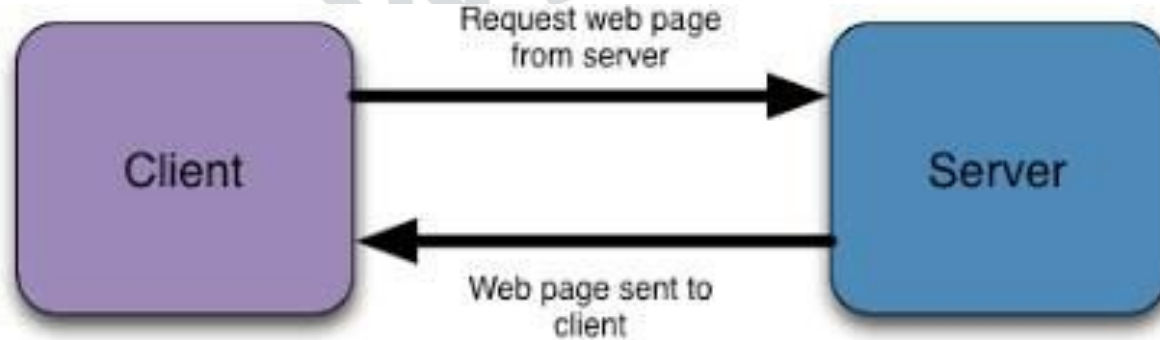
Ans. (a)

# Server Side Scripting

## Introduction to server side Scripting:

Server side scripting is defined as "web server technology in which the user's request is fulfilled by running a script directly on a web server to generate dynamic web pages"

Server side scripts runs on the server rather than our computer. When we visit a website, the script will create the webpage on dynamically, In case of server side scripting, the client Needs the request on the machine and then sends the result to the client machine



Popular server-side scripting languages include Perl PHP, ASP, JSP, Ruby, Cold fusion and Python.

### **Advantages:**

1. Load times are generally faster than client side scripting.
2. It does not require the user to download plugins like JAVA or flash.
3. We can create a single website template for the entire website. Each new dynamic page we create will automatically use it.
4. Our scripts are hidden from view, Users only see HTML output, even when they view the source

PHP scripts are executed on

- (a) Client computer
- (b) Server computer
- (c) ISP computer
- (d) It depends on PHP objects.

RPSC VPITI 2018 (CS)



PHP scripts are executed on

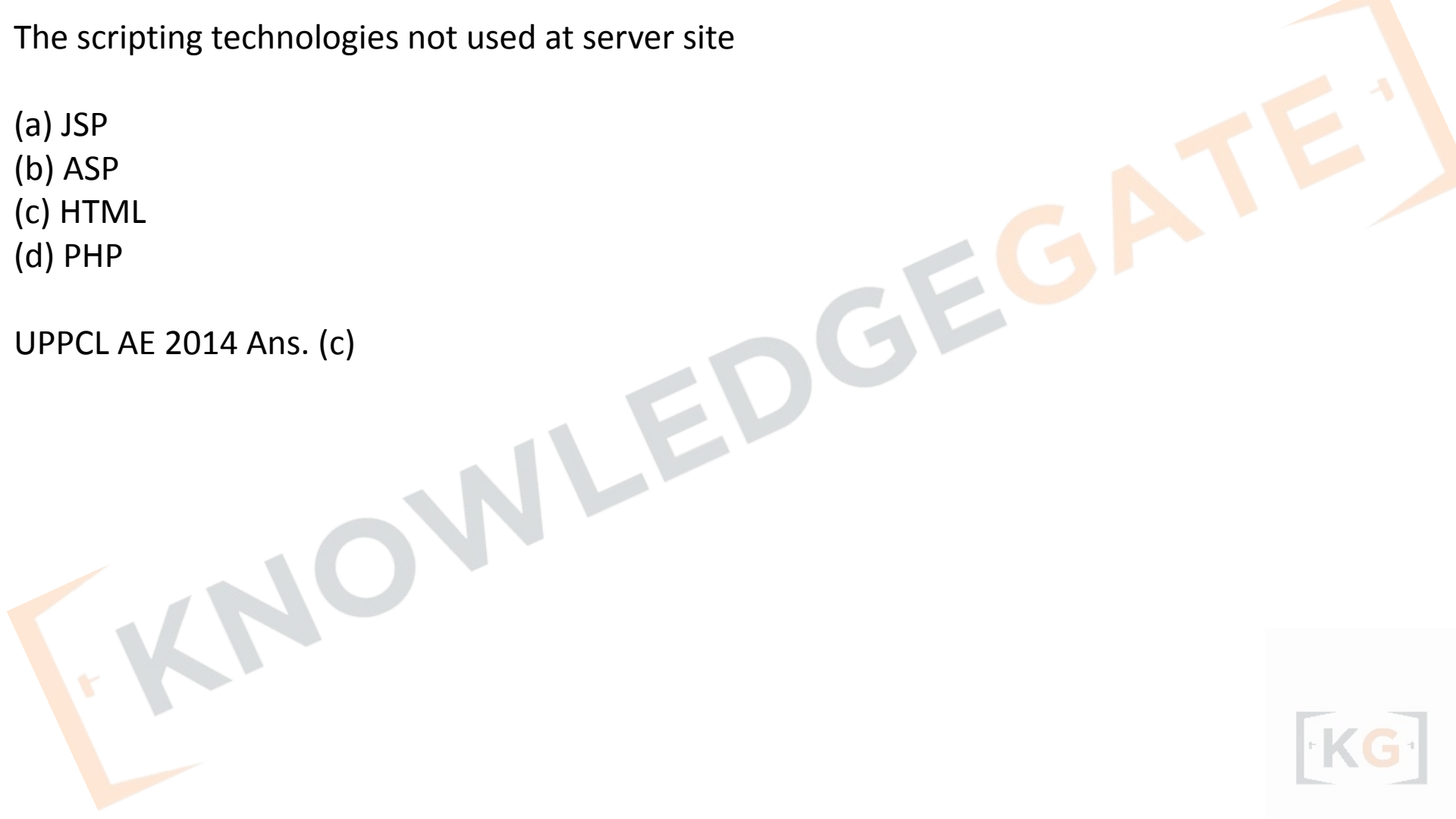
- (a) Client computer
- (b) Server computer
- (c) ISP computer
- (d) It depends on PHP objects.

RPSC VPITI 2018 (CS) Ans. (b)

The scripting technologies not used at server site

- (a) JSP
- (b) ASP
- (c) HTML
- (d) PHP

UPPCL AE 2014 Ans. (c)



The scripting technologies not used at server site

- (a) JSP
- (b) ASP
- (c) HTML
- (d) PHP

UPPCL AE 2014



# Components of server side Scripting:

The Components of server-side Scripting are:

**1. Web Server:** A web server is a software that continuously monitors for any request that has been made from client. It reads any request; it retrieves the particular file or values from the database and sends any the results to the client machine.

e.g. PWS (Personal Web Server), IIS (Internet Information Server), Apache

**2. Scripting Files:** The second part of the server-side scripting is the scripting files that are executed on the request of the client. These files are developed in any of the server-side scripting languages that are available with us: ASP (Active Server Pages), JSP (Java Server Pages), PHP (Personal Home Pages), Servlets.

**3. Database Files :** The third part of working of the server side scripting is the database files. These database files can be designed in any of the available database such that: MS Access, SQL Server, My SQL, ORACLE.

**4. SQL:** SQL is Structured Query Language that is used to interact with the databases. It is any Relational Database integral part of Management System (RDBMS) package system. SQL uses some syntaxed statements predefined to perform various database operations like.

**5. Scripting Methods :** The main scripting methods that are used in server side scripting are mentioned as under: (a) ASP (b) JSP

# Difference between CSS and SSS

| Server side scripting                                                                                            | Client side scripting                                                                                          |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Used to create dynamic pages based a number of conditions when the user's browser makes a request to the server. | used when the user's browser already has all the code and the page is altered on the basis of the users input. |
| The Web Server executes the server-side that produces the page to be sent to the browser.                        | The Web Browser executes the client scripting that resides at the user's computer                              |
| Server executes server-side scripts to send out a page but it does not execute client-side scripts               | The browser receives the page sent by the server and executes the client-side scripts.                         |
| Used to connect to the databases that reside on the web server.                                                  | Cannot be used to connect to the databases on the web server.                                                  |
| an access the file system residing at the web server.                                                            | can't access the file system that resides web server. at the web server                                        |
| The settings that belong to Web server can be accessed using server-side                                         | The files and settings that are local at the user's computer can be accessed                                   |
| can't be blocked by the user                                                                                     | possible to be blocked by the user.                                                                            |
| Response from a server-side script is slower as compared to a client-side script                                 | Response from a client-side script is faster as compared to a server-side script                               |
| Examples: PHP, JSP, ASP. Net, Ruby, Perl.                                                                        | Examples: Javascript, VB script, etc.                                                                          |

The scripting technologies not used at server site

- (a) JSP
- (b) ASP
- (c) HTML
- (d) PHP

UPPCL AE 2014

Ans. (c)

The scripting technologies not used at server site

- (a) JSP
- (b) ASP
- (c) HTML
- (d) PHP

UPPCL AE 2014

Ans. (c)



## Server-side Scripting method

some of the server side scripting methods are:

1. **(Common Gateway Interface):** Simple scripting mechanisms supported by practically web servers. Most commonly written in Perl, C, C++, or Python.
2. **Java Servlets:** Ability to execute Java objects on the server. Newer, less common technology but has dramatic performance advantages over CGI.

## JavaScript on server

JavaScript extends the capabilities of the server. By providing a scripting language the software can do more without calling an external program. This makes it easier for Webmasters to add features to their sites that browsers can take advantage of. And it can reduce the load on the server by keeping the processing within the server software.

1. **Request Object:** Each time a browser wants more information it sends a request to a server. Data about these requests are available on the server from the request object in JavaScript. If we have written CGI scripts.

1. **Client Object:** Each time a new client accesses the server application, a new client object is created. However, there are no built-in properties for the client object. If we need information retained about the client, then we create a property.
2. **JavaScript and CGIs:** When we are building websites, we have dealt with Common Gateway Interface (CGI) scripts. Prior to JavaScript, this was the primary means of creating interactive applications. Libraries of CGI scripts include counters, e-mailers, message boards, and many other functions.  
LiveWire can replace CGI programming. Instead of calling external programs, the server software runs applications that are closely integrated to it. JavaScript is the language of these applications. Applications are developed with three tools. We build these applications using LiveWire's Site manager. The source files for the applications are developed using the same HTML editors used to build browser JavaScript pages. The applications run in response to requests from Netscape Navigator.

**3. The JavaScript Balancing Act :** when interactivity was only done with a CGI, all of the processing was done on the server. Because Cpt can run on both the server and the browser, writing a successful application requires we to properly allocate the processing between the two.

**4. Exchanging Information between the Browser and Server:**

**From Client to Server:** User responses are submitted just as we currently handle forms. The user completes the form and clicks the submit button. The data from the radio buttons, checkboxes, textboxes, and text area are sent to the server. The server then places this data into the request object. Each element of the form has a corresponding property.

**From Server to Client:** Usually a server only returns a static page in response to a browser request. In LiveWire, the response is still a page, but the contents of the page vary. User input can result in changes to default form values, new values of hidden form elements, or direct substitutions.

Example:

```
<INPUT TYPE="text" NAME="example" VALUE="request.agent">
```

**Q.**What is the default time in minutes when a page session ends and if user does not refresh or request page? **(UPLT - 2018)**

- A.** 10
- B.** 15
- C.** 20
- D.** 25

